Transcendence of a planar reflection–group growth series and its relaxed companion

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Abstract

Let $W = \langle R_0, R_1, R_2 \rangle$ be the group generated by the reflections in the three sides of a right triangle in $\mathbb{Q}^2$ with positive unequal legs, and let $U(x) = \sum_{n \geq 0} u_n x^n$ be its geodesic word-length growth series (the integer sequence OEIS A396406) with relaxed companion V. We prove:

(i) the catalytic building-block q-series $\Sigma_0, \Sigma_1, S_0, S_1$ are transcendental over $\mathbb{Q}(q)$, *unconditionally*: their integer coefficients grow super-polynomially along the squares, so $|q| = 1$ is a natural boundary by the Pólya–Carlson theorem;

(ii) the common growth rate is $\beta_2 = 1.4916177871\ldots$ *exactly*, and $\frac{3}{2}$ is excluded;

(iii) the relaxed series V is transcendental over $\mathbb{Q}(x)$; the required pole set is supplied by the rectangle bound of Lemma 6 and the non-vanishing of the denominator block at the travel poles is proved by an elementary explicit chain (Appendix A): $|S_e(q_m)| \geq 0.35\sqrt{\tau_m}$ at every travel pole, with all constants numerical and no cited asymptotic theorem on this path. For the true series U we prove: U is transcendental over $\mathbb{Q}(x)$. The required amplitude estimate $(\star)$ ($|P_{12}(q_m)|w_m^3 < 2$ at every travel pole) is proved: the gate block is the symmetric second difference of the q-cosine at its own zero, a two-interval uniform certificate bounds it by $0.619\,\tau^{3/2} < \tau^{3/2}/\sqrt{2}$ for every pole with $\tau \leq 5 \cdot 10^{-3}$, and the six poles above that threshold are checked directly at 120-digit precision. An exact identity, $S_e = \frac{qZ}{2}\mathfrak{s}(Z) - P_{12}$, turns the same certificate into the denominator bound $|S_e| \geq 0.63\sqrt{\tau}$ and the sign input $b_0 > 0$ that close the whole transcendence gate. The turning-point amplitude atom on the U-side, long the special-function obstacle, is an explicit rational inequality *formally verified in Lean 4 / Mathlib* (Lemma 41);

(iv) an orthogonal, analysis-free route via Christol's theorem would settle U without any of the above: the p-kernel of ($u_n \bmod p$) is machine-verified to be maximally non-automatic at $p = 3, 5$, though a proof for dense-support series is open.

Thus both V and U rest on self-contained analysis: every coefficient is derived elementarily and machine-verified, the remainders are at the standard simple-saddle level, and the gate ($(\star)$, the denominator bound, and $b_0 > 0$) is proved; the precise rigor standing, including the arithmetic model of the committed certificates, is stated in the text. In sharp contrast with the rational growth of hyperbolic groups and the algebraic growth series of Parry's wreath products, the geometric word metric places this virtually-lamplighter group outside every algebraic class.

1 Introduction

A right triangle $T \subset \mathbb{Q}^2$ with legs $a \neq b$ has three sides; the reflections across them generate a group $W = \langle R_0, R_1, R_2 \rangle \leq \mathrm{Aff}(\mathbb{R}^2)$. Its *growth series* $U(x) = \sum_n u_n x^n$ counts group elements by

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minimal word length in $\{R_0, R_1, R_2\}$. Because the legs are unequal the dihedral angles are irrational multiples of π (Niven), so W is a generic triangle group [8] rather than a Coxeter group, and; unlike the rational Coxeter case governed by Steinberg's formula [21]; its growth is not a priori rational. The sequence $u_n = 1, 3, 5, 8, 13, 21, 34, 55, 89, \ldots$ is OEIS A396406; it coincides with the Fibonacci numbers F_{n+3} for $1 \le n \le 9$ and first deviates at $n = 10$. This paper settles the arithmetic nature of both series: the relaxed companion V is transcendental over $\mathbb{Q}(x)$ with every analytic input it consumes proved self-contained, and the true series U is transcendental over $\mathbb{Q}(x)$: the amplitude estimate ($\star$) and the full gate (numerator bound, denominator lower bound, and sign) are proved at every travel pole in Appendix B (Theorem 59, Lemma 60, Corollary 61).

Context. For word-growth series of groups, rationality is the norm in the classical families: hyperbolic groups have rational growth series with respect to every finite generating set (Cannon [3]), Coxeter systems are governed by Steinberg's rational formula, and for wreath products (including lamplighter-type groups) Parry computed the growth series for natural generating sets and found them *algebraic* [19]. Provably transcendental growth series are rare; the classical example is Stoll's [22]: the higher Heisenberg groups admit both rational and transcendental growth series, the transcendental one arising from a carefully constructed generating set. The group studied here is virtually the lamplighter $\mathbb{Z} \wr \mathbb{Z}$ [1], and its generating set is not constructed but imposed by Euclidean geometry; the three side-reflections of a right triangle. The results below show that for this natural geometric system the growth series is transcendental, by an explicit mechanism (a natural boundary for the building blocks; an accumulation of poles for the series themselves). This places the example in sharp contrast with the algebraicity of Parry's wreath-product series: the geometric word metric moves a virtually-lamplighter group outside every algebraic class. The higher-dimensional picture is subtler than a clean dimensional dichotomy [2]: the abstract right-angled Coxeter envelope of an n-orthoscheme has rational growth (rate $r_n = 1 + 2\cos\frac{2\pi}{n+3}$), but the geometric group is an amenable proper quotient that matches this rational envelope only to a finite depth before deviating below it. The plane is the one dimension in which the deviation has been computed past its onset; there it is the transcendental series treated here; and whether every higher dimension is likewise eventually transcendental is an open problem.

The relaxed/true dichotomy. The word length $\ell_T(g)$ is realised by strand walks in a finite transition system; permitting free isolated cycles yields a smaller, analytically tamer *relaxed* length $\ell_R(g)$, with $\ell_T = \ell_R + 2c(g)$, $c(g) \in \mathbb{Z}_{\ge 0}$ a cycle count. The relaxed counting function is the companion V; one has $v_n \ge u_n$, equality for $n \le 7$, first strict at $n = 8$. Both are governed by a linear q-difference (dilation $t \mapsto q^2 t$) *catalytic functional equation* (§2) that is q-holonomic and not of polynomial-kernel type.

Method. The catalytic transfer is assembled from four integer q-series $\Sigma_0, \Sigma_1, S_0, S_1$ with j^2-gapped coefficients. Their coefficients grow super-polynomially, so Pólya–Carlson yields unconditional transcendence over $\mathbb{Q}(q)$ (§3). Both V and U acquire poles at the *travel poles* q_m, the roots of $\Sigma_1^T(q) = 1$; transcendence follows from accumulation of these poles at $q = 1$ once a certain auxiliary integral T_2 is shown to be small (§4). What the theorems consume is only $|T_2| \to 0$, and that is proved self-contained in Lemma 6 by an absolute bound on a rectangle: no saddle, no path through a saddle, no cited asymptotic theorem. It is the single analytic input of this kind, shared by V (§5) and U (§6). The sharp size of T_2, with the constant $\sqrt{2}/36$, is recorded separately in Remark 5 as a numerically certain statement whose proof is incomplete (Remark 7); nothing here depends on it.

The U-side carries one extra ingredient. Writing the cocycle off-diagonal P_{12} in exact closed form as a Hahn–Exton q-Bessel function $J^{(3)}_{3/2}$, the transcendence gate reduces to one amplitude bound on that series at the travel poles. This bound is *proved* (Appendix B): P_{12} is the symmetric second difference of the q-cosine at its own zero (Proposition 52), so the gate is a second-derivative estimate on a window of width τ, delivered by a two-interval uniform certificate; an exact identity then converts the same certificate into the denominator lower bound and the sign input $b_0 > 0$, closing the gate in full (Corollary 61). The sharp constants behind the picture are *derived* in Remark 11: the pole–zero offset $\delta w = \frac{\sqrt{2}}{8}\tau^{3/2}$ and $|P_{12}|/\tau^{3/2} \to 1/(4\sqrt{2})$, a fourfold margin below the gate threshold. (The turning-point atom feeding that analysis, the rational envelope $h(X) = (1+\frac{3}{2X^2})/(1+\frac{1}{X^2}) \in [1, \frac{3}{2}]$ of Lemma 41, is formally verified in Lean 4.) U and V therefore rest on the same self-contained analysis; no special-function input separates them.

Section 7 records an independent arithmetic route: by Christol's theorem, if $(u_n \bmod p)$ is not p-automatic for one prime p then U is transcendental over $\mathbb{Q}(x)$; the p-kernel is maximally non-automatic at $p = 3, 5$.

2 The catalytic equation and its building blocks

Write $q = x^2 = e^{-\tau}$, $w = \sqrt{2/\tau}$. The relaxed length statistic satisfies a linear q-difference equation with dilation $t \mapsto q^2 t$; telescoping it produces four integer power series in q,

$$\Sigma_1(q) = \sum_{i\ge 1} (-1)^{i-1} \frac{w^{2i}}{(2i)!}\,\eta_i, \quad S_e(q) = \sum_{j\ge 0} \frac{(-2(1-q))^j q^{j(j+1)}}{(q;q)_{2j}}, \tag{1}$$

and their companions Σ_0, S_0, where $\eta_j = e^{-B_j}$ is the Gaussian-decaying form factor of §4 (equivalently $\eta_j = q^{j^2+j}[2(1-q)]^j(2j)!/[(q;q)_{2j}W^{2j}]$; the identification of this display with $1 - \Sigma_1 = \cos(Z; q^2)$ of Proposition 12 is the collapse used in Lemma 8). We write $\Sigma_1^{\mathrm{T}} = \Sigma_1$ when the travel rôle is to be emphasised. Both V and U are rational functions of these blocks over $\mathbb{Q}(x)$; concretely the relaxed travel resolvent is $G_0^T = \Sigma_0/(1-\Sigma_1)$, and U inherits its denominator $1 - \Sigma_1$ (Proposition 1). The *travel poles*

$$q_1 < q_2 < \cdots \uparrow 1, \qquad \Sigma_1^T(q_m) = 1,$$

form an infinite set accumulating at $q = 1$ (Lemma 8, proved in §4 once the collapse machinery is in place).

Proposition 1 (Invariance of the travel block)**.** *The denominator* $1 - \Sigma_1$ *is the identical, cycle-marker-free object in the generating functions of* U *and of* V*; hence* U *and* V *share the travel poles* q_m*.*

The proof is a strand-walk argument (every pure-travel element has cycle count 0, so the two length gradings agree on it); see the supplement [20].

3 Unconditional transcendence of the blocks

Theorem 2. *Each of* $\Sigma_0, \Sigma_1, S_0, S_1$ *is transcendental over* $\mathbb{Q}(q)$*.*

Proof. Each block is an integer power series with radius of convergence exactly 1. Along the squares its coefficients grow super-polynomially: for Σ_1 one has $|[q^{(j+1)^2}]\Sigma_1| \ge 2^{j+1}$, from the j^2-gapped catalytic telescoping (the block's coefficients $2, -4, 14, -52, 178, -856, 4626, \ldots$ at $q^{(j+1)^2}$, and the bound, are reproduced from the block's A/C recursion and machine-checked for $j \le 9$ in Lean 4, `PolyaCarlson.lean`, theorem `coeff_bound`). A power series with integer coefficients and radius 1 is, by the Pólya–Carlson theorem [5], either a rational function whose poles are roots of unity or has the unit circle as a natural boundary. Super-polynomial coefficient growth rules out the rational alternative, so $|q| = 1$ is a natural boundary; a function with a natural boundary is transcendental over $\mathbb{Q}(q)$. $\square$

This is the unconditional core: it uses no analytic gate. It does not, however, transfer directly to V or U, which are *ratios* of blocks in which the boundary singularities can cancel; those require the pole-accumulation argument below.

4 The steepest-descent estimate

The one analytic input for both V and U is the smallness of an auxiliary integral. Let $\varphi(y) = \log \frac{\sinh(y/2)}{y/2} = \sum_{n\geq 1} \varphi_n y^{2n}$ with $\varphi_n = \frac{(-1)^{n+1}\zeta(2n)}{n(2\pi)^{2n}}$. The form factor B_s is *defined* by the closed form (3) of Lemma 3; the familiar expansion $B_s \sim \sum_{n\geq 1} \varphi_n \tau^{2n}(P_n(2s) - s)$, $P_n(M) = \sum_{m=1}^{M} m^{2n}$ the Faulhaber sum, is its Euler–Maclaurin *asymptotic* series (equality at integer $2s$, divergent otherwise; Lemma 3(v)) and is used only truncated. The block S_e decomposes as $S_e = \cos W - T_2$ with $W = we^{-\tau/2}$ and

$$T_2 = \sum_{n\geq 1} (-1)^n \, g_n \, \frac{W^{2n}}{\Gamma(2n+1)}, \qquad g_s = 1 - e^{-B_s}. \tag{2}$$

Everything hinges on $T_2 \to 0$, and on its sharp size $|T_2| = O(\sqrt{\tau})$.

Lemma 3 (amplitude regularity)**.** *For* $\operatorname{Re} s > -\frac{1}{2}$ *set* $M = 2s$ *and*

$$B_s \;=\; \tfrac{\tau}{4} M(M{+}1) - M\log\tau - \log\Gamma(M{+}1) + \log(q;q)_\infty - \log(q^{M+1};q)_\infty - s\,\varphi(\tau), \tag{3}$$

$\varphi(y) = \log\frac{\sinh(y/2)}{y/2}$ *as above, each q-Pochhammer logarithm meaning the sum of principal logarithms* $\sum_{j\geq 0} \operatorname{Log}(1 - q^{M+1+j})$*: every factor has* $|q^{M+1+j}| < 1$*, hence* $\operatorname{Re}(1 - q^{M+1+j}) > 0$*, so this is the analytic branch on the half-plane (the logarithm of the* product *would differ by multiples of* $2\pi i$*). Then* B_s *is analytic on* $\operatorname{Re} s > -\frac{1}{2}$*, and at every integer* $n \geq 0$

$$B_n \;=\; \sum_{i=1}^{2n} \varphi(i\tau) \;-\; n\,\varphi(\tau), \tag{4}$$

so (3) *is the analytic interpolation of the collapse coefficients* $\eta_n = e^{-B_n}$ *of* (1)*. On the strip*

$$\mathcal{S} = \{-\tfrac{1}{2} < \operatorname{Re} s \leq 2w,\ |\Im s| \leq 2w\}, \qquad \tau \leq 2/\pi^2,$$

one has the explicit bounds

$$|B_s| \;\leq\; 30.3\sqrt{\tau}, \qquad |B_s'| \;\leq\; 7.6\,\tau, \tag{5}$$

and, on the smaller region $|M| \leq 2w$ *which contains the saddles, the truncation bound that Remark 5 consumes:*

$$\big|B_s - \varphi_1\tau^2\big(P_1(M) - s\big)\big| \;\leq\; 0.02\,\tau^{3/2}, \qquad \varphi_1 = \tfrac{1}{24}. \tag{6}$$

Consequently $|e^{-B_s}| \leq e^{30.3\sqrt{\tau}}$*, the amplitude* $A := 1 - e^{-B_s}$ *satisfies* $|A| \leq 1 + e^{30.3\sqrt{\tau}} \to 2$*, and along any rectifiable* $\Gamma \subset \mathcal{S}$ *of length* $\ell(\Gamma) \leq cw$*,*

$$V_\Gamma(A) \;\leq\; \sup_\Gamma |e^{-B_s}| \int_\Gamma |B_s'|\,|ds| \;\leq\; e^{30.3\sqrt{\tau}} \cdot 7.6\,\tau \cdot cw \;=\; O(\sqrt{\tau}) \;\longrightarrow\; 0,$$

using $\tau w = \sqrt{2\tau}$. The strip contains both saddles $s^* = -\frac{1}{4} \pm i(\frac{w}{2} + O(\sqrt{\tau}))$ of Remark 5 and the $O(w)$-length portion of its steepest-descent contour through them (which is all the $\ell(\Gamma) \le cw$ clause above requires, the Gaussian width being $O(\sqrt{w})$) and, being built from the larger argument $w = We^{\tau/2}$, it contains the corresponding W-strip as well.

Proof. (i) Closed form, and (4). Unwinding $\eta_n = q^{n^2+n}[2(1-q)]^n(2n)!/[(q;q)_{2n}W^{2n}]$ with $W^2 = (2/\tau)e^{-\tau}$ gives $\eta_n = e^{-\tau n^2}[(1-q)\tau]^n(2n)!/(q;q)_{2n}$. Now $1-q^i = i\tau\, e^{-i\tau/2}\,\frac{\sinh(i\tau/2)}{i\tau/2}$, i.e. $\log(1-q^i) = \log(i\tau) - \frac{i\tau}{2} + \varphi(i\tau)$; summing over $1 \le i \le M$,

$$\log(q;q)_M = \log\Gamma(M{+}1) + M\log\tau - \tfrac{\tau}{4}M(M{+}1) + \sum_{i=1}^{M}\varphi(i\tau). \tag{7}$$

Since $\log[(1-q)\tau] = -\frac{\tau}{2} + 2\log\tau + \varphi(\tau)$, substituting (7) into $-\log\eta_n$ collapses every elementary term and leaves exactly (4); and rewriting $\log(q;q)_M = \log(q;q)_\infty - \log(q^{M+1};q)_\infty$ in (7) turns (4) into (3) for all s. Every ingredient of (3) is analytic in s for $\operatorname{Re} s > -\frac{1}{2}$: there $|q^{M+1}| < 1$ keeps $(q^{M+1};q)_\infty$ zero-free and the branch above canonical, and $\operatorname{Re}(M{+}1) > 0$ keeps $M{+}1$ off the poles of Γ. (Both singular factors do in fact cancel at $M = -1$, so B_s continues past $\operatorname{Re} s = -\frac{1}{2}$; we never need that.)

(ii) Splitting into Gamma blocks with exact cancellation. By the Weierstrass product $\frac{\sinh(y/2)}{y/2} = \prod_{k\ge1}(1 + \frac{y^2}{4\pi^2k^2})$ we have $\varphi(y) = \sum_{k\ge1}\log(1 + y^2/(4\pi^2k^2))$, so with $R_k := 2\pi k/\tau$ and $\prod_{n=1}^{M}(1 + n^2a^2) = a^{2M}\,\Gamma(M{+}1{-}\frac{i}{a})\Gamma(M{+}1{+}\frac{i}{a})/\big[\Gamma(1{-}\frac{i}{a})\Gamma(1{+}\frac{i}{a})\big]$ at $a = 1/R_k$, (4) becomes

$$B_s = \sum_{k\ge1} G(M;R_k) - s\,\varphi(\tau), \qquad G(M;R) := \int_0^M \Big[\psi(1{+}u{+}iR) + \psi(1{+}u{-}iR) - 2\log R\Big]du, \tag{8}$$

the integral along the segment $[0,M]$ fixing the branch (legitimate since $|M| < R_1$ puts the segment far from the poles of $\psi(1{+}u \pm iR)$ at $u = -1 - j \mp iR$). *The $-2\log R$ is not a normalisation: it is the exact cancellation of the two leading Stirling terms*, and it is what makes the k-sum converge; without it each block would grow like $2M\log R$.

Identity (8) is proved at integers, where the i-sum is finite and the interchange of $\sum_i$ and $\sum_k$ is trivial; it extends to all s as follows.

Domain. Both sides are analytic on the half-plane $\operatorname{Re} M > -1$: the left by (i), the right because the segment $[0,M]$ then stays in $\operatorname{Re} u > -1$ while the poles of $\psi(1{+}u \pm iR)$ lie in $\operatorname{Re} u \le -1$, and because $\sum_k G$ converges locally uniformly there, all but finitely many k having $R_k > 2|M|$ and hence $|G(M;R_k)| = O(|M|^3/R_k^2)$. (The disc $|M| < R_1$ of step (iii) is where the *quantitative* majorant lives; it is not the domain of analyticity, and the recursion below would walk out of it.)

Recursion. Write $\widetilde{B} = B_s + s\varphi(\tau)$ for either side with the elementary term restored. Both obey the *same* unit recursion in M: on the left $\frac{\tau}{4}[(M{+}1)(M{+}2) - M(M{+}1)] - \log\tau - \log(M{+}1) + \log(1 - q^{M+1}) = \varphi((M{+}1)\tau)$ term by term, and on the right $G(M{+}1;R) - G(M;R) = \log(1 + (M{+}1)^2/R^2)$, which sums over k to the same $\varphi((M{+}1)\tau)$ by the Weierstrass product. (For B_s itself both increments are $\varphi((M{+}1)\tau) - \frac{1}{2}\varphi(\tau)$, the extra term being common to the two sides.)

Conclusion. Hence the difference D is 1-periodic on $\operatorname{Re} M > -1$, so it continues to a 1-periodic *entire* function. Its Fourier coefficients satisfy $|c_m| \le e^{2\pi m y}\sup_{0\le x\le1}|D(x{+}iy)|$ for every real y, and D grows at most polynomially: on the left by Stirling, on the right by splitting the k-sum at $R_k = 2|M|$: the $O(|M|\tau)$ low terms are each $O(|M|\log|M|)$ by Stirling and the remainder is $O(|M|^3\sum_{R_k>2|M|}R_k^{-2}) = O(|M|^2\tau)$. Letting $y \to -\infty$ for $m > 0$ and $y \to +\infty$ for $m < 0$ gives $c_m = 0$ for $m \ne 0$, so D is constant; and $D(0) = 0$ by (4) at $n = 0$, so $D \equiv 0$.

(iii) The integrand. For $|z| \geq 1$ with $|\arg z| \leq \frac{3\pi}{4}$ one has $|\psi(1+z) - \log z - \frac{1}{2z}| \leq |z|^{-2}$ (the constant is 0.13 on that sector; the bound fails without limit as $\arg z \to \pi$, which is why the sector, not a distance to the cut, is the right hypothesis). On $\mathcal{S}$, $|M| \leq 4\sqrt{2}\,w$ and $|M|/R_1 = \frac{4\sqrt{2}\,w\tau}{2\pi} = \frac{4}{\pi}\sqrt{\tau} \leq 0.58$, so every $R_k > |M|$; and along the segment $|\arg(u \pm iR_k)| \leq 120^\circ < 135^\circ$ with $|u \pm iR_k| \geq R_1 - |M| > 13$, so the estimate applies throughout. Adding the two conjugate terms, the logarithms combine with $-2\log R$ to $\log(1 + u^2/R^2)$ and the $\frac{1}{2z}$ terms to $u/(u^2+R^2)$, whence for $|u| \leq |M|$

$$\left|\psi(1+u+iR) + \psi(1+u-iR) - 2\log R\right| \;\leq\; \frac{|M|^2 + |M|}{R^2 - |M|^2} + \frac{2}{(R - |M|)^2} \;=:\; \gamma(R).$$

(iv) Summation. Hence $|G(M;R_k)| \leq |M|\,\gamma(R_k)$ and, from (8),

$$|B_s| \;\leq\; |M| \sum_{k\geq 1} \gamma(R_k) + |s|\,\varphi(\tau), \qquad |B_s'| \;\leq\; 2\sum_{k\geq 1} \gamma(R_k) + \varphi(\tau),$$

the second because $\partial_M G = \psi(1+M+iR) + \psi(1+M-iR) - 2\log R$ and $M = 2s$. Both sums are dominated by $\sum_k R_k^{-2} = \tau^2/24$; carrying the finite corrections explicitly at $|M| = 4\sqrt{2}\,w$, $\tau \leq 2/\pi^2$ gives $|M|\sum_k \gamma(R_k) + |s|\varphi(\tau) \leq 30.3\sqrt{\tau}$ and $2\sum_k \gamma(R_k) + \varphi(\tau) \leq 7.6\,\tau$, which is (5). Both ratios are *increasing* in τ on the range (30.20, 25.14, 23.26, 22.21, 21.64 and 7.55, 6.28, 5.81, 5.55, 5.41 at $\tau = 2/\pi^2$, 0.1, 0.05, 0.02, 0.005, decreasing to the limits $\frac{64}{3}$ and $\frac{16}{3}$ as $\tau \to 0$), so each attains its maximum at the right endpoint $\tau = 2/\pi^2$, and the stated constants hold throughout.

For (6): expand the integrand of G one order further, now *keeping the third Stirling term* $-\frac{1}{12z^2}$, which is what supplies the linear part of P_1: its integral is $\frac{M}{6(M^2+R^2)} = \frac{M}{6R^2} + O(|M|^3/R^4)$. With $\int_0^M \log(1 + u^2/R^2)du = \frac{M^3}{3R^2} + O(|M|^5/R^4)$ and $\int_0^M u/(u^2+R^2)du = \frac{M^2}{2R^2} + O(|M|^4/R^4)$ this gives $R^2 G(M;R) \to P_1(M) = \frac{M^3}{3} + \frac{M^2}{2} + \frac{M}{6}$, so with $\sum_k R_k^{-2} = \tau^2/24$ exactly,

$$B_s = \varphi_1 \tau^2 \big(P_1(M) - s\big) + O(|M|^5 \tau^4) + O(|s|\tau^4).$$

On $|M| \leq 2w$ every remainder is $O(\tau^{3/2})$, and the constant is 0.02 (worst 0.0177 at $\tau = 2/\pi^2$, decreasing).

(v) What is not *used.* The τ-expansion $B_s = \sum_{n\geq 1} \varphi_n \tau^{2n}(P_n(2s) - s)$, P_n the Faulhaber sum, is the Euler–Maclaurin *asymptotic* expansion of (3): it is *divergent* at non-integer $2s$, since $P_n(2s)$ carries a Bernoulli part of size $(2n)!\, e^{2\pi|\Im 2s|}/(2\pi)^{2n+1}$. Nothing above sums it; it is used only in the truncated form "leading term plus remainder", as in Remark 5, where the $n = 1$ term at the saddle supplies the constant $\sqrt{2}/36$. ☐

The algebraic skeleton of this proof is machine-checked in Lean 4 (`BboundedSkeleton.lean`, ten theorems, no `sorry`, axioms `propext`, `Classical.choice`, `Quot.sound`). Five carry the arithmetic content: the Gauss sum, the summation (7) built on it, the telescoping cancellation (4), the scaling identity $(\tau^2 w^3)^2 = 8\tau$ (exact for w, only $e^{-3\tau/2}$-approximate for W), and the strip condition $|M|/R_1 < 1$. The other five (the per-factor rewriting, the collapse of $\log[(1-q)\tau]$, the cancellation $2M\log a + 2M\log R = 0$ at $aR = 1$, the w-definition, and the bounded-variation arithmetic) are *rearrangements of their hypotheses*, verified as such: they certify that the paper's algebra is what it claims, not the transcendental facts fed into them. Those facts (the Weierstrass product, the digamma remainder, the convergence of the k-sum, and the continuation argument of step (ii)) are not formalised.

Remark 4 (sharpness). The majorant of step (iv) is lossy by about a factor 6: the true extremes on $\mathcal{S}$ are $\operatorname{Re} B_s \geq -5.03\sqrt{\tau}$ and $|B_s'| \leq 5.46\,\tau$, and $\sup_{\mathcal{S}} |A| = 9.02$ at $\tau = 2/\pi^2$ (`star_gate_certificate.py`, part (7)). Only the proved bounds (5) are used anywhere.

The sketch exhibits the saddles explicitly and derives the constants by Cauchy's theorem, Stirling's formula and elementary Gaussian integration, and it would end by invoking a second-order remainder estimate for a simple non-degenerate saddle, [17, Ch. 4, Thm. 7.1]. Its hypotheses are *not* verified here (the saddle is flat rather than non-degenerate, and the amplitude's bounded variation is available only on a bounded piece of the path), which is one of the reasons the sketch is not a proof (Remark 7). No theorem of this paper rests on it.

Remark 5 (the sharp size of T_2; HEURISTIC). Numerically, $T_2 = \frac{\sqrt{2}}{36}\sqrt{\tau}\,\sin X - \frac{161}{1296}\,\tau\cos X + O(\tau^{3/2})$ uniformly in the phase, for any argument X with $\tau X^2 \in [2e^{-\tau}, 2]$; the constant and the exponent of the remainder are certain to eight digits over $m \le 1024$. The sketch below indicates why, by steepest descent, but it does *not* constitute a proof, for the reasons set out in Remark 7. Accordingly this statement carries no weight anywhere in this paper: the only property of T_2 that the theorems consume is the crude bound $|T_2| \to 0$ of Lemma 6, which is proved, and which is what Lemma 8 cites.

Sketch, not a proof; see Remark 7. Since $\cos W = \sum_{n\ge 0}(-1)^n W^{2n}/(2n)!$, (2) gives the exact alternating sum $S_e = \cos W - T_2 = \sum_{n\ge 0}(-1)^n e^{-B_n} W^{2n}/(2n)!$ (with $B_0 = 0$). Let $b(z) = e^{-B_z} W^{2z}/\Gamma(2z+1)$, analytic for $\operatorname{Re} z > -\frac{1}{2}$ (by (3) and Lemma 3; *not* by termwise polynomiality, the expansion being divergent off z). As $\pi\csc\pi z$ has residue $(-1)^n$ at each integer, Cauchy's theorem gives $S_e = \frac{1}{2\pi i}\oint b(z)\,\pi\csc(\pi z)\,dz$ over a contour enclosing $\{0, 1, 2, \dots\}$. Write the integrand as $e^{h(z)}\pi\csc\pi z$ with $h(z) = 2z\log W - \log\Gamma(2z+1)$; including the csc phase, the exponent is stationary where $2\log W - 2\psi(2z+1) \pm i\pi = 0$, so by $\psi(w) = \log w + O(1/w)$ there are two simple, non-degenerate saddles

$$z_\pm^* = \pm\tfrac{i}{2}W\big(1 + O(1/W)\big), \qquad h''(z_\pm^*) = \pm\tfrac{4i}{W} + O(W^{-2}) \neq 0.$$

Deform the contour to the steepest-descent paths through $z_\pm^*$; this is legitimate because b decays super-exponentially as $\operatorname{Re} z \to +\infty$ and csc suppresses $|\Im z| \to \infty$. A direct Stirling evaluation shows that at the modulation-free level $B \equiv 0$ the two saddles sum to $\cos W$ *exactly*: the divergent factors $e^{\pm\pi W/2}$ from $\csc(\pi z_\pm^*)$ and from $1/\Gamma(\pm iW + 1)$ cancel identically, each saddle contributing $\frac{1}{2}e^{\pm iW}$. Restoring the modulation scales each saddle by $e^{-B_{z_\pm^*}}$; since $P_1(2z) = \frac{2z(2z+1)(4z+1)}{6}$ and $\varphi_1 = \frac{1}{24}$,

$B_{z_+^*} = \varphi_1\tau^2\big(P_1(iW) - \frac{iW}{2}\big) + O(\tau)$ (the $\tau^{3/2}$ part of the remainder is (6); the $O(\tau)$ also absorbs $B_{z_+^*} - B_{iW/2}$ a

(using $\tau^2 W^3 = 2\sqrt{2}\,\sqrt{\tau}\,e^{-3\tau/2}$, the factor $e^{-3\tau/2}$ being absorbed in the $O(\tau)$), and $B_{z_-^*} = \overline{B_{z_+^*}}$. Hence

$$S_e = \tfrac{1}{2}e^{-B_{z_+^*}}e^{iW} + \tfrac{1}{2}e^{-B_{z_-^*}}e^{-iW} + E = \cos W - \tfrac{\sqrt{2}}{36}\sqrt{\tau}\,\sin W + E.$$

The remainder E is $O(\tau)$ by the classical second-order estimate for a simple saddle [17, Ch. 4, Thm. 7.1]: parametrising the descent path by arclength s, the integrand equals $e^{h(z^*)}e^{-s^2}(1 + O(sW^{-1/2}) + O(\tau))$, the next saddle coefficient is $\propto h'''(z^*) = O(W^{-2}) = O(\tau)$, and the bounded-variation hypothesis on the amplitude e^{-B_z} along the path is supplied by Lemma 3. This citation is the proof's one non-elementary step. The saddle prediction matches S_e to $O(\tau)$ over $\tau = 0.02, 0.01, 0.005$ numerically (`saddle_verify.py`). Thus $T_2 = \frac{\sqrt{2}}{36}\sqrt{\tau}\,\sin W + O(\tau)$, with $\sin W = \sin w + O(\sqrt{\tau})$. □

Lemma 6 (absolute bound on T_2)**.** *For $\tau \le 0.02$, $|T_2| \le 0.17\,\tau^{1/4}$; in particular $|T_2| \le 0.064 < 1$ there, and $|T_2| \to 0$ as $\tau \to 0$. The argument uses no saddle, no path through a saddle, and no cited asymptotic theorem: only the Mellin–Barnes residues, a rectangle contained in the strip $\mathcal{S}$*

of Lemma 3, and the amplitude bounds proved there. The same holds at any argument X with $\tau X^2 \in [2e^{-\tau}, 2]$; the constant above is the one for $X = w$, which is the case Lemma 8 uses and the worse of the two.

Proof. Since $\operatorname{Res}_{s=n} \pi/\sin \pi s = (-1)^n$ and $h(s) = X^{2s} g_s/\Gamma(2s+1)$ is analytic on $\operatorname{Re} s > -\frac{1}{2}$ (Lemma 3(i)), the terms of (2) are the residues of $h(s)\pi/\sin \pi s$ at $s = 1, 2, \ldots$. Split the series at $n = \lfloor 2w \rfloor$.

The tail. At integers, $B_n = \sum_{i \le 2n} \varphi(i\tau) - n\varphi(\tau) \ge n\varphi(\tau) \ge 0$ by (4) and the monotonicity of φ, so $0 \le g_n = 1 - e^{-B_n} \le 1$ and $|\sum_{n>2w} (-1)^n g_n X^{2n}/(2n)!| \le \sum_{n>2w} X^{2n}/(2n)!$, which is below 10^{-9} for $\tau \le 0.02$ and decreases superexponentially (it is $7.5 \cdot 10^{-10}$ at $\tau = 0.02$ and $1.9 \cdot 10^{-11}$ at $\tau = 0.0127$).

The head. The remaining residues are enclosed by the rectangle $R = \{\frac{1}{2} \le \operatorname{Re} s \le 2w,\ |\Im s| \le X/2\}$, which lies *wholly inside* $\mathcal{S}$, so the amplitude bounds of Lemma 3 apply at every point of ∂R; no bound on B_s outside $\mathcal{S}$ is needed anywhere. On $\operatorname{Re} s = \frac{1}{2}$ we have $|\sin \pi s| = \cosh \pi t \ge 1$, so no pole is grazed. On the horizontal edges the two exponentials cancel: $|\pi/\sin \pi s| \le \pi/\sinh(\pi X/2) \le 2.01\pi e^{-\pi X/2}$ (valid once $X \ge 1.69$, hence throughout $\tau \le 0.02$, where $X \ge 9.9$), while $1/|\Gamma(2\sigma + 1 + iX)|$ carries the compensating $e^{+\pi X/2}$; the resulting kernel is $O(\sqrt{2\pi/X})$, attaining 0.793 at $\sigma = \frac{1}{2}$ against $\sqrt{2\pi/X} = 0.797$ when $\tau = 0.02$.

For the amplitude: where $|2s| \le 2w$, (6) gives $|B_s| \le \frac{\tau^2}{24}\left|\frac{M^3}{3} + \frac{M^2}{2} - \frac{M}{3}\right| + 0.02\,\tau^{3/2}$ with $M = 2s$, hence $|g_s| = |1 - e^{-B_s}| \le |B_s| e^{|B_s|}$; elsewhere on ∂R, $|g_s| \le 1 + e^{30.3\sqrt{\tau}}$ by (5). The quantitative content is the resulting contour integral, evaluated by adaptive quadrature piecewise between the region breakpoints (certificate `t2abs_certificate.py`); the prose above fixes the geometry, not the constant. The ratio $|T_2|/\tau^{1/4}$ comes out 0.1624, 0.0428, 0.0308, 0.0226, 0.0173, 0.0136, 0.0115 at $\tau = 2 \cdot 10^{-2},\ 1.27 \cdot 10^{-2},\ 10^{-2},\ 5 \cdot 10^{-3},\ 10^{-3},\ 10^{-4},\ 10^{-5}$: increasing in τ, so its supremum on $\tau \le 0.02$ is the endpoint value 0.1624, and 0.17 bounds it. At $\tau = \tau_4 = 0.012665$, the largest value Lemma 8 uses, the bound is 0.0144. □

Remark 7 (why Remark 5 is not used). The steepest-descent sketch is recorded because it produces the sharp constant, but it is not a proof, on three counts. (i) It asserts that the two saddles sum to $\cos W$ *exactly*; they do not: the leading saddle term carries a phase $e^{-i/(24W)}$, so the two sum to $\cos W + \frac{1}{24W} \sin W + O(W^{-2})$. (ii) It sizes the first saddle correction by $h'''(z^*) = O(W^{-2})$; but the saddle is *flat* ($h'' = 4i/W \to 0$), so the expansion parameter is W^{-1} and the correction is $c_1 = i/(24W) + O(W^{-2})$. Errors (i) and (ii) are each about three quarters of the quantity being computed and cancel exactly, which is why the stated constant is nevertheless right; taken literally the chain yields $\sqrt{2}/144$, four times too small. (iii) The deformation is justified by the claim that csc suppresses $|\Im z| \to \infty$, which is false for the exact B, as the numbers in the previous proof show; and the uniform remainder is attributed to a steepest-descent error estimate whose hypotheses (a parameter-independent phase and amplitude, an explicit path, an explicit truncation order) are not verified here. A complete proof needs the two-term flat-saddle expansion together with an error estimate on the curved path; that is not written, and nothing in this paper waits on it.

Lemma 8 (infinitude and accumulation of the travel poles)**.** *The set $\{q \in (0,1) : \Sigma_1^T(q) = 1\}$ is infinite, and together with Lemma 53 it accumulates exactly at $q = 1$. In particular there is a travel pole with $w \in (m\pi, (m+1)\pi)$ for every large m.*

Proof. By Proposition 12, $1 - \Sigma_1^T = \cos(Z; q^2)$ with $Z^2 = z_0^2 e^{\tau}$. The collapse above defines the radius-free coefficients $e^{-B_n} = q^{n^2+n}[2(1-q)]^n (2n)!/[(q;q)_{2n} W^{2n}]$ through $S_e = \cos(z_0; q^2) = \sum_{n \ge 0} (-1)^n e^{-B_n} W^{2n}/(2n)!$; replacing z_0^2 by $Z^2 = z_0^2 e^{\tau}$ multiplies the n-th coefficient by $e^{n\tau}$, i.e.

replaces W by $We^{\tau/2} = w$ exactly:

$$1 - \Sigma_1^T \;=\; \sum_{n\geq 0}(-1)^n e^{-B_n}\,\frac{w^{2n}}{(2n)!} \;=\; \cos w - T_2(w),$$

with $T_2(\cdot)$ the series (2) evaluated at argument w in place of W; the coefficients e^{-B_n} are the *same* ones, so no re-derivation of the amplitude is involved. Only the argument moves, and it moves inside the region where the two estimates are already proved: Lemma 3 is stated on a strip built from the larger argument w (and containing the W-strip), so it applies verbatim, and Lemma 6 is stated for any argument with $\tau X^2 \in [2e^{-\tau}, 2]$, hence at $X = w$, where $\tau w^2 = 2$ identically. Thus $|T_2(w)| \leq 0.064$ once $\tau \leq 0.02$, which along the extreme-phase sequence $w = m\pi$, $\tau^{(m)} = 2/(m\pi)^2$, holds for every $m \geq 4$ (there $\tau \leq \tau_4 = 0.012665$ and the bound is 0.0144). Along that sequence

$$1 - \Sigma_1^T\big|_{w=m\pi} \;=\; (-1)^m - T_2(m\pi),$$

which has the sign of $(-1)^m$ for all $m \geq m_0$, with $m_0 = 4$ from Lemma 6 (only $|T_2(m\pi)| < 1$ is used, a far weaker input than the sharp size recorded in Remark 5). By continuity of $q \mapsto \Sigma_1^T(q)$ the intermediate value theorem yields a zero of $1 - \Sigma_1^T$ with $w \in (m\pi, (m+1)\pi)$ for every $m \geq m_0$: infinitely many travel poles, with $\tau \downarrow 0$, i.e. $q \uparrow 1$. Since every travel pole has $q > \frac{1}{10}$ and the pole set is locally finite in $(0,1)$ (Lemma 53), its only accumulation point is $q = 1$. □

The alternation is in fact immediate: $|T_2(\pi)| \leq 0.027$ by direct evaluation and $|T_2(m\pi)| = O(\tau_m)$ thereafter, so $m_0 = 1$ and there is a travel pole in every window $(m\pi, (m+1)\pi)$, $m \geq 1$. The dominant pole q_1 is the one below this ladder, in $w \in (0, \pi)$. The transport identity and the sign alternation (checked for $m \leq 40$, worst $|T_2(m\pi)| = 0.0267$) are verified in `star_gate_certificate.py`.

Remark 9 (the Hubbard–Stratonovich integral representation). Writing $q^{j(j+1)} = q^j e^{-j^2\tau}$ and linearising the Gaussian through $e^{-j^2\tau} = (4\pi\tau)^{-1/2}\int_{\mathbb{R}} e^{-u^2/(4\tau)} e^{iju}\,du$, the denominator block admits the *exact* real integral

$$S_e = \frac{1}{\sqrt{4\pi\tau}}\int_{-\infty}^{\infty} e^{-u^2/(4\tau)}\,\Psi(u,q)\,du, \qquad \Psi(u,q) = \sum_{j\geq 0}\frac{\left(-2(1-q)q\,e^{iu}\right)^j}{(q;q)_{2j}}, \tag{9}$$

the interchange of sum and integral being justified by absolute convergence ($\sum_j |Z|^j/(q;q)_{2j} < \infty$, $Z = -2(1-q)q\,e^{iu}$, against the integrable Gaussian). Because the coefficients of Ψ in $\zeta = e^{iu}$ are real, $\Psi(-u,q) = \overline{\Psi(u,q)}$, so (9) is real and may be written over a single variable, $S_e = (4\pi\tau)^{-1/2}\int_{\mathbb{R}} e^{-u^2/(4\tau)}\,\mathrm{Re}\,\Psi\,du$; the exact Gaussian moment underlying it, $\int_{\mathbb{R}} e^{-u^2/(4\tau)} e^{iju}\,du = \sqrt{4\pi\tau}\,e^{-\tau j^2}$, is machine-checked in Lean 4 (`GaussHS.lean`, theorem `gaussian_moment`; axioms `propext`, `Classical.choice`, `Quot.sound`, no `sorry`).

This representation, though exact and convenient, is *not* an analytic shortcut. The amplitude $\Psi(\cdot,q)$ is entire in ζ, but it is *not* uniformly bounded as $q \to 1^-$: on any fixed window it grows like $\cosh\!\big(w\sin(u/2)\big)$ (numerically $|\Psi|$ already exceeds 300 at $u = 1$, $\tau = 10^{-2}$), and its termwise limit $\cos(w\,e^{iu/2})$ *differs* from Ψ by an $O(1)$ factor at the dominant scale $j \sim \tau^{-1/2}$ (where the subleading part of $(q;q)_{2j}$ is itself $O(1)$; numerically $\Psi/\cos(we^{iu/2}) \approx 0.62$). The size of S_e is therefore governed by the balance between this exponential growth and the Gaussian; a simple *complex* saddle at $u^* \approx -\sqrt{2\tau}$, of the same order of difficulty as the direct sum, *not* a bounded-amplitude stationary-phase integral with trivial hypotheses. In particular the bounded-variation input of Lemma 3 is *not* eliminated by passing to (9); it, together with the rectangle bound of Lemma 6, remains the analytic core. The value of (9) is representational (one real variable, an entire integrand, an exact Lean-checked moment), not a reduction of hypotheses; the numerical agreement of its saddle value with S_e is recorded in the supplement (`gaussint_verify.py`).

Remark 10 (an elementary derivation of the coefficients of Remark 5). The leading term and the constant $\sqrt{2}/36$ of Remark 5 can be obtained without the steepest-descent theorem, by an exact algebraic reduction. Writing $1-e^{-k\tau}=2e^{-k\tau/2}\sinh(k\tau/2)$ throughout, the j-th term of $S_e=\sum_{j\geq 0}(-2(1-q))^j q^{j(j+1)}/(q;q)_{2j}$ collapses to

$$S_e=\sum_{j\geq 0}(-1)^j e^{-j\tau}\,\frac{\sinh^j(\tau/2)}{\prod_{k=1}^{2j}\sinh(k\tau/2)}=\sum_{j\geq 0}(-1)^j\,\frac{(2/\tau)^j}{(2j)!}\,e^{-j\tau+E_j},\quad E_j=\tfrac{j\tau^2}{24}-\tfrac{\tau^2}{144}(2j)(2j+1)(4j+1)+O(\tau^4 j^5),$$

E_j being the exact sinh-curvature defect. Two elementary facts finish the extraction. First, the factor $e^{-j\tau}$ resums *in closed form*: $\sum_j(-1)^j(2e^{-\tau}/\tau)^j/(2j)!=\cos W$ with $W=\sqrt{2/\tau}\,e^{-\tau/2}$, which is exactly the frequency of Remark 5; so the phase correction needs no estimate at all. Second, the entire-function moment identity $\sum_j(-1)^j y^j j^m/(2j)!=(y\,d/dy)^m\cos\sqrt{y}$ evaluates the curvature term through its leading part $-\frac{\tau^2}{9}(y\,d/dy)^3\cos\sqrt{y}=-\frac{\tau^2}{9}\cdot\frac{W^3}{8}\sin W=-\frac{\sqrt{2}}{36}\sqrt{\tau}\,\sin W$ (using $\tau^2W^3=2\sqrt{2}\,\sqrt{\tau}\,e^{-3\tau/2}$, the factor $e^{-3\tau/2}$ being absorbed in the $O(\tau)$), giving

$$S_e=\cos W-\tfrac{\sqrt{2}}{36}\sqrt{\tau}\,\sin W+O(\tau),\qquad W=\sqrt{2/\tau}\,e^{-\tau/2},$$

all constants explicit and verified to ten digits (`elemcos_verify.py`). Carried to higher moments the same reduction produces every subsequent coefficient as a finite combination of $W^k\sin W$, $W^k\cos W$, so it serves the $O(\tau^2)$-relative demand of the U-numerator in the same elementary terms. The one input it does *not* render elementary is the remainder estimate; that the alternating expansion is asymptotic with $O(\tau)$ error. That step is a *cancellation* statement: not only does the triangle inequality on $\sum_j(2/\tau)^j/(2j)!\,|\cdot|$ return $\cosh W\asymp e^{\sqrt{2/\tau}}$, but the partial sums of the $\cos W$ series themselves swing up to $\asymp e^{c\sqrt{2/\tau}}$ before cancelling to $O(1)$ (numerically $\max_J|S_J|=9$, 48, 1459, 7.6×10^4 at $\tau=0.1,0.05,0.02,0.01$). Hence *every* partial-summation method (Abel, summation-by-parts, Euler transformation) fails, since each is controlled by these exponentially large partial sums; only a global resummation captures the cancellation. This global resummation is exactly the two-saddle steepest-descent computation carried out self-contained in the proof of Remark 5: the saddles $z^*_\pm=\pm iW/2$ supply the cancellation the partial sums cannot. Together, the elementary extraction here and that saddle argument reduce Remark 5 to Cauchy's theorem, Stirling's formula, elementary Gaussian integration, the bounded-variation bound of Lemma 3, and *one* cited estimate: the classical remainder bound for a simple saddle [17, Ch. 4, Thm. 7.1], used once in the proof of Remark 5. It is not independent of that citation, and we do not claim otherwise; Appendix A provides the route that is.

Remark 11 (derivation of the U-numerator amplitude). The sharp picture behind Theorem 59: the asymptotic value $|\sum_k d_k|=\frac{3}{8\sqrt{2}}\tau^{5/2}(1+O(\sqrt{\tau}))$ at the travel poles, on the one explicit series $P_{12}=\frac{2q^3}{1-q^3}\sum_k d_k,\ d_k=(-2)^k(1-q)^k q^{k^2+3k}/[(q^2;q^2)_k(q^5;q^2)_k]$, which submits to the identical reduction. Writing $1-q^m=2e^{-m\tau/2}\sinh(m\tau/2)$ throughout collapses the terms *exactly* to

$$d_k=(-1)^k e^{-k\tau}\,\frac{\sinh^{k+1}(\frac{\tau}{2})\,\sinh(\frac{3\tau}{2})\,\sinh((k+1)\tau)}{\prod_{m=1}^{2k+3}\sinh(\frac{m\tau}{2})},$$

and the elementary expansion (exact linear absorption into the argument, Faulhaber sums for the sinh-curvature, the moment identities $k^m\mapsto(y\,d/dy)^m$) yields, with $\Phi(y)=(\sin W-W\cos W)/(2W^3)$, $W=\sqrt{y}$, $D=y\,d/dy$,

$$\sum_k d_k=6\Big[\Phi+c_3D^3\Phi+c_2D^2\Phi+\tfrac{c_3^2}{2}D^6\Phi+(c_5+c_2c_3)D^5\Phi+\tfrac{c_3^3}{6}D^9\Phi\Big](y_*)+O(\tau^3),$$

$y_* = \frac{2}{\tau}e^{-\tau-23\tau^2/36}$, $c_2 = -\frac{5}{12}\tau^2$, $c_3 = -\frac{1}{9}\tau^2$, $c_5 = \frac{1}{450}\tau^4$; *every constant an exact rational, derived, not fitted* (`elemY3_verify.py`; the error is verified $O(\tau^3)$ over $\tau \in [0.005, 0.08]$ and the formula reproduces the exact series at the poles $m = 7, \ldots, 33$ to the displayed accuracy). At the travel poles the expansion gives $|\sum_k d_k|/\tau^{5/2} \to 3/(8\sqrt{2}) = 0.2652$, i.e. exactly the observed gate value $|P_{12}|/\tau^{3/2} \to 1/(4\sqrt{2})$ with its fourfold margin: structurally, the pole sits at offset δw from the zero of the displayed expansion, and the constant $3/(8\sqrt{2})$ arises as $(3\tau/2)\,\delta w/\tau^{5/2}$.

The *phase-matching step* that fixes δw is carried out by the same machinery. The pole equation $\Sigma_1(q_m) = 1$ collapses exactly to $1-\Sigma_1 = \sum_{j\geq 0}(-1)^j \sinh^j(\frac{\tau}{2})/\prod_{m=1}^{2j}\sinh(\frac{m\tau}{2})$; the S_e-series stripped of its $e^{-j\tau}$ phase; so the identical expansion applies with $y_* = \frac{2}{\tau}e^{\tau^2/36}$, $c_2 = -\frac{1}{12}\tau^2$ and the *universal* $c_3 = -\frac{1}{9}\tau^2$, $c_5 = \frac{1}{450}\tau^4$. Solving the two phase conditions perturbatively in the common variable $w = \sqrt{2/\tau}$ near each grid point $(M+\frac{1}{2})\pi$ gives

$$\chi_{\text{pole}} = -\frac{\sqrt{2}}{36}\sqrt{\tau} + \frac{41\sqrt{2}}{2400}\tau^{3/2} + O(\tau^2), \qquad \chi_{\text{zero}} = -\frac{\sqrt{2}}{36}\sqrt{\tau} - \frac{259\sqrt{2}}{2400}\tau^{3/2} + O(\tau^2).$$

The $\sqrt{\tau}$ phases agree *identically*; both are produced by the universal cubic curvature c_3, and the constant $\sqrt{2}/36$ is that of Remark 5; this is precisely why the travel poles track the zeros of $Y_3(1)$. The $\tau^{3/2}$ terms differ by exactly

$$\delta w = \chi_{\text{pole}} - \chi_{\text{zero}} = \frac{\sqrt{2}}{8}\tau^{3/2} + O(\tau^2), \qquad \text{whence} \qquad |\sum_k d_k| = \frac{3\tau}{2}\,\delta w\,(1+O(\sqrt{\tau})) = \frac{3}{8\sqrt{2}}\tau^{5/2}(1+O(\sqrt{\tau})),$$

and therefore $|P_{12}|/\tau^{3/2} \to 1/(4\sqrt{2}) = 0.17678 < 1/\sqrt{2}$: *the numerator gate, derived, with its fourfold margin*. Verification: the two phase series match the exact pole and zero locations to seven digits at $m = 11, 17, 23, 29$, and $\delta w/\tau^{3/2} = 0.17699,\ 0.17687,\ 0.17683,\ 0.17681 \to \sqrt{2}/8 = 0.176777$ (`phase_match_verify.py`); the pole-side series independently reproduces the McMahon-type coefficients of the travel-pole expansion. The remainder discipline throughout is that of Remark 5: every coefficient extracted elementarily, the $O(\cdot)$ terms controlled by the same two-saddle argument.

4.1 The travel-pole equation as a q-trigonometric identity

The two series that govern the poles admit an exact identification with the q-cosine of Koornwinder and Swarttouw [4], $\cos(z;q^2) := {}_1\phi_1(0;q;q^2,q^2z^2) = \sum_{j\geq 0}(-1)^j q^{j^2+j}z^{2j}/(q;q)_{2j}$.

Proposition 12. *Let $z_0 = \sqrt{2(1-q)}$ and $Z = z_0/\sqrt{q}$. Then*

$$S_e = \cos(z_0;q^2), \qquad 1-\Sigma_1 = \cos(Z;q^2).$$

In particular the travel poles q_m are exactly the solutions of $\cos(Z;q^2) = 0$, and S_e is the same function evaluated one half q-step below its own zero.

Proof. Write $1-q^m = 2e^{-m\tau/2}\sinh(m\tau/2)$ in the sinh-forms of the two series (§4): the products collapse to $1-\Sigma_1 = \sum_j(-1)^j\big(2(1-q)\big)^j q^{j^2}/(q;q)_{2j}$ and $S_e = \sum_j(-1)^j\big(2(1-q)\big)^j q^{j^2+j}/(q;q)_{2j}$. Comparing with the series for $\cos(z;q^2)$ term by term forces $q^2z^2 = 2(1-q)q$, i.e. $z = Z$, in the first case and $q^2z^2 = 2(1-q)q^2$, i.e. $z = z_0$, in the second; both series converge for $|q|<1$, the factor q^{j^2} dominating. The per-term identity is machine-checked in Lean 4 (`GramCasoratian.lean`, theorem `sigma1_term`; axioms `propext`, `Classical.choice`, `Quot.sound`, no `sorry`). ☐

Remark 13. The identification is consistent with the pole data: at all 58 tabulated poles the computed value of $\cos(z_0;q^2)$ agrees with the stored S_e to at least thirty digits, and $|S_e|/\sqrt{\tau}$ lies in

$[0.698, 0.708]$, approaching $1/\sqrt{2}$ (`qtrig_verify.py`). The half-step structure accounts for the size of the gate quantity: $Z - z_0 = z_0(1-\sqrt{q})/\sqrt{q} = \tau^{3/2}/\sqrt{2} + O(\tau^{5/2})$, the pole condition annihilates $\cos(\,\cdot\,; q^2)$ at Z, and the q-derivative rule $(1-q)D_{q,z}\sin(z; q^2) = \cos(z; q^2)$ of [4] gives the function slope $1/(1-q) \sim 1/\tau$ near the zero, so the sampled value is of size $(\tau^{3/2}/\sqrt{2})(1/\tau) = \sqrt{\tau/2}$, the observed constant. Turning this observation into a proved lower bound on $|S_e(q_m)|$ independent of Remark 5 requires uniform control of $\cos'(\,\cdot\,; q^2)$ between z_0 and Z as $q \to 1^-$. That control is supplied by Appendix A (Lemma 50 bounds $\max|f''|$ on $[qZ, Z]$, and Proposition 46 converts it into the lower bound), independently of Remark 5.

5 The relaxed series V

Theorem 14. *The relaxed growth series V is transcendental over $\mathbb{Q}(x)$.*

Proof. V contains the travel resolvent $(1-\Sigma_1)^{-1}$ times a bulk-dressing factor holomorphic and non-vanishing at the q_m; thus V has a genuine pole at each travel pole q_m; by Lemma 8 these are infinitely many and accumulate at the boundary of the disc of convergence. An algebraic function over $\mathbb{Q}(x)$ has only finitely many poles in $|x| < 1$; an infinite pole set accumulating at the boundary is impossible for an algebraic function, so V is transcendental; *provided* the poles are not cancelled by the numerator. Non-cancellation is exactly the statement that $S_e(q_m) = \cos W - T_2 \neq 0$ for all large m, i.e. that $|T_2| < |\cos W|$ there; since $|\cos W| \asymp \sqrt{\tau/2}$ at the poles while $|T_2| = O(\sqrt{\tau})$ with the small constant $O(\sqrt{\tau})$ by Lemma 6, the inequality holds with a fixed margin; the phrase "for all large m" is licensed by Lemma 53, and the pole set itself by Lemma 8 via Lemma 6. Hence V is transcendental. The non-cancellation input is Proposition 46, proved in Appendix A by an elementary chain with explicit numerical constants and no cited asymptotic theorem; Remark 5 sharpens the constant to $\sqrt{2}/36$ but is not needed for the theorem. $\square$

Corollary 15. *The exponential growth rate of both u_n and v_n is $\beta_2 = 1.4916177871\ldots$, with β_2^2 the reciprocal of the smallest travel pole q_1; the value $\frac{3}{2}$ is excluded. This is unconditional (it needs only the location of the dominant travel pole, not Remark 5).*

Remark 16 (the number β_2: a value-transcendence question). Whether the *number* β_2 is transcendental is open, but the question has a clean reduction. Since $\beta_2 = 1/\sqrt{q^*}$ with $\Sigma_1^T(q^*) = 1$, the number β_2 is algebraic iff q^* is; if q^* were algebraic, the series Σ_1^T would take the *rational* value 1 at an algebraic interior point of its disc. For the Jacobi-theta class this is impossible: by Nesterenko's theorem on Eisenstein values [15], $\sum_n \alpha^{n^2}$ is transcendental at *every* algebraic α with $0 < |\alpha| < 1$ [9]. The blocks here are theta-*like*; their coefficients ride the squares $q^{(j+1)^2}$, which is the source of the natural boundary of Theorem 2; but their Euler-product denominators $(q;q)_{2k+1}$ spread the support densely, and no analogue of [15] is known off the modular class.

Concretely the reduction localizes on a single confluent basic hypergeometric function. Writing $\Sigma_1^T(q) = 1 - S(q)$,

$$S(q) \;=\; \sum_{k\geq 0} \frac{\big(-2(1-q)\big)^k q^{k^2}}{(q;q)_{2k}} \;=\; {}_1\phi_1(0; q; q^2,\, 2q(1-q)),$$

a q-Bessel-class function (using $(q;q)_{2k} = (q;q^2)_k(q^2;q^2)_k$); β_2 is algebraic iff the least positive zero $q^* = 0.449453630\ldots$ of S is. Certified exclusions at 430 digits (residual 10^{-442}; PSLQ with post-verification of every candidate): neither q^* nor β_2 satisfies an integer polynomial of degree ≤ 8 with height 10^{20}, degree ≤ 12 with height 10^{18}, or degree ≤ 16 with height 10^{14}; no bilinear relation ties q^* to the fold point q_c of the zero curve (height 10^{12}). The continued fraction of q^* (383 certified partial quotients, two-precision guard) is Gauss–Kuzmin generic: digit frequencies match, Khinchin mean 2.63 (2.685), Lévy exponent 3.21 (3.276), largest quotient $a_{170} = 8250$ (probability $\approx 6\%$ over 383 draws, consistent with chance). The number offers no arithmetic foothold of any tested kind. We record what the natural tools give here, since it locates the difficulty precisely. First, S is *neither* q-holonomic *nor* a Mahler function in q (verified to $O(q^{48})$): the diagonal specialization $z = 2q(1-q)$ ties the argument to the base and destroys the single-variable q-difference structure, so the methods of Mahler–Nishioka and of holonomy do not apply. Second, two elementary refinements fail *decisively*, not for want of effort: an ultrametric argument is void; the v-adic and archimedean sums of $S(\alpha)$ are unrelated numbers, so $|S(\alpha)|_v = 1$ does not bound the real value; and a Liouville/height bound is beaten by the q-Pochhammer denominators, whose growth over the conjugates of modulus > 1 is cubic ($\sim K^3$) against the merely quadratic ($\sim K^2$) tail, Kronecker-sharply (equality would need Mahler measure 1, i.e. a root of unity). A more promising reformulation uses two variables. Set $G(q,z) = {}_1\phi_1(0;q;q^2,z) = \sum_k (-1)^k q^{k(k-1)} z^k/(q;q)_{2k}$, so $S(q) = G(q, 2q(1-q))$; unlike S, the function G *is* q-holonomic in z, satisfying the second-order q-difference equation

$$q\,G(q,z) \;-\; (q+1-qz)\,G(q,q^2z) \;+\; G(q,q^4z) \;=\; 0.$$

One checks that $z^* := 2q^*(1-q^*) = 0.4948901\ldots$ is the *smallest positive zero* of $G(q^*,\cdot)$. Hence β_2 algebraic would force an *algebraic zero of a q-Bessel-class function at an algebraic base*; and the classical analogue is false: the zeros of the Bessel function J_ν are transcendental (Siegel). A q-analogue of Siegel's theorem for the zeros of ${}_1\phi_1$ would therefore settle β_2. The right framework is arithmetic. Writing the equation as a system $Y(q^2z) = A(z)Y(z)$ with $Y = (G(z), G(q^2z))^\top$, the companion matrix

$$A(z) = \begin{pmatrix} 0 & 1 \\ -q & q+1-qz \end{pmatrix}$$

is *polynomial* in z with $\mathbb{Z}[q]$ entries and $\det A = q$, so the iterated matrices carry no denominators and the Casoratian obeys $W(q^2z) = qW(z)$, i.e. $W(z) = c\,z^{1/2}$: the two solutions form a q-Bessel pair with Frobenius exponents $0, \frac{1}{2}$. The second solution is explicit, $\widetilde{G}(z) = z^{1/2}H(z)$ with $H(z) = \sum_k (-1)^k q^{k^2}(1-q)z^k/(q;q)_{2k+1}$, and the pair satisfies the exact q-Wronskian identity

$$q\,G(z)\,H(q^2z) - G(q^2z)\,H(z) \;=\; q-1,$$

so at the zero z^* one has the clean relation $G(q^2z^*)H(z^*) = 1-q$. Moreover $G(q,\cdot)$ is *entire of order zero*; a q-E-function, not a q-G-function. Hence β_2 transcendental would follow from a q-analogue of the Siegel–Shidlovskii theorem for this q-E-function: such a theorem gives $G(q,\alpha)$ transcendental at every algebraic $\alpha \neq 0$ off the singular locus, whereas $G(q^*,z^*) = 0$ is algebraic; forcing q^*, and with it β_2, transcendental (and by the shared q-connection core, likewise U). All the structural hypotheses hold here (polynomial companion matrix, constant determinant, entire order-zero solution); the sole open point is whether the available q-Siegel–Shidlovskii results (André; Di Vizio–Hardouin; and the Siegel-method theory of [12]) extend to a ${}_1\phi_1$ that is *irregular* at $z = \infty$; a precise, expert-checkable question, not an absence of theory. (Note the contrast with the *function-level* results of this paper, which say nothing about individual interior values.)

For *irrationality* (weaker than transcendence) the classical architecture transports verbatim. G and H are Hahn–Exton (third Jackson) q-Bessel functions: $G(q,z) \propto J_{-1/2}(\sqrt{z}/q; q^2)$ and $H(q,z) \propto J_{+1/2}(\sqrt{z/q}; q^2)$ (verified; the zeros correspond exactly), i.e. the q-analogues of cos and sin; and z^* is the q-analogue of $(\pi/2)^2$. Niven's irrationality proof for π runs on the Lommel-polynomial ladder $J_{m+1/2}(r) = R_{m,1/2}(r)J_{1/2}(r) - R_{m-1,3/2}(r)J_{-1/2}(r)$ evaluated at the zero of

$J_{-1/2} = \sqrt{2/\pi r}\cos$; the exact q-analogue of every ingredient exists (Koelink–Swarttouw): the ladder $J_{\nu+1}(x;q^2) = ((1-q^{2\nu})/x + x)J_\nu - J_{\nu-1}$, the q-Lommel connection $J_{\nu+m} = R_{m,\nu}J_\nu - R_{m-1,\nu+1}J_{\nu-1}$ with $x^m R_{m,\nu} \in \mathbb{Z}[q, x^2]$, and the second-kind functions carrying $q^{m^2/2}$ prefactors; all verified numerically for our pair. Moreover, since a rational zero would make $G(q^*, z^*) = 0$ a *rational value* at rational base and argument, β_2 is irrational as soon as this Hahn–Exton function omits the value 0 at rational points: a plain *value-irrationality* statement, strictly weaker than every transcendence target above. What blocks even this is now a single quantified invariant: the *arithmetic ratio* ρ of the series (b-adic decay exponent of the terms over b-adic growth exponent of the cleared denominators, $q = s/t$ in lowest terms). Every proven q-irrationality theorem (Tschakaloff, q-exponential, the Bézivin–Bundschuh–Väänänen classes) lives at $\rho \geq 1$, where the Pochhammer denominator itself supplies the decay; our G has an independent q^{k^2} numerator against the double Pochhammer $(q;q)_{2k}$, giving $\rho = \frac{1}{2}$. All three classical constructions fail by exactly this margin, measured: the z-Padé remainder $q^{3.5n^2}$ against height q-degree $\approx 9n^2$; the z-direction continued fraction (convergents in $\mathbb{Z}[q, z]$, Pincherle limit the H-ratio) converges only geometrically against t^{n^2} heights; and the ν-ladder at the zero has $J_{m+1/2}(x_0) = R_{m,1/2}(x_0)J_{1/2}(x_0) \sim Cx_0^m$; geometric, because the q-factorial is geometric where Niven's $m!$ is super-exponential; against t^{m^2}. The ν-ladder is in fact fully costed: the q-Lommel coefficients satisfy $\deg_q p_m = m^2$ *exactly* (immediate from the recurrence, verified symbolically), so their denominators at $q = s/t$ are the pure power t^{m^2}: carrying *no* cyclotomic structure, so the lcm-compression that powers the little-q-Legendre proofs has nothing to act on; while by Poincaré–Perron the m-recurrence, whose coefficient tends to $x_0 + 1/x_0$ with *distinct* characteristic roots $x_0, 1/x_0$, admits *only geometric* solutions: no theta-type q^{m^2}-decaying solution exists in the ν-direction for either kind (the second kind's $q^{\nu^2/2}$ prefactor is exactly cancelled by the $(q^{-\nu+1};q)_\infty$ blow-up of its shifted argument). Hence the cleared ν-ladder forms grow like $t^{m^2}x_0^{-m} \to \infty$ for *every* rational $q = s/t$, including $s = 1$. The $b \leftrightarrow z$ transformation ${}_1\phi_1(0;q;q^2,z) = \frac{(z;q^2)_\infty}{(q;q^2)_\infty}{}_1\phi_1(0;z;q^2,q)$ redistributes but conserves ρ. The structural summary: the theta-type decay lives only in the z-direction, where the equation is *irregular* at $z = \infty$ (Newton polygon slopes $\{0, 1\}$) and every integral structure carries the double-Pochhammer weight $\rho = \frac{1}{2}$; the ν-direction is *regular* (Poincaré class) with clean $\mathbb{Z}[q, X]$ arithmetic but only geometric decay. Decay and arithmetic live in different directions and never meet; the same irregularity that obstructs q-Siegel–Shidlovskii creates the arithmetic deficit. The precise open door remains a q-irrationality criterion at $\rho = \frac{1}{2}$; one notch beyond the proven $\rho \geq 1$ theory.

Remark 17 (the β_2 family). q^* is the first member of an infinite family. On $(0, 1)$ the parabola $w(q) = 2q(1-q)$ falls like $1-q$ while every zero branch falls like $(1-q)^2$ (the confluent limit), so w overtakes each branch exactly once: S has infinitely many zeros $q_1^* < q_2^* < \cdots \to 1$, the j-th given by $w(q_j^*) = z_j(q_j^*)$ (verified at $j = 2$ to 43 digits: exactly one zero of $G(q_2^*, \cdot)$ lies below w). The accumulation obeys the classical cosine-zero law

$$1-q_j^* \sim \frac{8}{\pi^2(2j-1)^2}, \qquad \text{measured } (1-q_j^*)\,\frac{(2j-1)^2\pi^2}{8} = 0.679,\ 0.961,\ 0.986,\ 0.993,\ 0.9956,\ 0.9970,\ 0.9979,\ 0.998$$

The second member is $q_2^* = 0.913486638731047141696\ldots$, i.e. $\beta_2^{(2)} := 1/\sqrt{q_2^*} = 1.046282352687417855\ldots$; neither satisfies an integer polynomial of degree ≤ 8 with height 10^{10}, and no bilinear relation ties q_1^* to q_2^* (PSLQ, 100 digits). The family interpolates between the open head ($j = 1$, the wild β_2) and the classical tail (the π^2-law): the roots become "provably transcendental in the limit" while the first, the one of combinatorial interest, remains untouched; the confluent boundary of the step-ratio mechanism made visible inside a single object. (That the higher roots are genuine singularities of U itself is exactly the non-vanishing $\mathcal{N}_U(q_m) \neq 0$ proved for every travel pole in Theorem 43 via Corollary 61.)

Three refinements (all computations at 115 digits, 30 cosine and 10 sine members). First, the sine side has its own family, $H(q_j^{\rm s}, 2q(1-q)) = 0$ with $1-q_j^{\rm s} \sim 2/(j\pi)^2$ (law ratios $0.913, \dots, 0.9991$ for $j \leq 10$), and the two families *interlace*: $q_j^* < q_j^{\rm s} < q_{j+1}^*$ (verified $j \leq 9$); its head is $q_1^{\rm s} = 0.815032640290\dots$ Second, the accumulation law has an exact-looking rational second order, identical for both families:

$$1 - q_j^* = \frac{2}{\mu_j}\Big(1 - \frac{8}{9\mu_j} + \frac{B}{\mu_j^2} + O(\mu_j^{-3})\Big), \qquad \mu_j = \Big(\tfrac{(2j-1)\pi}{2}\Big)^2 \text{ resp. } (j\pi)^2,$$

with $A = -8/9$ matched to 12 digits (fit residual $3 \cdot 10^{-18}$ out of sample) and B derived below. The constant A is derived, and the derivation separates two regimes that must not be conflated. Writing $\varepsilon = 1 - q$, the logarithm of the ratio of the k-th term of $G(1-\varepsilon, \varepsilon^2 y)$ to $(-1)^k y^k/(2k)!$ equals $\frac{k}{2}\varepsilon + c_2(k)\varepsilon^2 + O(\varepsilon^3)$ with

$$c_2(k) = -\frac{k^3}{9} - \frac{k^2}{12} + \frac{23k}{72};$$

to first order G is $\cos\sqrt{y\, e^{\varepsilon/2}}$, so at *fixed* j the zero branch obeys $z_j = \varepsilon^2 \mu_j(1 - \frac{1}{2}\varepsilon + O(\varepsilon^2))$: the fixed-$j$ confluent correction is $-\frac{1}{2}$ (measured -0.5004 at $\varepsilon = 0.002$, converging linearly). The crossing, however, sits in the *uniform* regime $\mu\varepsilon \approx 2$, where the k^3-term acts as the operator $-\frac{\varepsilon^2}{9}D^3$, $D = (u/2)\, d/du$ on $\cos u$, $u = \sqrt{X}$; at the zeros $D^3 \cos u = \frac{u^3}{8} \sin u + O(u^2)$, while even powers of D are cos-weighted and drop out (parity protects the order). With $m = \varepsilon\mu$ the crossing equation $2(1-\varepsilon)e^{\varepsilon/2} = \varepsilon\mu(1 - \varepsilon^2\mu/36)$ becomes $m = 2 - \varepsilon + \varepsilon m^2/36 + O(\varepsilon^2)$, whence $m = 2 - \frac{8}{9}\varepsilon$:

$$A = -1 + \tfrac{1}{9} = -\tfrac{8}{9},$$

identically for the sine family (its ε^2-coefficient carries the same leading $-k^3/9$). The two regimes carry different constants ($-\frac{1}{2}$ versus the crossing's $-\frac{8}{9}$). One more order of the same calculus derives B exactly. The ingredients: the exact prefactor $2(1-\varepsilon)e^{\varepsilon/2} = 2 - \varepsilon - \frac{3}{4}\varepsilon^2 - \cdots$; the cubic coefficient $c_3(k) = -\frac{k^3}{9} - \frac{k^2}{12} + \frac{17k}{72}$ (the m^3-terms cancel in its defining sum, so c_3 shares the leading $-k^3/9$); the quartic $c_4(k) = \frac{k^5}{450} + O(k^4)$ together with $\frac{1}{2}c_2(D)^2 = \frac{D^6}{162} + \frac{D^5}{108} + \cdots$; and one subtlety: at order $\varepsilon^6 u^9$ the cube $\frac{1}{6}(c_2(D)\varepsilon^2)^3$ contributes $+\frac{u^9}{2239488}\sin u$ through the leading term of D^9, cancelling the $\delta^3/6$ of the cosine expansion *identically*; truncating the exponential before the cube gives a spurious m^5-term. With that cancellation the crossing equation closes as $m = 2 - \frac{8}{9}\varepsilon - \frac{1891}{8100}\varepsilon^2 + O(\varepsilon^3)$, and inversion gives

$$B = 2m_2 + \tfrac{64}{81} = \frac{1309}{4050} = 0.3232098765432\dots,$$

confirmed by a five-order free fit to the computed roots: $B = 0.3232098765429\dots$ agrees to 13 digits (with $A = -8/9$ simultaneously reproduced to 17), and the (A, B)-constrained fit has out-of-sample residuals below $4 \cdot 10^{-16}$. The accumulation law of the β_2 family therefore begins

$$1 - q_j^* = \frac{2}{\mu_j}\Big(1 - \frac{8}{9\mu_j} + \frac{1309}{4050\,\mu_j^2} + O(\mu_j^{-3})\Big),$$

with both corrections rational and derived. Three closing observations. (a) *Universality*: the sine family obeys the *same* law; a constrained fit to 22 sine members gives $B_{\rm sin} \cdot 4050 = 1309.000000$ to the fit precision $2 \cdot 10^{-8}$; so through second order the two interlaced families share one accumulation series, with only μ_j distinguishing them. (b) Every ingredient of the crossing calculus is rational (the $c_n(k)$ are rational polynomials; $D^n \cos u$ has odd u-powers on sin and even on cos, an invariant of $D = (u/2)d/du$ preserved by the one-line induction; the δ-elimination therefore stays in odd powers and the m-equation in rational coefficients), so we conjecture the expansion is rational to all orders. The third constant resists identification: $C = -0.4745357072870\ldots$ is stable to 10^{-14} across model orders and members through $j = 55$, excludes every denominator up to $\sim 10^6$ (PSLQ), and the fitted higher coefficients grow ($F \approx 4.3$), marking the series as asymptotic-divergent; which caps the extractable precision and leaves C's (conjecturally rational, large-denominator) value to the third-order calculus. (c) At $j = 1$ the same series becomes a rational-coefficient expansion in $4/\pi^2$ evaluated at the wild head: divergent, hence a resummation question. If a Borel-type summation of the rational series could be shown to reproduce q^*, the arithmetic of π^2 would enter the problem for the first time through the family; we record this as a direction and have tested its carrying capacity twice. A Borel–Padé resummation of the seven known coefficients reproduces the four accessible samples (both family heads and both second members) to residuals $7.9 \cdot 10^{-3}$, $1.7 \cdot 10^{-6}$, $4.6 \cdot 10^{-9}$, $7.4 \cdot 10^{-11}$ at $\mu = 2.47$, 9.87, 22.2, 39.5. An attempted transseries extraction from these residuals then *refuted its own premise*: the log-slopes between consecutive samples are 1.14, 0.48, 0.24, matching the truncation signature $7/\mu = 1.17$, 0.47, 0.24 exactly (three points), not an exponential $e^{-\beta\mu}$; the residuals are the x^7-tail of the divergent series, not a nonperturbative signal. At the head this is the standard resurgence degeneracy: seven coefficients place the evaluation at optimal truncation, where truncation floor and first transseries term coincide in scale ($e^{-t_0\mu_1} \approx 0.01$, with $t_0 \approx 0.75 \pm 1.70i$ the Padé–Borel singularities), so the head residual $-0.008(4)$ *cannot* be attributed to a nonperturbative sector on present data. That cost has now been paid: the crossing calculus has been automated in exact rational arithmetic (weight-graded formal Newton for the zero shift; parity and residual controls exact), producing the law's coefficients through order thirty; *all rational*, e.g. $A_3 = -1695089/3572100$ (whose denominator $2^2 3^6 5^2 7^2$ explains the failed PSLQ) and $A_4 = 10199641/535815000$. The coefficients are computed by running the calculus over $\mathbb{F}_p$ and recovering each A_n by rational reconstruction from twenty 62-bit primes; the values are independently confirmed by extracting A_n from the accumulation law applied to the roots themselves, computed to 105 digits.

The Borel geometry is only partly determined by these coefficients. The signs of A_n follow a three-positive, three-negative pattern from $n \approx 12$, indicating a conjugate pair of singularities near 60°; the pattern is not exact over the full range. This period-6 oscillation dominates the naive growth diagnostics, and must be removed before the Gevrey order can be read off. Setting

$$G_n := \tfrac{1}{6}\left(\log|A_n| - \log|A_{n-6}|\right),$$

which is phase-matched and therefore free of the oscillation, the model $|A_n| \sim C^n(n!)^s$ gives $G_n \sim \log C + s\log n$, so s is the slope of G_n against $\log n$. Least squares yields $s = 0.844$ on $16 \le n \le 30$, 0.897 on $23 \le n \le 30$ and 1.033 on $26 \le n \le 30$: the estimate rises towards 1 as the fitting window advances, and the data are consistent with Gevrey-1. (The unnormalised ratio $\log|A_n|/(n\log n)$, which equals $s + (\log C - s)/\log n$, is 0.313 at $n = 14$ and 0.411 at $n = 30$; its corrections are of order $1/\log n$ and it is therefore useless for extrapolation at these lengths.)

For the singularity modulus three independent estimates are available and agree only in order of magnitude: the phase-matched growth constant gives $|t_0| = 1/C = 3.59$–3.98 over $24 \le n \le 30$; a three-term recurrence fit, appropriate to a conjugate pair, gives $|t_0| = 3.285$, 3.394, 3.493, 3.558 at $n_0 = 22, 25, 28, 30$, extrapolating in $1/n_0$ to 4.30, with $\arg t_0 = 60.8°$, 61.7°, 62.4°, 62.3°; and the Padé–Borel approximants give $|t| = 2.61$, 2.56, 2.96, 3.09, 3.33 for [6/6] through [14/14]. Thus $|t_0|$ lies between 2.9 and 4.3 and is not pinned. Gevrey-1 growth, and with it the $n!$-Borel framework, is supported but not established; the location of the singularities is not established; and the Borel representation below remains a conjecture whose analytic premise is open. One structural constraint is available: by the weight lemma, only Bernoulli numbers of index $\lesssim n$ reach A_n (a B_j occurring in c_m sits at k-degree $\deg c_m - j$ and reaches A_n only if $j \le 2n - m + 1$), which excludes the

Gevrey-2 growth that B_{2n} would otherwise force, but does not pin the class. Where testable the resummation succeeds: at the sine head the Padé–Borel ladder converges $1.7 \cdot 10^{-6} \to 3.7 \cdot 10^{-11}$ (any transseries prefactor $\lesssim 10^{-6}$ unless phase-suppressed), while at the cosine head the plain ladder reaches its approximation wall at $\sim 10^{-3}$ with residuals oscillating about zero. We therefore pose the *Borel representation conjecture*: $1 - q_j^* = (2/\mu_j)\,\mathcal{S}(1/\mu_j)$ for both families and all j, where $\mathcal{S}$ is the Borel–Laplace sum of the explicit rational series $\sum A_n x^n$; in particular

$$q^* \;=\; 1 - \frac{8}{\pi^2}\,\mathcal{S}\Big(\frac{4}{\pi^2}\Big),$$

the first exact (conjectural) bridge between q^* and π^2-arithmetic. Numerically the head representation is confirmed to $\sim 10^{-4}$ by conformal-map Borel resummation (the residual crosses zero smoothly with no plateau above the estimator scatter), and the *same* resummation machinery reproduces the three lower-x heads to 10^{-13}–10^{-26}; the coefficients are now computed through A_{22}, still rational (e.g. A_{21}, A_{22} have denominators $\sim 10^{93}$), consistent with the rationality of the expansion at every order.

Two honest boundaries frame the bridge. *Analytically*, its proof is *not* a cite-the-theorems exercise: it reduces to a single hard lemma; a q-uniform, turning-point (Olver/Airy-grade) factorial remainder bound $|A_n| \leq C\sigma^n n!$ for the Hahn–Exton crossing as $q \to 1$; which has no citable form in the literature and is plausibly the program's recurring degenerate-saddle obstruction in Borel-summability costume. *Arithmetically*, the Borel transform $b(t) = \sum (A_n/n!)t^n$ carries a sharp, exceptionless prime law: for every odd prime $p \leq 43$, $p \mid \mathrm{den}(b_n)$ exactly when $2n+1 \geq p$, with p-adic valuation exactly 2 at the onset $n = (p-1)/2$, and thereafter non-vanishing with valuation growing linearly *on average* (least-squares slopes $2.38, 1.17, 0.73, \ldots$ for $p = 3, 5, 7$; the valuation is non-monotone locally, e.g. v_5 dips $6 \to 5$ and v_{17} dips $2 \to 1$). The clean-before-onset half is a theorem for all p. (i) The polynomials $c_n(k)$ built from the calculus are p-integral for $n \leq p-2$ (von Staudt–Clausen at the Faulhaber layer). (ii) A_n depends only on the calculus data through order $2n$; this is now proved, in two steps. First, by series reversion A_n is a function of $m_1, \ldots, m_n$ alone (the order-n inverse coefficient uses only the first n forward coefficients; verified by perturbing $m_{>n}$). Second, m_i depends only on $c_1, \ldots, c_{2i}$: in the turning-point grading, where $k \sim \varepsilon^{-1/2}$ assigns $\varepsilon^a u^b$ the weight $2a-b$ (physical order = weight/2) and every operation is additive on weights, the polynomial c_m enters at minimum weight $2m - \deg_k c_m$, attained by its top-degree term. Here $\deg_k c_m = m + [m \text{ even}]$: the degree-$m$ coefficient of the ε^m log-ratio $\ell_m(i)$ equals $(-1)^m [x^m] \log\big((e^x-1)/x\big)$, which vanishes for every odd $m \geq 3$ because $\log\big((e^x-1)/x\big) = \frac{x}{2} + \log\big(2\sinh(x/2)/x\big)$ and the last term is *even* in x; the Faulhaber sum then raises the k-degree by one. Hence c_m has minimum weight $m - [m \text{ even}]$ and first reaches physical order i only when $m \leq 2i$, so m_i (and therefore A_n) sees no c_m with $m > 2n$. (iii) The assembly introduces no prime p at output order $n < p$: the Bernoulli denominators of the Faulhaber sums acquire p only when $(p-1) \mid 2m$, i.e. $2m \geq p-1$, and all factorial and division indices stay below p. For $n < (p-1)/2$ one has $2n \leq p-3$, so every ingredient feeding A_n is p-integral and $p \nmid \mathrm{den}(A_n)$.

The matching lower direction; that p *does* appear at $n = (p-1)/2$, with valuation exactly 2; reduces to a unit. At $n = (p-1)/2$ the sole prime-p source is c_{p-1} (all lower c are p-integral), and it enters A_n *linearly*, since c_{p-1}^2 has weight $2(p-2) > 2n$. Only the top three coefficients of c_{p-1} (degrees $\geq p-2$) survive the weight cutoff; of the two that carry p^2; the k^1 and k^p monomials (von Staudt–Clausen at the Faulhaber layer, $\deg_k c_{p-1} = p$); only k^p reaches A_n, as k^1 sits at weight $2p-3 > 2n+1$. The coefficient of that top monomial in $A_{(p-1)/2}$ is a *unit*, $\partial A_{(p-1)/2}/\partial c_{p-1}[k^p] = \pm 1$, and this now has a leading-symbol proof. The monomial perturbs the confluent expansion of G by $\varepsilon^{p-1}k^p$; the turning-point map (the D^p-action, $D = (u/2)\,d/du$) sends k^p to $a_p u^p \sin u + b_p u^{p-1} \cos u$ with $a_p = (-1)^{n-1}2^{-p}$ the leading sine coefficient and $b_p = (-1)^n n(2n+1)2^{-p}$ the leading cosine coefficient ($p=2n+1$; both closed forms follow from the recursion $(A,B)\mapsto(\frac{u}{2}(A'-B),\frac{u}{2}(B'+A))$). At a simple zero u_* of cos the cosine term vanishes and every sine power below u^p contributes only at higher order in ε, so the zero shift is $\delta=\varepsilon^{p-1}a_p u_*^p$ (the factor $\sin u_*=\pm 1$ cancels against $G'=-\sin$). With $m=\varepsilon u^2$ and the leading balance $u_*^2=2/\varepsilon$ (from $\varepsilon u^2=2(1-\varepsilon)e^{\varepsilon/2}\to 2$),

$$\delta m = 2\varepsilon u_*\,\delta = 2a_p\,\varepsilon^p u_*^{p+1} = a_p\,2^{\,n+2}\,\varepsilon^n, \qquad \text{so}\quad \frac{\partial m_n}{\partial c_{p-1}[k^p]} = a_p 2^{\,n+2} = \pm 2^{\,1-n},$$

and with the reversion Jacobian $\partial A_n/\partial m_n = 2^{n-1}$ (from $\varepsilon\sim 2/\mu$) the product is ±1. Only the top sine power u^p reaches order ε^n, so the magnitude is exact; the sign is $(-1)^n$ (verified for all $n\le 10$, independently of p). Since $c_{p-1}[k^p]$ has p-adic valuation -2 while the other reaching monomials (k^{p-1},k^{p-2}) have valuation ≥ -1 against p-integral multipliers, $v_p(\operatorname{den} A_{(p-1)/2})=2$ exactly. Thus the onset law; $p\mid \operatorname{den}(A_n) \iff 2n+1\ge p$, with valuation 2 at $n=(p-1)/2$; is *proved for all odd* p. The average-linear (Legendre/factorial) growth; not the logarithmic growth an $\operatorname{lcm}(1,\dots,2n)$ or ζ-type denominator would give; makes $\operatorname{den}(A_n)$ super-geometric ($\log\operatorname{den}(A_n)/n$ rises $5.4\to 10.2$ over $n\le 20$, unbounded). Consequently $b(t)$ is *not* a G-function (super-geometric denominators alone suffice, independent of holonomy), so $\sum A_n x^n$ is not of André arithmetic-Gevrey type and lies outside the G-function, E-function, and E-operator (Fischler–Rivoal) classes entirely. The closest existing frame, the E-operator theory, has exactly one blocking gap; the hypothesis that b is a G-function; which the prime law refutes rather than supplies. The missing tool is therefore genuinely new, not an incremental hypothesis-removal: a value theory for Borel–Laplace sums whose Borel transform is non-holonomic with factorial denominators, evaluated at a point ($4/\pi^2$) transcendental over $\mathbb{Q}$. This constant is a canonical first target for such a theory. Third, the family sum converges to

$$T:=\sum_{j\ge1}(1-q_j^*)=0.7357490877157094\ldots\ (\pm3\cdot10^{-17},\ \text{tail summed via the fitted law and Hurwitz zeta}),$$

for which no relation with $\{1,\pi^{-2},\pi^{-4},\log 2,\gamma,\pi^2/6\}$ exists at height 10^4 within the stated uncertainty (and $T\ne 2/e$): the classical tail is π^2-rational piece by piece, but the wild head shifts the sum off every tested constant; a sum rule, if one exists, must come from global structure not yet visible.

The failure pattern above is not an artifact of particular constructions; it is forced by exact arithmetic facts, which we record as a theorem-level ledger.

Proposition 18 (Arithmetic no-go ledger at $\rho=\frac12$)**.** *Let $q=s/t\in(0,1)$ with $\gcd(s,t)=1$, and let $g_k=(-1)^k q^{k(k-1)}/(q;q)_{2k}$ be the Taylor coefficients of $G(q,\cdot)$. Then:*

- (i) (exact denominators) *in lowest terms $g_k=(-1)^k\,s^{k^2-k}\,t^{k^2+2k}/\prod_{i=1}^{2k}(t^i-s^i)$; in particular $\operatorname{den}(g_k)=\prod_{i=1}^{2k}(t^i-s^i)=t^{k(2k+1)(1+o(1))}$ exactly, while $|g_k|=(s/t)^{k^2(1+o(1))}$, so the arithmetic ratio is $\rho=\log(t/s)/(2\log t)\le\frac12$, with equality iff $s=1$.*

- (ii) (ν-ladder arithmetic) *the q-Lommel polynomials of Koelink–Swarttouw satisfy $\deg_q p_m=m^2$ exactly, with leading q-coefficient $(-1)^m$; hence at $q=s/t$ their denominators are the pure power t^{m^2}, carrying no cyclotomic structure.*

- (iii) (ν-direction regularity) *the ladder recurrence $u_{m+1}=((1-q^{2m+1})/x_0+x_0)u_m-u_{m-1}$ has coefficients converging exponentially to x_0+1/x_0 with distinct characteristic roots $x_0^{\pm1}$; by Poincaré–Perron every nonzero solution satisfies $|u_m|^{1/m}\to x_0^{\pm1}$. No theta-type ($q^{cm^2}$-decaying) solution exists in the ν-direction, for either kind. Consequently the cleared ν-ladder forms are $\gg t^{m^2}x_0^{-m}\to\infty$ for every rational q, including $s=1$.*

(iv) (cyclotomic floor) *any linear form built over the coefficients* $g_0, \dots, g_M$ *has decay at best* $q^{M^2(1+o(1))}$*, while its cyclotomic support is* $\{\Phi_d(t,s) : d \le 2M\}$*; even if every* Φ_d *appeared at most* once*; the idealized best case of any lcm/acceleration argument; the denominator exceeds* $t^{\sum_{d\le 2M}\varphi(d)} = t^{(12/\pi^2+o(1))M^2}$*, and* $12/\pi^2 = 1.2159\ldots > 1$*. Hence no cyclotomic-compression (Rhin–Viola-type) refinement of the Padé, continued-fraction, or ladder families can produce integer forms tending to* 0*. Moreover the true solved Padé denominators contain growing* non-cyclotomic *irreducible factors (at* $n = 3$*, one of degree* 26 *with coefficients up to* 4*; no integer* d *has* $\varphi(d) = 26$*), which no lattice-of-cyclotomes argument addresses.*

Proof. (i) $q^{k(k-1)} = s^{k(k-1)}/t^{k(k-1)}$ and $1/(q;q)_{2k} = t^{k(2k+1)}/\prod_{i\le 2k}(t^i-s^i)$; the numerator $s^{k^2-k}t^{k^2+2k}$ is coprime to each factor since $\gcd(st,\, t^i-s^i) = 1$. (ii) Induction: $p_{m+1} = (X+1-q^{2m+1})p_m - Xp_{m-1}$, and the top q-degree term of p_{m+1} is $-q^{2m+1}\cdot$ (top of p_m), of degree $m^2+2m+1 = (m+1)^2$, which cannot cancel against Xp_{m-1} (degree $(m-1)^2$). (iii) The perturbation $(1-q^{2m+1})/x_0 - 1/x_0$ decays geometrically, so the Poincaré–Perron theorem applies with the limit equation $u_{m+1} = (x_0+1/x_0)u_m - u_{m-1}$, whose roots $x_0 \ne 1/x_0$ are simple. (iv) A form using $g_0,\dots,g_M$ vanishes to order at most M at $z=0$, so its value at a fixed z_0 inside the disc is $\gg |g_{M+1}z_0^{M+1}| = q^{M^2(1+o(1))}$ in the truncation-based families; its denominators involve (t^i-s^i) for $i \le 2M$, i.e. $\Phi_d(t,s)$ for $d \le 2M$, and $\sum_{d\le N}\varphi(d) = \frac{3}{\pi^2}N^2 + O(N\log N)$ gives the floor with $N = 2M$. The degree-26 factor is exhibited by the exact $n=3$ computation (`code/zeta_probe/route_b/beta2_qniven_ladder.py`). □

Remark 19 (scope). Proposition 18 closes every construction whose coefficients live in the natural $\mathbb{Z}[q^{\pm1}, z]$-lattices of the q-difference module; Taylor Padé and Hermite–Padé, the argument-direction continued fraction, the ν-ladders of both kinds; together with all cyclotomic-compression refinements of them. It does *not* exclude a fundamentally different family of forms; producing one is exactly what a $\rho = \frac{1}{2}$ criterion would mean. The ledger thereby locates β_2's arithmetic at a precise frontier of q-Diophantine approximation.

The remaining freedom can be narrowed further: the finite places of $\mathbb{Q}$ are provably useless here. The series $S(q) = \sum_k g_k z^{*k}$, $z^* = 2q(1-q)$, has rational terms, so besides its real sum it possesses s-adic and t-adic *avatars*; the same series summed in $\mathbb{Q}_s$ and $\mathbb{Q}_t$, where it also converges (the valuations of the terms tend to $+\infty$).

Lemma 20 (adelic units)**.** *Let* $q = s/t \in (0,1)$*,* $\gcd(s,t) = 1$*, and* $z^* = 2q(1-q)$*. For every* $k \ge 1$ *and every prime* p*: if* $p \mid t$ *with* $e = v_p(t)$*, then*

$$v_p(g_k z^{*k}) \;=\; k^2 e + k\,[\,p=2\,] \;\ge\; k^2,$$

(exact, verified symbolically), and if $p \mid s$ *then* $v_p(g_k z^{*k}) \ge k^2$*. Consequently the series converges in* $\mathbb{Q}_p$ *for every* $p \mid st$*, and the finite-place avatars satisfy* $S_p \equiv 1 \pmod{p^{v_p(st)}}$*; in particular* $S_t \equiv 1 \pmod{t}$ *and* $S_s \equiv 1 \pmod{s}$*: they are* units*, hence nonzero; regardless of whether the real avatar vanishes.*

Proof. Each factor $t^i - s^i$ of $\mathrm{den}(g_k)$ is a p-unit for every $p \mid st$ (it is $\equiv -s^i$ or $\equiv t^i \bmod p$), so by Proposition 18(i), $v_p(g_k) = (k^2+2k)e$ for $p \mid t$ and $v_p(g_k) = (k^2-k)\,v_p(s)$ for $p \mid s$. From $z^* = 2s(t-s)/t^2$: for $p \mid t$, $v_p(t-s) = 0$ and $v_p(z^*) = [\,p=2\,] - 2e$, giving $v_p(g_k z^{*k}) = (k^2+2k)e + k([\,p=2\,]-2e) = k^2 e + k[\,p=2\,]$; for $p \mid s$, $v_p(z^*) = v_p(2s) \ge v_p(s) \ge 1$, giving $v_p(g_k z^{*k}) \ge (k^2-k)+k = k^2$. All are ≥ 1 for $k \ge 1$, so each avatar is 1 plus terms of positive valuation. □

Remark 21 (exact adelic balance, and the shape of any future criterion). Lemma 20 has two consequences. First, it extends the no-go ledger from the module's coefficient lattices to the *values*:

no divisibility or smallness can be sourced from the finite places, since the finite-place avatars are units; a hypothetical rationality of β_2 would be a maximally place-asymmetric event, the real avatar vanishing while every finite avatar equals a unit $\equiv 1$. Second, it makes the $\rho = \frac{1}{2}$ balance exact in the sense of the product formula: the term $g_k z^{*k}$ carries archimedean decay $(s/t)^{k^2(1+o(1))}$, s-adic and t-adic numerator content $s^{k^2} t^{k^2}$, and denominator $t^{2k^2(1+o(1))}$; the books balance with *zero margin* at the k^2 scale, so any successful construction must be lossless at all three places simultaneously and win at the geometric scale. Every mechanism surveyed above loses a factor t^{ck^2} somewhere; a $\rho = \frac{1}{2}$ criterion is therefore forced to be a purely *archimedean cancellation* phenomenon. This suggests the precise principle that would settle β_2, which we record as a conjecture: for the irregular rank-two q-difference module with polynomial companion matrix and $\det A = q$, the vanishing locus is *adelically rigid*: the entire function $G_v(q, \cdot)$ cannot vanish at a rational point z_0 at which all finite-place avatars are units (as at $z_0 = z^*$, by Lemma 20) at one place v of $\mathbb{Q}$ while the avatars at all other convergent places are units. The finite-place half of this rigidity is exactly Lemma 20; its archimedean half implies $\beta_2 \notin \mathbb{Q}$. This is a q-analogue, for irregular q-difference modules, of the place-uniformity enjoyed by E- and G-function values, where the exceptional sets of Siegel–Shidlovskii–André theory are independent of the place.

Further analysis of the archimedean obstruction yields several exact structural facts and a faithful reduction. All are verified numerically to 40–60 digits (`code/zeta_probe/route_b/beta2_*.py`).

Proposition 22 (structure of the zero curve)**.** *Write $Q = q^2$ and let $0 < z_1(q) < z_2(q) < \cdots$ be the positive zeros of $G(q, \cdot)$.*

(i) (base change) $G(q,z) = \dfrac{(q^2;q^2)_\infty}{\sqrt{q}\,(q;q^2)_\infty}\, z^{1/4}\, J^{(3)}_{-1/2}(\sqrt{z}/q;\, q^2)$, *where $J^{(3)}_\nu$ is the Hahn–Exton (third Jackson) q-Bessel function; thus G is, up to an explicit prefactor, the* Hahn–Exton q-cosine*, and its zeros are those of a classical, well-studied object.*

(ii) (zero lattice) *$z_k(q)\, Q^k \to 1$ as $k \to \infty$, with theta-suppressed dressing $\delta_k := z_k Q^k - 1 = O(c^k Q^{k^2})$ (numerically $\delta_0 = -0.5051,\ \delta_1 = -1.17 \cdot 10^{-2},\ \delta_2 = -1.03 \cdot 10^{-5}, \ldots$); $z^* = z_1(q^*)$ is the maximally dressed member, displaced from the bare lattice point $Q^0 = 1$ by an $O(1)$ correction; it is* not *a torsion/lattice point of the connection curve.*

(iii) (power sums are rational) *the symmetric functions $\sigma_p(q) := \sum_{n\geq 1} z_n(q)^{-p}$ lie in $\mathbb{Q}(q)$ for every $p \geq 1$ (Newton's identities on the $\mathbb{Q}(q)$-coefficients g_k); e.g. $\sigma_1 = 1/((1-q)(1-q^2))$. Consequently $z^*(q) = \lim_{p\to\infty} \sigma_p(q)/\sigma_{p+1}(q)$ with geometric rate $(z_1/z_2)^p$, giving an all-orders $\mathbb{Q}(q)$-approximation of the zero curve.*

Two of the no-go's ingredients also sharpen. First, the *cause* of $\rho = \frac{1}{2}$ is exactly the *double* Pochhammer: $\rho = \log(t/s)/(2\log t)$, and the factor 2 in the denominator is the $2k$ in $(q;q)_{2k}$; a single Pochhammer would give $\rho > 1$ (whence Liouville) when $t > s^2$. The intractability is intrinsic to the base-q^2 Hahn–Exton structure, not to the choice of construction. Second, the finite-place primes are *disjoint* from st: $\prod_{i\leq 2N}(t^i - s^i)$ is coprime to st (only the prime 2, via lifting-the-exponent when s, t are both odd, contributes an $O(N)$ term), so the arithmetic of every partial-sum denominator is supported off $\{s, t\}$; orthogonal to the unit residues of Lemma 20. The Hankel/determinantal escape is likewise closed exactly: $\det(g_{i+j})_{0\leq i,j\leq n}$ has denominator of *cubic* archimedean size ($\log|\mathrm{den}| \sim c\, n^3$), one power worse than the quadratic remainder.

Finally, the Nesterenko route is blocked more precisely than by non-holonomicity alone. Nesterenko's method needs a finitely generated δ-closed ($\delta = q\, d/dq$) $\mathbb{C}(q)$-algebra containing S, i.e. that S be *differentially algebraic*; non-holonomicity does not suffice (the quasimodular E_2 is non-holonomic yet

δ-algebraic). One computes $\delta S = \sum_k a_k(k^2 - \frac{kq}{1-q} + L_k)$ with $L_k = \sum_{i=1}^{2k} iq^i/(1-q^i) \to (1-E_2)/24$: S carries *every* finite Eisenstein truncation L_k, not merely the limit, which is precisely why it escapes the modular differential ring. Whether S is differentially algebraic is itself open, but the exclusion is now proof-grade over a wide window: S satisfies no algebraic differential equation in the δ-jets at any of the profiles (order, degree, $\deg_q$) $= (2,12,3)$, $(3,8,3)$, $(4,5,6)$, $(5,4,8)$, $(6,3,15)$, $(8,2,35)$, $(12,1,120)$; exact linear algebra on 2000 coefficients modulo a prime, every matrix of full column rank; since a primitive integer relation survives reduction, full rank modulo one prime excludes a relation over $\mathbb{Q}$ at that profile outright. (The pipeline is validated on E_2, for which it recovers the q-Chazy equation $\delta^3 E_2 = E_2\delta^2 E_2 - \frac{3}{2}(\delta E_2)^2$ at its exact profile.) The same exclusion holds verbatim for the zero branch $z_1(q)$. The Nesterenko route is blocked *modulo the differential transcendence of* S, which is itself open; and, with no functional equation in q available for S (no shift, q-difference, or Mahler equation exists), no current method can even attack it.

Remark 23 (reduction to a theta non-resonance). All routes converge on the same faithful reformulation: β_2 is irrational iff $G(q^*, z^*) \neq 0$ is forced by the arithmetic, and (Ramis–Sauloy–Zhang) this is a *theta non-resonance* statement; the connection ratio $R_q(z) = [P_{21}\, y_{\mathrm{irr}}]/[P_{11}\, y_{\mathrm{reg}}]$, an explicit θ_{q^2}-quotient, must satisfy $R_q(2q(1-q)) \neq -1$ at every algebraic q. The evidence points toward truth: in the confluent limit $q \to 1$ this is exactly Lindemann's theorem that cos has no algebraic zero. But the reduction is not yet shown *easier*: $R_q(z) = -1$ is a functional-equation-consistent value that a nonconstant theta-quotient must attain somewhere, and on the diagonal $z = 2q(1-q)$ the theta-arguments appear algebraically dependent, so no current independence theorem (Bertrand, Nesterenko, Schneider–Lang) forbids it. The archimedean obstruction is thus the exact mirror of the proven finite-place balance: units ($\equiv 1$) at every $p \mid st$, and a putative -1 at ∞, with neither forbidden. The required connection-entry computation is carried out below.

Proposition 24 (the connection entries). *Write $Q = q^2$, $\theta_b(x) = \sum_{n\in\mathbb{Z}} b^{n(n-1)/2}x^n$. The equation for G has, at $z = \infty$, the formal solution basis $f_1(z) = \theta_Q(-z)\,\hat{g}(z)$ and $f_2(z) = h(z)/\theta_Q(-z/q)$, where $h(z) = \sum_k c_k z^{-k}$ is* convergent *(entire in $1/z$, $c_k = O(Q^{k^2/2})$) while $\hat{g}(z) = \sum_k b_k z^{-k}$ is q-Gevrey divergent, with the closed form*

$$b_k = \frac{q^{\,k-k(k-1)/2}}{(q;q)_k}$$

(equivalent to the recursion $b_k = [-(q+1)Q\, b_{k-1} - Q^{3-k} b_{k-2}]/(q(Q^k - 1))$; verified symbolically for $k \le 8$). By Euler's theorem $\sum_j x^j/(q;q)_j = 1/(x;q)_\infty$, the q-Borel transform is therefore explicit:

$$\widehat{B}(\xi) = \sum_{k\ge 0} b_k\, q^{k(k+1)/2}\xi^{-k} = \sum_{k\ge 0} \frac{(q^2/\xi)^k}{(q;q)_k} = \frac{1}{(q^2/\xi;\, q)_\infty},$$

meromorphic on $\mathbb{C}^*$ with poles exactly on $\xi \in q^{2+\mathbb{N}_0}$, and satisfying $\widehat{B}(\xi) + \frac{q+1}{\xi}\widehat{B}(q\xi) + (\xi^{-2} - 1)\widehat{B}(q^2\xi) = 0$ identically (direct computation from the product form; in particular $\widehat{B}(-1) = (q+1)\widehat{B}(-q)$). The theta-kernel q-Laplace

$$g^{[\lambda]}(z) = \frac{\sum_{m\in\mathbb{Z}} q^{m(m-1)/2}(\lambda/z)^m\, \widehat{B}(\lambda q^m)}{\theta_q(\lambda/z)}, \qquad f_1^{[\lambda]} = \theta_Q(-z)\, g^{[\lambda]}(z),$$

is then singular only for $\lambda q^{\mathbb{Z}}$ meeting the pole set, i.e. for the single class $[\lambda] \in q^{\mathbb{Z}}$ of $E_q = \mathbb{C}^*/q^{\mathbb{Z}}$ (the positive-real lattice class, near which the zeros of G lie); every other class is regular. For $\lambda \notin q^{\mathbb{Z}}$ the bilateral sum converges absolutely on compacts of $\mathbb{C}^* \setminus (-\lambda q^{\mathbb{Z}})$ and $Lf_1^{[\lambda]} = 0$ exactly*: the* *theta shifts* $\theta_q(q^{-2}x) = q^{-3}x^2\theta_q(x)$ *and* $\theta_q(q^{-4}x) = q^{-10}x^4\theta_q(x)$ *turn* $f_1^{[\lambda]}(Qz)$ *and* $f_1^{[\lambda]}(Q^2z)$ *into the same kernel sum with* $\widehat{B}(\lambda q^m)$ *replaced by* $\widehat{B}(\lambda q^{m+2})$, $\widehat{B}(\lambda q^{m+4})$*; collecting the coefficient of* $q^{m(m-1)/2}(\lambda/z)^m/\theta_q(\lambda/z)$ *in* $Lf_1^{[\lambda]}/\theta_Q(-z)$ *then yields* $q\big[\widehat{B}(\xi) + \frac{q+1}{\xi}\widehat{B}(q\xi) + (\xi^{-2}-1)\widehat{B}(q^2\xi)\big]$ *at* $\xi = \lambda q^m$*, identically zero. (Each step was verified numerically to* 40 *digits; the kernel normalization was validated on convergent input to* 34 *digits, and* $Lf_1^{[\lambda]} \approx 10^{-61}$ *numerically.) Writing* $G = C_1 f_1^{[\lambda]} + C_2^{[\lambda]} f_2$*, numerically to* 59–60 *digits:*

(i) *since* $\det A = q$*, every Casoratian* $\mathrm{Cas}(u,v)(z) := u(z)v(Qz) - u(Qz)v(z)$ *of solutions satisfies* $\mathrm{Cas}(Qz) = q\,\mathrm{Cas}(z)$ *(immediate from* $u(Q^2z) = a\,u(Qz) - q\,u(z)$*), so* $C_1 = \mathrm{Cas}(G, f_2)/\mathrm{Cas}(f_1^{[\lambda]}, f_2)$ *and* $C_2^{[\lambda]} = \mathrm{Cas}(f_1^{[\lambda]}, G)/\mathrm{Cas}(f_1^{[\lambda]}, f_2)$ *are* Q*-elliptic; proven, and verified to* 10^{-60}*;*

(ii) C_1 *is a* constant*, independent of both* z *and the resummation direction* λ*: this is proven in Theorem 25 below (the limit argument applies along every orbit class* $\neq [1]_Q$*, and ellipticity extends the value); the slope-*1 *component of* G *is Stokes-invariant, and all* q*-Stokes ambiguity is carried by* $C_2^{[\lambda]}$*, nonconstant elliptic with* λ^2 *in its zero divisor;*

(iii) (dressing-product formula, verified to 59 digits)

$$C_1 = \frac{1}{(Q;Q)_\infty \prod_{k\ge1} z_k Q^{\,k-1}} \ :$$

the Stokes-invariant connection constant is the regularized product of the zero dressings;

(iv) *at* $q = q^*$ *the entire apparatus evaluates:* $C_1(q^*) = 2.69898382616391375\ldots$*, and the zero condition* $G(q^*, z^*) = 0$ *takes the pinned form*

$$C_2^{[\lambda]}(z^*) \;=\; -\,C_1\,\frac{f_1^{[\lambda]}(z^*)}{f_2(z^*)} \;=\; 0.011592994120624266\ldots \qquad (\lambda = \tfrac{3}{2}),$$

verified to 30 *digits. This is the explicit instance of the theta non-resonance: every object in it is now constructed.*

The constant C_1 is in fact *modular*, and this upgrades the dressing-product formula into an exact evaluation of the zero product.

Theorem 25 (the connection constant, and the zero product of the Hahn–Exton q-cosine)**.** *For* $0 < q < 1$,

$$C_1 = \frac{1}{(q;q)_\infty}, \qquad \textit{and consequently} \qquad \prod_{k\ge1} z_k(q)\,Q^{\,k-1} \;=\; (q;q^2)_\infty,$$

where $z_1 < z_2 < \cdots$ *are the positive zeros of* $G(q,\cdot)$*. Likewise, for the second solution's underlying entire function* H *with zeros* $w_1 < w_2 < \cdots$*,* $\prod_{k\ge1} w_k(q)\,q\,Q^{\,k-1} = (q^3;q^2)_\infty$*.* (Verified at three bases including q^*, to 41, 23, 26 digits for C_1 and 58 digits for the product.)

Proof. (1) An exact identity. From $(q;q)_{2k} = (q;q)_\infty/(q^{2k+1};q)_\infty$ and Euler's expansion $(x;q)_\infty = \sum_{j\ge0}(-1)^j q^{j(j-1)/2}x^j/(q;q)_j$ at $x = q^{2k+1}$, together with $k(k-1)+2kj = (k+j)(k+j-1)-j(j-1)$ and the substitution $m = k+j$, absolute convergence of the double sum permits the rearrangement

$$(q;q)_\infty\,G(z) \;=\; \sum_{m\ge0}(-1)^m q^{m(m-1)} z^m\,P_m(1/z), \qquad P_m(w) := \sum_{j=0}^{m} c_j w^j, \quad c_j = \frac{q^{\,j-j(j-1)/2}}{(q;q)_j}$$

(the c_j coincide with the b_j of Proposition 24; identity checked symbolically through z^8 and numerically to 41 digits).

(2) The constant. Fix $u<0$ and put $z=uQ^{-N}$. The moduli $q^{m(m-1)}|z|^m$ peak at $m_0=N+O(1)$ and decay like $q^{(m-m_0)^2+O(1)}$ away from it. For $1\le j\le m\le 2N$, $2Nj-j(j-1)/2\ge Nj$, so $|c_j z^{-j}|\le Cq^{Nj}|u|^{-j}$ and $|P_m(1/z)-1|\le C'q^N$ uniformly; the terms with $m>2N$ contribute a relative $O(q^{N^2})$. The negative-m part of $\theta_Q(-z)$ is $O(|z|^{-1})$, negligible against the peak. Hence $(q;q)_\infty G(z)/\theta_Q(-z)\to 1$ along $z=uQ^{-N}$, for each fixed $u<0$.

(3) Geometric zero dressing. On the circle $|z-Q^{1-k}|=\varepsilon q^k Q^{1-k}$ the identity of step (1) and the bounds of step (2) give $|(q;q)_\infty G-\theta_Q(-z)|\le Cq^k|\theta_Q(-z)|_{\mathrm{peak}}$, while $\theta_Q(-z)$ has a simple zero at Q^{1-k} with derivative bounded below there; Rouché gives $z_k=Q^{1-k}(1+O(q^k))$, i.e. $\log(z_kQ^{k-1})=O(q^k)$, absolutely summable.

(4) The product. $G(z)=\prod_k(1-z/z_k)$ (order 0, genus 0, $G(0)=1$) and $\theta_Q(-z)=(Q;Q)_\infty(z;Q)_\infty(Q/z;Q)_\infty$ (Jacobi). Along $z=-R$, $R\to\infty$,

$$\frac{G(-R)}{\theta_Q(R)}=\frac{1}{(Q;Q)_\infty(-Q/R;Q)_\infty}\prod_{k\ge1}\frac{1+R/z_k}{1+RQ^{k-1}},$$

and for $x,y>0$ the function $R\mapsto \log\frac{1+Rx}{1+Ry}$ is monotone from 0 to $\log(x/y)$, so each factor is dominated by $|\log(z_kQ^{k-1})|$, summable by step (3). Dominated convergence gives $G(-R)/\theta_Q(R)\to 1/\big[(Q;Q)_\infty\prod_k z_kQ^{k-1}\big]$ as $R\to\infty$; evaluating along the subsequence $R=|u|Q^{-N}$ by step (2), the limit equals $1/(q;q)_\infty$, whence $\prod_k z_kQ^{k-1}=(q;q)_\infty/(Q;Q)_\infty=(q;q^2)_\infty$ by Euler's $(q;q)_\infty=(q;q^2)_\infty(q^2;q^2)_\infty$. Finally, in the basis of Proposition 24: along the negative axis (a regular class, $[-1]\notin q^{\mathbb{Z}}$) one has $g^{[\lambda]}(z)\to 1$ and $f_2(z)/\theta_Q(-z)=O\big(1/(\theta_Q(R)\theta_Q(R/q))\big)\to 0$, so the limit just computed is the entry C_1; hence $C_1=1/(q;q)_\infty$. The H-statement is identical with comparison function $\theta_Q(-qz)$ and ratio $(1-q)/(q;q)_\infty$. Finally, steps (1)–(2) apply verbatim along $z=uQ^{-N}$ for any $u\in\mathbb{C}^*$ whose orbit avoids the zero class $[1]_Q$ of $\theta_Q(-\,\cdot)$, using the standard theta lower bound $|\theta_Q(-z)|\ge c([u])\cdot$(peak term) off that class; along each such orbit $C_1(z)$ is constant (ellipticity) and $f_2/f_1\to 0$, $g^{[\lambda]}\to 1$, so $C_1([u])=1/(q;q)_\infty$ on every class $\neq[1]_Q$, and by meromorphy $C_1\equiv 1/(q;q)_\infty$: for every regular λ. In particular C_1 is z- and λ-independent. ☐

Remark 26 (Diophantine consequences at z^*). At $q=q^*$ the identity splits off the candidate rational zero:

$$\prod_{k\ge2} z_k(q^*)\,Q^{\,k-1}\;=\;\frac{(q;q^2)_\infty}{z^*}\;=\;0.988299884628746\ldots\qquad\text{(verified to 30 digits).}$$

Were β_2 rational, the *modular* value $(q;q^2)_\infty$ (transcendental at algebraic q by Nesterenko's theorem applied to the Euler products) would equal the rational z^* times the tail dressing product $\prod_{k\ge2} z_kQ^{k-1}$. Thus the entire modular sector of the problem is now explicit; the Stokes-invariant connection constant is $1/(q;q)_\infty$, the zero product is $(q;q^2)_\infty$; and *all* remaining unknown arithmetic is provably confined to the tail dressing product, equivalently to the q-Stokes cocycle $C_2^{[\lambda]}$: the non-modularity of the β_2 problem *is* its q-Stokes data. A transcendence statement for that tail (an algebraic independence of the higher-zero dressings from the modular values, or a rigidity theorem for the q-Stokes cocycle) would settle β_2; no current theorem provides it.

The cocycle itself is now explicit. Writing $g=g^{[\lambda]}$ for the resummed series and h for f_2's convergent one,

$$C_2^{[\lambda]}(z)\;=\;\theta_Q(-z/q)\,\frac{g(z)\,G(Qz)+z^{-1}g(Qz)\,G(z)}{-(z/q)\,g(z)\,h(Qz)+z^{-1}g(Qz)\,h(z)}$$

This is an identity: it is the ratio of the Casoratian evaluations $\mathrm{Cas}(f_1,G)=\theta_Q(-z)\big[g(z)G(Qz)+z^{-1}g(Qz)G(z)\big]$ and $\mathrm{Cas}(f_1,f_2)=\frac{\theta_Q(-z)}{\theta_Q(-z/q)}\big[-(z/q)g(z)h(Qz)+z^{-1}g(Qz)h(z)\big]$, both instances of $\theta_Q(Qw)=w^{-1}\theta_Q(w)$ (and verified numerically to 10^{-61}); with divisor now fully described: an *explicit* zero lattice $z\in q^{-1}Q^{\mathbb{Z}}$ from the theta factor (the computed zero at $z=1/q$, exact to 10^{-63}), a direction-dependent zero at $z=\lambda^2$ arising from the exact *resummation resonance* $g^{[\lambda]}(\lambda^2)G(Q\lambda^2)+\lambda^{-2}g^{[\lambda]}(Q\lambda^2)G(\lambda^2)=0$ (verified to 10^{-62} at $\lambda\in\{\frac{3}{2},2,\frac{13}{5}\}$; a proof for general regular λ is open), and poles at the zeros of the h-bracket. Substituting into the pinned zero condition at z^* and cancelling the common theta factor reduces the entire β_2 question to *one* relation between the two modular values $\theta_Q(-z^*)$, $(q;q)_\infty$ and five values of the non-modular pair $(g^{[\lambda]},h)$ on the orbit of z^*. The remaining input is a transcendence/independence statement for those five values; supplying it is one of three research programs (a non-modular extension of Nesterenko's method via the deformation of this connection, a $\rho<\frac{1}{2}$-regime irrationality criterion, or adelic rigidity of the vanishing locus); none is currently available.

The first of those programs now has its foundation: the deformation of the module in q closes finitely.

Proposition 27 (the deformation module)**.** *Let $L=q-a\sigma+\sigma^2$ with $a=q+1-qz$, $\sigma f(z)=f(Qz)$, and let $\theta=z\partial_z$, $\delta=q\partial_q$ (acting on all q-dependence, including the base $Q=q^2$, so that $\delta\sigma=\sigma\delta+2\sigma\theta$). Then*

$$L(\theta G)=-qz\,\sigma G,\qquad L(\delta G)=-qG+q(1+3z)\,\sigma G+4q\,\theta G-2a\,\sigma\theta G$$

(two commutator computations; verified to 51 *digits). Consequently the $\mathbb{Q}(q,z)\langle\sigma\rangle$-module generated by $\{G,\theta G,\delta G\}$ has finite rank ≤ 6, with explicit σ-companion relations $\sigma^2G=a\sigma G-qG$, $\sigma^2\theta G=a\sigma\theta G-q\theta G-qz\,\sigma G$, and $\sigma^2\delta G=a\sigma\delta G-q\delta G-qG+q(1+3z)\sigma G+4q\theta G-2a\,\sigma\theta G$.*

This finiteness is of the first jet only*: the module is* not *closed under a second application of θ or δ. Both θ^2G and δ^2G escape* $\mathrm{span}_{\mathbb{Q}(q,z)}\{G,\sigma G,\theta G,\sigma\theta G,\delta G,\sigma\delta G\}$ *(verified by exact modular linear algebra at $q=\frac{1}{3}$, z-series to* 70 *terms, coefficient degrees ≤ 6: the augmented rank strictly exceeds the plain rank). The two derivations are moreover linked by the exact identity*

$$\delta G=(\theta^2-\theta)G+C,\qquad C=\sum_{k\ge 0}(-1)^k\Big(\textstyle\sum_{i=1}^{2k}\frac{iq^i}{1-q^i}\Big)\frac{q^{k(k-1)}z^k}{(q;q)_{2k}}$$

($=z^2G_{zz}+C$; immediate from g_k's logarithmic q-derivative), whose weight C carries the truncated Eisenstein sums $\sum_{i\le 2k}iq^i/(1-q^i)$ term by term; the same off-modular data that obstructs the differential closure. In particular, differentiating the pinned relation of Proposition 24 in q does not *close: the second q-derivative leaves the module, consistent with Proposition 35. The value of the rank-*6 *structure is at the* function *level; it is the module whose difference Galois group is computed in Theorem 32; not as a Nesterenko precondition in the descent variable, which Proposition 35 shows does not exist.*

Corollary 28 (equation of the zero curve)**.** *Implicit differentiation of $G(q,z_1(q))=0$ gives*

$$z_1'(q)\;=\;-\frac{z_1\,(\delta G)(q,z_1)}{q\,(\theta G)(q,z_1)},$$

with $\theta G,\delta G$ the first-jet generators of Proposition 27 (formula checked against a finite-difference derivative to 20 *digits at $q=0.3$). The right-hand side is not a closed vector field; iterating d/dq leaves the rank-*6 *module (Proposition 27); so this is one exact relation, not an autonomous system. The number q^* is the crossing of the zero curve with the diagonal: $z_1(q^*)=2q^*(1-q^*)$ (residual $<10^{-27}$).*

Lemma 29 (irreducibility of the σ-module)**.** *For $0 < q < 1$ the rank-2 σ-module of G is irreducible over $\mathbb{C}(z)$: the Riccati equation*

$$u(z)\,u(Qz) - a(z)\,u(z) + q = 0, \qquad a = q + 1 - qz,$$

has no solution $u \in \mathbb{C}(z)$.

Proof. Suppose $u = P/R$ with $P, R \in \mathbb{C}[z]$ coprime, of degrees p, r, leading coefficients α, β. Clearing denominators,

$$F(z) \;:=\; P(Qz)P(z) - a(z)P(z)R(Qz) + q\,R(z)R(Qz) \;\equiv\; 0.$$

At $z = 0$: $F(0) = P(0)^2 - (q+1)P(0)R(0) + qR(0)^2 = 0$; if $R(0) = 0$ then $P(0) = 0$, contradicting coprimality, and symmetrically; so $P(0)R(0) \neq 0$ and $u(0) \in \{1, q\}$. *Top degree:* the three terms have degrees $2p$, $p + r + 1$, $2r$. If $p = r$, the top degree $2p + 1$ is attained by the middle term alone, with coefficient $q\alpha\beta Q^r \neq 0$; if $|p - r| \geq 2$, the top degree is attained by the first or third term alone, with coefficient $\alpha^2 Q^p$ or $q\beta^2 Q^r$. Both are impossible, so $|p - r| = 1$. *Root pairing:* let $\zeta \neq 0$ be a root of R of multiplicity m, so $P(\zeta) \neq 0$. At $z = \zeta/Q$ the three terms of F have orders k, $k + m + [\,a(\zeta/Q) = 0\,]$, $m_0 + m$, where $k = \operatorname{ord}_{\zeta/Q} P$ and $m_0 = \operatorname{ord}_{\zeta/Q} R$; since $F \equiv 0$ the minimum order cannot be attained by a single term, forcing $k \geq m_0 + m \geq m$. Symmetrically, a root $\xi \neq 0$ of P of multiplicity k forces $\operatorname{ord}_{Q\xi} R \geq k$ (coprimality gives $\operatorname{ord}_{Q\xi} P = 0$, so the first term has order k at ξ, the third $\operatorname{ord}_{Q\xi} R$, and the middle at least $k + \operatorname{ord}_{Q\xi} R$; again no single minimum). The two maps $\zeta \mapsto \zeta/Q$, $\xi \mapsto Q\xi$ are mutually inverse on the nonzero root sets, and chains close at length two by coprimality. Summing multiplicities, $p \leq r \leq p$, so $p = r$; contradicting $|p - r| = 1$. $\square$

Proposition 30 (the symmetric square)**.** *Products $w = y\tilde{y}$ of solutions of L satisfy the third-order equation*

$$w(Q^3 z) \;=\; \frac{A_1(A_0A_1 - q)}{A_0}\,w(Q^2 z) \;+\; q\,(q - A_0A_1)\,w(Qz) \;+\; \frac{q^3 A_1}{A_0}\,w(z),$$

with $A_0 = a(z)$, $A_1 = a(Qz)$, obtained by eliminating the mixed product $m = y\,\tilde{y}(Q\,\cdot) + \tilde{y}\,y(Q\,\cdot)$ through $m(Qz) = 2A_0\,w(Qz) - q\,m(z)$. Its characteristic roots at $z = 0$ are $\{1, q, q^2\}$; verified on the solutions G^2, $z^{1/2}GH$, zH^2 to 41–43 digits.

Lemma 31 (divisor orientation)**.** *Let $r \in \mathbb{C}(z)^\times$ satisfy the rational-quotient relation of Proposition 30, $r_0r_1r_2 = c_2r_0r_1 + c_1r_0 + c_0$ with $r_k = r(Q^k z)$, $c_2 = A_1(A_0A_1 - q)/A_0$, $c_1 = q(q - A_0A_1)$, $c_0 = q^3A_1/A_0$. On every Q-orbit in $\mathbb{C}^*$ disjoint from the singular orbit $z_aQ^{\mathbb{Z}}$ ($z_a = (q+1)/q$, the zero of a), the maximal-modulus zero-or-pole of r cannot be a pole, and the minimal-modulus zero-or-pole cannot be a zero.*

Proof. Off $z_aQ^{\mathbb{Z}}$ the c_i are finite and $c_0 \neq 0$. If the maximal-modulus zero-or-pole ζ on such an orbit were a pole, evaluate at $z = Q^{-2}\zeta$: there r_0, r_1 are finite and nonzero (all higher-modulus orbit points carry no zero or pole), r_2 has a pole, so the left side has a pole while the right side is finite. If the minimal-modulus zero-or-pole ζ' were a zero, rewrite the relation for $s = 1/r$ as $c_0s_0s_1s_2 + c_1s_1s_2 + c_2s_2 = 1$ and evaluate at $z = \zeta'$: s_0 has a pole, s_1, s_2 are finite and nonzero, $c_0(\zeta') \neq 0$, so the left side has a pole while the right side is 1. $\square$

Theorem 32 (the difference Galois group contains SL_2)**.** *For every $0 < q < 1$, the difference Galois group over $\mathbb{C}(z)$ of the module of G contains SL_2.*

Proof. For $N \geq 1$ suppose $\operatorname{Sym}^N M$ has a rank-1 submodule, i.e. there is a nonzero solution w of the N-th symmetric-power equation with $r := w(Q\,\cdot)/w \in \mathbb{C}(z)$. The products $f_1^{[\lambda]\,N-k} f_2^{k}$ ($0 \leq k \leq N$) form a solution basis (their Casoratian is a power of $\operatorname{Cas}(f_1, f_2) \neq 0$; checked numerically for $N = 2$), so $w = \sum_k c_k f_1^{N-k} f_2^k$ with constants c_k, and the case split is λ-robust since Stokes jumps preserve $k_0 := \min\{k : c_k \neq 0\}$.

Case $k_0 < N$. Along a ray in a regular class, f_2/f_1 decays super-polynomially, so the rational function r has the Poincaré asymptotic expansion of the quotient of the dominant component:

$$r \;\sim\; (-1/z)^{N-k_0}(-z/q)^{k_0}\left(\tfrac{\hat{g}(Qz)}{\hat{g}(z)}\right)^{N-k_0}\left(\tfrac{h(Qz)}{h(z)}\right)^{k_0}.$$

The order-j coefficient d_j of $\hat{g}(Qz)/\hat{g}(z)$ satisfies $d_j \sim -b_j$ with $b_j = q^{(3j-j^2)/2}/(q;q)_j$ (verified: $d_j/(-b_j) = 0.99986,\ 0.99998,\ 0.999998$ at $j = 10, 12, 14$, $q = \frac{1}{3}$, converging to 1). Since this factor enters r to the power $N - k_0 \geq 1$ against rational, hence geometrically bounded, prefactors, the coefficients of r inherit the growth $b_j \sim q^{-j^2/2}$, which is super-geometric, whereas a rational function's expansion at ∞ has geometrically bounded coefficients. Since matching asymptotic expansions are unique, no rational r exists.

Case $k_0 = N$. Then $r = (-z/q)^N \rho^N$ with $\rho := h(Q\,\cdot)/h$ meromorphic on $\mathbb{C}^*$; $\rho^N \in \mathbb{C}(z)$ and single-valuedness force $\rho \in \mathbb{C}(z)$ (an N-th root of a rational function is single-valued on $\mathbb{C}^*$ only if every divisor multiplicity, including at 0 and ∞, is divisible by N). Then $f_2(Qz)/f_2(z) = -(z/q)\rho(z)$ is rational, giving a rational solution of the rank-2 Riccati equation; contradicting Lemma 29.

Hence no $\mathrm{Sym}^N M$ has a rank-1 submodule. $N = 1$ is irreducibility; $N = 2$ excludes the imprimitive (dihedral) case; and a projectively finite irreducible subgroup (A_4, S_4, A_5) fixes a line in some Sym^N with $N \leq 12$ (the classical binary polyhedral invariants), which is likewise excluded. By the classification of algebraic subgroups of GL_2, the group contains SL_2. ☐

Proposition 33 (no $\sigma\delta$-integrability)**.** *For every $0 < q < 1$ there is no $B \in \mathrm{gl}_2(\mathbb{C}(z))$ and no $c \in \mathbb{C}(z)$ with $\theta(A) = \sigma(B)A - AB + cA$, where $A = \left(\begin{smallmatrix} 0 & 1 \\ -q & a \end{smallmatrix}\right)$ and $\theta = z\partial_z$.*

Proof. Writing $B = \left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right)$, the entries give $\gamma = -q\beta_1$, $\delta = \alpha_1 + a\beta_1 - c$, and eliminating α (the central terms c cancel identically; verified symbolically) leaves the necessary condition

$$qa\,\beta_3 \;=\; a_1(aa_1 - q)\,\beta_2 \;-\; a(aa_1 - q)\,\beta_1 \;+\; qa_1\,\beta \;+\; q\,z(a_1 + q^2 a),$$

$\beta_k = \beta(Q^k z)$, $a = a(z)$, $a_1 = a(Qz)$. A polynomial part of degree $m \geq 0$ contributes top coefficient $\propto \beta_m(1 - q^{2+2m}) \neq 0$ at degree $m + 3$, above the inhomogeneity: so β has no polynomial part. If β has a pole off the orbit $z_a Q^{\mathbb{Z}}$, the maximal-modulus pole appears in exactly one term (the slot $z = \zeta Q^{-3}$, coefficient $qa \neq 0$ there), and the minimal-modulus pole likewise (slot $z = \zeta'$, coefficient qa_1): both impossible; so the poles lie in the window $\{z_a Q^{-1}, \ldots, z_a Q^3\}$ ($a(z_a Q^k) = (q+1)(1 - Q^k)$ vanishes only at $k = 0$). Sweeping the slots $z = z_a Q^{-4}, \ldots, z_a Q^{-1}$ kills the orders at $z_a Q^{-1}, \ldots, z_a Q^2$ one by one (each pole appears in a single term with nonvanishing coefficient), and the order at $z_a Q^3$ is killed at the slot $z = z_a Q$; coefficient $a_1(aa_1 - q)$ with $(aa_1 - q)(z_a Q) = (q+1)^2(1 - q^2)(1 - q^4) - q$; or, where that vanishes (one root $q \approx 0.78027$), at the slot $z = z_a Q^2$; coefficient $a(aa_1 - q)$ with $(aa_1 - q)(z_a Q^2) = (q+1)^2(1 - q^4)(1 - q^6) - q$ (one root $q \approx 0.86753$); the two polynomials are coprime, so for every $q \in (0,1)$ at least one slot applies. Hence $\beta \equiv 0$, contradicting $z(a_1 + q^2 a) = z\big[(q+1)(1 + q^2) - 2q^3 z\big] \not\equiv 0$. (Cross-check: at $q = \frac{1}{3}$ the exact linear system for poles of order ≤ 3 in the window has $\mathrm{rank}(A) = 15$, $\mathrm{rank}[A|b] = 16$.) ☐

Corollary 34 (differential transcendence of G)**.** *For every $0 < q < 1$ the function $G(q, \cdot)$ satisfies no algebraic differential equation over $\mathbb{C}(z)$: the Hahn–Exton q-cosine is differentially transcendental.*

Proof, via parametrized difference-Galois theory. We apply the parametrized (σ, θ)-difference Galois theory with derivation $\theta = z\partial_z$. The determinant equation $\sigma(y) = \det(A)\,y = q\,y$ has the solution

$y = z^{1/2}$ (indeed $\sigma(z^{1/2}) = (Qz)^{1/2} = q\, z^{1/2}$, as $Q = q^2$), which is algebraic over $\mathbb{C}(z)$; hence the parametrized difference Galois group of $\sigma(y) = \det(A)y$ lies in $\mathbb{C}^\times$. This is exactly the hypothesis of Dreyfus–Hardouin–Roques [13, Prop. 2.9], which for a rank-2 system whose difference Galois group contains SL_2 (here Theorem 32) gives the dichotomy: the parametrized group Gal^θ either is conjugate into $\mathrm{GL}_2(\mathbb{C})$ (all solutions differentially algebraic) or contains $\mathrm{SL}_2(\widetilde{\mathcal{C}})$ (all nonzero solutions differentially transcendental), and the *first* alternative holds if and only if there exists $B \in \mathrm{gl}_2(\mathbb{C}(z))$ with $\theta(A) = \sigma(B)A - AB$ (their integrability condition (7); $\theta(\det A) = 0$ since $\det A = q$ is constant in z, so no trace term appears). By Proposition 33 no such B exists; indeed none exists even allowing a scalar twist cA. Therefore the second alternative holds: $\mathrm{Gal}^\theta \supseteq \mathrm{SL}_2(\widetilde{\mathcal{C}})$, and every nonzero solution, in particular $G(q,\cdot)$, is differentially transcendental over $\mathbb{C}(z)$. $\square$

The transcendence of the *number* q^* is a value-level statement, and the natural route (Nesterenko's multiplicity method fed by Corollary 34) meets a precise structural obstruction, which we isolate.

Proposition 35 (where the differential closure fails in the descent variable)**.** *Write $\delta = q\partial_q$, $\theta = z\partial_z$, and let $u_j(q) := G(q, Q^j z^*(q))$ denote the orbit values along the diagonal $z^*(q) = 2q(1-q)$.*

(i) (orbit collapse) *The σ-orbit is* not *an obstruction to finiteness: the z-recurrence evaluated on the diagonal has coefficients $a(Q^j z^*) = q + 1 - qQ^j z^* \in \mathbb{Q}(q)$, so $u_{j+2} = a(Q^j z^*)\, u_{j+1} - q\, u_j$ and every orbit value lies in* $\mathrm{span}_{\mathbb{Q}(q)}\{u_0, u_1\}$ (verified for $j \le 3$ to 32 digits).

(ii) (the E_2-coupled jet structure) *The q-derivative of G obeys the exact identity*

$$\delta G = (\theta^2 - \theta)G + \varphi(q)\, G - T, \qquad \varphi(q) = \sum_{i\ge1} \frac{iq^i}{1-q^i}, \qquad T = \sum_{k\ge0} (-1)^k \Big(\sum_{i>2k} \tfrac{iq^i}{1-q^i}\Big) \frac{q^{k(k-1)} z^k}{(q;q)_{2k}}$$

(verified to 41 digits)*, where φ is the quasimodular Eisenstein weight* ($24\varphi = 1 - E_2$ *up to normalization) and the correction T has coefficients smaller than G's by the factor $\sum_{i>2k} iq^i/(1-q^i) = O(kq^{2k})$. Expanding these weights as Lambert series and interchanging the (absolutely convergent) sums exhibits T as an Eisenstein-weighted trace of the module over the forward σ-orbit:*

$$T(q,z) \;=\; \sum_{d\ge1} \frac{2q^d}{1-q^d}\, (\theta G)(q, q^{2d} z) \;+\; \sum_{d\ge1} \frac{q^d}{(1-q^d)^2}\, G(q, q^{2d} z)$$

(verified to 41 digits)*: the entire non-modular content of δG is an orbit convolution of $(G, \theta G)$ against E_2-type weights. On the diagonal the orbit values collapse by* (i) *into four generators u_0, u_1, θ-analogues v_0, v_1 (the v's obey the inhomogeneous transported recurrence $v_{j+2} = a_j v_{j+1} - qv_j - qQ^j z^* u_{j+1}$), but the collapse coefficients accumulate into transfer-product series $A, B, C, D \in \mathbb{Z}[[q]]$ (integrality is immediate: transfer data and Lambert weights lie in $\mathbb{Z}[[q]]$), which are* not *of any classical type: they satisfy no relation over* $\mathrm{span}\{1, \varphi, \sum \sigma_3(n) q^n\}$ *with rational-function coefficients of degree ≤ 8, and no linear q-ODE of order ≤ 3 and degree ≤ 6 (exact linear algebra on 45 coefficients). The identity relocates, and does not remove, the obstruction. Since δ and θ commute, each application of δ raises the z-jet order by two: $\delta^n S$ involves $\theta^{2n} G|_{z=z^*}$ with unit leading coefficient.*

(iii) (the actual obstruction) *The z-jets do not close: $\theta^2 G$ and $\delta^2 G$ escape the rank-*6 *module of Proposition 27, and by Corollary 34 the tower $\{\theta^m G\}_m$ satisfies no algebraic relation over $\mathbb{C}(z)$. A finitely generated δ-closed $\mathbb{C}(q)$-algebra containing S would force an algebraic relation over $\mathbb{C}(q)$ among the diagonal restrictions $\{\theta^m G(q, z^*(q))\}_{m\le M}$ for some finite M; a statement* *equivalent to S being differentially algebraic in q, for which there is no supporting structure and proof-grade negative evidence across a wide window: no algebraic differential equation at profiles up to* $(2,12,3)$*,* $(3,8,3)$*,* $(4,5,6)$*,* $(5,4,8)$*,* $(6,3,15)$*,* $(8,2,35)$*,* $(12,1,120)$ *in (order, degree,* $\deg_q$*), by full column rank on* 2000 *coefficients modulo a prime (*`beta2_diffalg_search.py`*,* `beta2_ade_sweep.py`*;* E_2/q*-Chazy control recovered).*

Thus the Nesterenko route closes if and only if *S is differentially algebraic in q; the obstruction is carried by the z-jet tower; whose function-level independence is exactly Corollary 34; and not by the σ-orbit, which collapses to rank two.*

Remark 36 (the value-level descent as a joint multiplicity estimate)*.* Nesterenko's method derives transcendence of a value from three ingredients: a finite ring closed under the relevant operator, a *multiplicity estimate* bounding $\operatorname{ord} P(\text{generators}) \le c\,(\deg P)^{\kappa}$, and routine analytic size/Siegel-lemma constructions. Theorem 32 and Proposition 33 furnish precisely the nondegeneracy a multiplicity estimate consumes (no invariant subvariety; no first integral). But Proposition 35 shows the first ingredient fails in the descent variable q: the finiteness of the β_2 module is intrinsically *joint* in (σ_z, δ_q), and σ_z is a non-infinitesimal q-lattice shift, outside the Pfaffian/differential multiplicity theories of Nesterenko and Binyamini. Thus the transcendence of $q^* = \beta_2^{-2}$ reduces to a *joint q-difference–differential multiplicity estimate* for the rank-≤ 6 module at (q^*, z^*): a bound $\operatorname{ord}_{(q^*,z^*)} P(G, \theta G, \delta G) \le c\, N^{\kappa}$ ($N = \deg P$) in the local ring, uniform along the σ-orbit. Such an estimate for finite (σ, δ)-modules is not available in the literature; it is the sole non-routine input remaining, and the precise open problem on which the (ir)rationality of β_2 now turns.

Remark 37 (the obstruction inside q-Siegel–Shidlovskii theory)*.* The same wall appears, quantified, in the value-transcendence theory built on Siegel's method. The approach of Amou–Matala-aho–Väänänen [12] (and the Matala-aho-school Baker-type refinements) proves non-vanishing of values of Poincaré-type q-difference solutions at algebraic points under two hypotheses: the system is Poincaré-type at 0 (met here: $A(0)$ is invertible, $\det A = q$), and a *Siegel margin* $\gamma < \Gamma$ holds, where γ is the ratio of the logarithmic height of the cleared denominators to that of the analytic decay. For the Hahn–Exton class that ratio is exactly $\gamma = \rho = \log(t/s)/(2\log t) \le \frac{1}{2}$ of Proposition 18: the double Pochhammer $(q;q)_{2k}$ is the factor 2; so the margin *fails*, and the theory does not apply. Thus Conjecture Z (transcendence of the Hahn–Exton zeros at algebraic base, which by $z^* = z_1(q^*)$ implies β_2 transcendental) reduces to extending q-Siegel– Shidlovskii past the margin for a class that is additionally irregular at ∞. The connection theory of this paper supplies the irregular-∞ data explicitly; the margin itself is the value-theoretic face of the multiplicity estimate of Remark 36, and is equal in strength to β_2-transcendence, not weaker. A detailed program is recorded in the repository (`paper/journal/qSS_program.md`).

One further mechanism deserves record, because it isolates which half of the Siegel scheme is actually blocked. Under the rationality hypothesis all tail power sums $\tau_p = \sum_{k\ge 2} z_k^{-p} = \sigma_p - z^{*-p}$ are rational, and they are the moments of a positive measure with infinite support (unit atoms at $1/z_k$); hence every Hankel determinant $\det[\tau_{i+j+2}]_{i,j\le n}$ is *strictly positive for free*; no auxiliary nonvanishing argument is needed; and, being a positive rational, is bounded below by the reciprocal of its denominator ($\le t^{8n^3/3+O(n^2)}$). Analytically these determinants decay like $Q^{n^3/3(1+o(1))}$ (atoms at $\approx Q^{k-1}$; verified: strictly positive with super-quadratic exponent growth at q^*). The comparison $2\log(t/s) > 8\log t$ fails for every rational q: the positivity route conserves the ledger at a factor 4. The instructive point is that *nonvanishing was never the bottleneck* of any construction in this ledger; positivity supplies it gratis here; the entire difficulty of the problem resides in the size-versus-height comparison. Nor is there an Apéry-type denominator miracle hiding in that comparison: the *measured* reduced denominators of the partial sums $S_N(s/t)$ equal 96–100% of the naive clearing

$\prod_{i\le 2N}(t^i-s^i)$ in logarithmic size, with the ratio tending to 1 as N grows (exact rational arithmetic, $N\le 12$, at $q=\frac{1}{2}$ and $\frac{2}{5}$); the cancellation is an $O(N)$ boundary effect in an $O(N^2)$ exponent. The arithmetic of the ledger is exactly its worst-case costing.

Remark 38 (backward-orbit normal form of the zero condition). The zero condition admits a compression into a single explicit rational sequence. If $G(q,z_1)=0$, the three-term recurrence along the backward σ-orbit, $u_{-m}=\big(a(Q^{-m}z_1)u_{-m+1}-u_{-m+2}\big)/q$ with $u_n=G(q,Q^nz_1)$, loses one solution constant: $u_{-m}=u_1\,r_m(q)$ with $r_0=0$, $r_1=-1/q$, and $r_m\in\mathbb{Q}(q,z_1)$ computable by the same recurrence. The asymptotics of Theorem 25 then give the exact evaluation

$$\lim_{m\to\infty} r_m(q)\,Q^{m(m+1)/2}(-z_1)^{-m} \;=\; \frac{\theta_Q(-z_1)}{(q;q)_\infty\,G(q,Qz_1)},$$

with geometric rate Q^m and q-Gevrey correction series $\sum_j b_j z_1^{-j}Q^{mj}$ (verified to 17 digits at q^* by $m=24$). Under the hypothesis $q^*=s/t$ the normalized terms are rationals of height $(st)^{m^2+O(m)}$ approximating the limit to $O(Q^m)$; accelerated forms that cancel J correction terms gain decay Q^{mJ} but pay the quadratic heights of the b_j, and the arithmetic ratio stays $\le\frac{1}{2}$; the balance of Proposition 18 once more. The normal form is recorded as the sharpest single-sequence statement of what any future method must exploit: the entire rationality hypothesis is equivalent to the arithmetic of this one explicit sequence. The θ-jet admits the same normal form one level up: along the backward orbit,

$$\frac{(\theta G)(q,Q^{-m}z_1)}{G(q,Q^{-m}z_1)}-m \;\longrightarrow\; -\frac{z_1\,\theta_Q'(-z_1)}{\theta_Q(-z_1)} \qquad (\text{rate } Q^m;\ \text{verified to 10 digits at } q=0.3),$$

an exact theta log-derivative: each backward step increments $\theta\log\theta_Q(-z)$ by exactly one (quasi-periodicity), and the $\hat{g}$-correction vanishes in the limit.

The *forward* orbit admits the complementary normal form, in the language of rational dynamics. The Riccati relation gives the explicit continued fraction $q/(a(z)-q/(a(Qz)-\cdots))$, which by Pincherle's theorem converges to the minimal-solution ratio $q\,H(Qz)/H(z)$; the zero condition $G(q,z_1)=0$ forces the *dominant* ratios $w_n:=G(q,Q^{n+1}z_1)/G(q,Q^nz_1)$ to obey $w_1=a(z_1)$ exactly and thereafter the rational map

$$w_{n+1}=a(Q^nz_1)-\frac{q}{w_n},$$

so that under the hypothesis $q,z_1\in\mathbb{Q}$ the entire forward orbit is an explicit $\mathbb{Q}$-orbit converging to the attracting fixed point 1 with $w_n-1=\Theta(Q^n)$ (verified to 25 digits at q^*). Its heights compound quadratically ($t^{O(n^2)}$), so the Liouville comparison Q^n vs. t^{-n^2} is again consistent: both normal forms; backward theta-growth and forward rational dynamics; conserve the balance of Proposition 18, and together they exhaust the one-orbit reformulations of the zero condition.

Combining the two orbits with the q-Wronskian makes the relation-count of the rationality hypothesis explicit. Under $G(q,z_1)=0$ the unknown transcendental data reduces to three numbers ($g_1:=G(q,Qz_1)$, $H(q,z_1)$, and the backward limit ρ_∞), bound by exactly two relations: $g_1H(q,z_1)=1-q$ (the q-Wronskian at the zero) and $\rho_\infty g_1=\theta_Q(-z_1)/(q;q)_\infty$ (the backward normal form). Eliminating g_1,

$$\frac{\rho_\infty}{H(q,z_1)}=\frac{\theta_Q(-z_1)}{(1-q)\,(q;q)_\infty}$$

(verified to 32 digits at q^*), with modular right-hand side. Three unknowns, two relations: the rationality hypothesis is consistent at this level, and a proof requires precisely one further independent relation; the multiplicity statement of Remark 36, in yet another equivalent form.

Remark 39 (truncation roots, and the exact deficit). A scheme not covered by the lattice ledger prices the wall in its simplest form. Clearing the truncation of S gives integer polynomials $P_N = (q;q)_{2N} \sum_{k\le N}(-2)^k q^{k^2}(1-q)^k/(q;q)_{2k} \in \mathbb{Z}[q]$ of degree $2N^2+N$, whose heights are only $e^{O(N)}$ (the pentagonal sparsity of q-Pochhammer coefficients; measured $\log_2 H = 2.6, 4.6, \ldots, 12.1$ for $N = 2, \ldots, 7$), and whose root nearest q^* satisfies $|q_N - q^*| \asymp q^{*\,(N+1)^2}$ (verified $N \le 7$). If $q^* = s/t$ then $|P_N(q^*)| \ge t^{-(2N^2+N)}$, while analytically $|P_N(q^*)| = e^{O(N)} q^{*\,(N+1)^2}$; contradiction would require $q^* < t^{-2}$, impossible for $q^* = s/t \ge 1/t$. The deficit is exactly the factor 2 in the clearing degree; the double Pochhammer once more, now in s-independent form. The same computation shows that an equivalent representation of the root condition with *single*-Pochhammer denominators (clearing degree $\sim N^2/2$) would yield $q^* \ne s/t$ for all $s < \sqrt{t}$. No such representation exists within the Rogers–Ramanujan/Bailey framework: those transformations require the argument to be a q-power, and the argument here is the catalytic point $z = 2q(1-q)$. In this precise sense the obstruction to β_2's irrationality is the non-q-power (catalytic) argument itself.

The catalytic locus is moreover *seat-independent*. Writing $S(q) = {}_1\phi_1(0; q; Q, z^*)$ ($Q = q^2$), the parameter–argument transformation ${}_1\phi_1(0; b; Q, x) = \frac{(x;Q)_\infty}{(b;Q)_\infty}\, {}_1\phi_1(0; x; Q, b)$ gives the equivalent zero condition

$$T(q) \;:=\; {}_1\phi_1(0;\, 2q(1-q);\, q^2,\, q) \;=\; \sum_{k\ge 0} \frac{(-1)^k q^{k^2}}{(q^2;q^2)_k\,(2q(1-q);q^2)_k} \;=\; 0$$

(verified to 51 digits; the prefactor cannot vanish since $z^* < 1$), in which the argument is the q-power q and the catalytic value sits in the *parameter*; equivalently, q^* is the point where the Hahn–Exton q-Bessel of q-dependent order $\nu(q) = \log_{q^2}(2q(1-q)) - 1$ vanishes at the fixed argument $q^{-1/2}$. The clearing degree of T's truncations is again $\sim 2N^2$: the deficit is invariant under the swap, and in either seat it is the non-special value $2q(1-q)$ that lies outside every Rogers–Ramanujan/Bailey lattice. The well-poised generalization does not help: a WP-Bailey specialization producing a single-Pochhammer form would require $2q(1-q)$ to be a q-power (the only special-value type such summations admit), whereas at $q = q^*$ one has $q^* < 2q^*(1-q^*) < 1$, strictly between consecutive q-powers (the two loci where $2q(1-q)$ is a q-power, $q = \frac{1}{2}$ and $q = \frac{2}{3}$, both miss q^*). The catalytic argument is thus q-power–free in every seat, which is the transformation-theoretic content of the $\rho = \frac{1}{2}$ wall.

Proposition 40 (effective non-rationality of q^*)**.** *The least positive root q^* of S admits no rational representation s/t with $t \le 10^{1044}$. Consequently, if $\beta_2 = u/v$ in lowest terms then $u > 10^{522}$.*

Proof. All steps are certified interval arithmetic (outward rounding; `mpmath.iv`; script `beta2_effective_irrationa` with the series tail bounded by $(q;q)_{2k} \ge (1-2q)/(1-q) > 0$ for $q < \frac{1}{2}$ and term ratios $< \frac{1}{2}$ beyond the truncation. (i) $S > 0$ on $[0, 0.40]$ (400 certified chunks). (ii) $S' \le -1$ on $[0.40, 0.46]$ (600 certified chunks; S' evaluated term-by-term with the logarithmic-derivative weights), so S has at most one zero in $[0.40, 0.46]$. (iii) with c the 2090-digit truncation of q^*, $a = c - 2\cdot 10^{-2090}$ and $b = c + 2\cdot 10^{-2090}$, certified evaluation gives $S(a) > 0 > S(b)$ (working precision 2130 digits, series tail enclosed in $\pm 10^{-2125}$); by (i)–(ii), $q^* \in [a,b]$. (iv) the interval $[a,b]$ contains the rational s_0/t_0 (Stern–Brocot construction) with $9.5\cdot 10^{1044} < t_0 < 10^{1045}$, verified by exact rational comparison, and certified $S(s_0/t_0) \ne 0$. If $m/n \in [a,b]$ with $n \le 10^{1044}$ and $m/n \ne s_0/t_0$, then $|m/n - s_0/t_0| \ge 1/(nt_0) > 10^{-2089} > b - a = 4\cdot 10^{-2090}$, impossible; so $m/n = s_0/t_0$, whence $n = t_0 > 10^{1044}$, a contradiction. Thus no rational with denominator $\le 10^{1044}$ lies in $[a,b]$ at all, in particular none equals q^*. For $\beta_2 = u/v$ in lowest terms, $q^* = v^2/u^2$ in lowest terms, so $u^2 > 10^{1044}$. ☐

6 The true series U

U inherits the travel poles of Proposition 1; it is transcendental once its numerator is non-zero at infinitely many q_m. The Bousquet-Mélou–type transfer expresses the relevant residue through a Möbius function of a cocycle variable, and non-vanishing reduces to a *gate*

$$b_0 > 0 \qquad \text{and} \qquad s := \tfrac{q}{1-q}\, t_1 < 1, \qquad t_1 = \frac{P_{12}}{S_e}, \tag{10}$$

on the off-diagonal cocycle entry P_{12} (the supplement [20], § lifting). Here $\mathcal{N}_U(q_m)$ denotes the assembled numerator of U's residue at q_m (the Möbius numerator of the transfer), and b_0 is the y-free bulk-resolvent sum, with closed form $b_0 = \frac{2q}{1-q}\, S_o/S_e$ (the supplement's telescoped first-order system; Lemma 51 identifies S_o). The reduction is proved in the supplement by exact Riccati and telescope identities, and additionally verified numerically at all tabulated poles. Both columns of the cocycle satisfy the scalar three-term recursion $y_{n+1} = (1 + q^3 - 2(1-q)q^{2n+2})y_n - q^3 y_{n-1}$; setting $x = q^n$ turns it into the linear q-difference equation

$$Y(qx) + q^3 Y(x/q) = (1 + q^3 - 2(1-q)q^2x^2)Y(x), \tag{11}$$

whose regular solution is, in closed form, a Hahn–Exton q-Bessel function,

$$Y_3(x) = \sum_{k\geq 0} d_k x^{2k+3}, \qquad d_k = \frac{(-2)^k(1-q)^k q^{k^2+3k}}{(q^2;q^2)_k(q^5;q^2)_k} = J^{(3)}_{3/2} \text{ normalised}, \tag{12}$$

and $P_{12} = \frac{2q^3}{1-q^3} Y_3(1)$. The denominator S_e is the block of (1), controlled by Remark 5. It remains to bound the numerator residual against its confluent limit; this is where the two-Bessel identity enters.

6.1 The amplitude atom, formally verified

As $q \to 1$, equation (11) degenerates to $\tau Y'' - (2\tau/x)Y' + 2Y = 0$, solved by $x^{3/2}J_{3/2}(Wx)$. The q-corrected connection coefficient across the turning point $X = \nu = \frac{3}{2}$ is governed by the following identity, whose point is that the confluence is *entire*; it passes *through* the turning point, bypassing the Liouville–Green singularity where asymptotic methods stall.

Lemma 41 (two-Bessel confluence; amplitude normalisation)**.** *The regular solution of* (11) *has the confluent form*

$$Y_3(x) = \gamma_{\rm cl}\, x^{3/2}\Big[J_{3/2}(X) + \tfrac{\tau X}{2} J_{5/2}(X)\Big] + O(\tau^2), \qquad X = wx,\ \gamma_{\rm cl} = \tfrac{3\sqrt{\pi}}{4}\big(\tfrac{w}{2}\big)^{-3/2},$$

and its single-solution envelope satisfies $A/A_{\rm cl} = 1 + \tau\, h(X) + O(\tau^2)$ *with the* explicit rational

$$h(X) = \frac{1 + 3/(2X^2)}{1 + 1/X^2} \ \in\ [1, \tfrac{3}{2}] \qquad (X > 0).$$

Consequently $\gamma/\gamma_{\rm cl} = 1 + O(\tau)$ *with explicit constant* $\frac{3}{2}$.

Proof. From the q-Pochhammer confluence $d_k = \tau^{-k} D_k(1 - k\tau + O(\tau^2))$ with $D_k = (-1)^k/[2^k k!\,(\frac{5}{2})_k]$, the leading sum is $\gamma_{\rm cl} x^{3/2} J_{3/2}(X)$ by the Bessel representation of ${}_0F_1$, and the $-k\tau$ correction contributes $\frac{\tau X}{2}\gamma_{\rm cl} x^{3/2} J_{5/2}(X)$ via ${}_0F_1(;\frac{7}{2};\cdot)$. Both series are entire in X. The envelope of $J_{3/2} +$

$\frac{\tau X}{2} J_{5/2}$ follows from $A^2 = M_{3/2}^2 + (\frac{\tau X}{2})^2 M_{5/2}^2 + \tau X (J_{3/2}J_{5/2} + Y_{3/2}Y_{5/2})$ and the closed half-integer cross-term identity

$$J_{3/2}(X)J_{5/2}(X) + Y_{3/2}(X)Y_{5/2}(X) = \frac{4}{\pi X^2}\Big(1 + \frac{3}{2X^2}\Big),$$

which with $M_{3/2}^2 = \frac{2}{\pi X}(1 + \frac{1}{X^2})$ gives $h(X) = (1 + \frac{3}{2X^2})/(1 + \frac{1}{X^2})$; this is monotone decreasing from $h(0^+) = \frac{3}{2}$ to $h(\infty) = 1$, so $1 \le h \le \frac{3}{2}$. □

Remark 42 (machine verification). The cross-term identity and the bound $1 \le h(X) \le \frac{3}{2}$ have been formalised and checked in Lean 4 against Mathlib (file `AtomN.lean`); the proof terms depend only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`, with no `sorry`. The identity is proved by a single `linear_combination` off $\sin^2 + \cos^2 = 1$, and the bound by `nlinarith`.

Theorem 43. *The true growth series U is transcendental over $\mathbb{Q}(x)$. The numerator amplitude estimate $(\star)$ (at every travel pole q_m, $|P_{12}(q_m)|\, w_m^3 < 2$) is Theorem 59 (Appendix B): $|P_{12}| \le 0.619\,\tau^{3/2}$ for $\tau \le 5 \cdot 10^{-3}$ and $|P_{12}|w^3 \le 0.75$ at the six poles above that threshold. The full gate (10) ($b_0 > 0$ and $s < 1$) is Corollary 61: $|s| \le 0.983$ on the uniform range and $|s| \le 0.296$ at the six exceptional poles, with the denominator lower bound $|S_e| \ge 0.63\sqrt{\tau}$ supplied by Lemma 60. The reduction of non-vanishing to the gate is the supplement's Möbius factorisation ([20], § lifting), proved there by exact Riccati and telescope identities and additionally verified numerically at all tabulated poles. The leading-order picture of Remark 11 (the phase-matched zero giving $|P_{12}|/\tau^{3/2} \to 1/(4\sqrt{2})$, a fourfold margin below the threshold $1/\sqrt{2}$) is consistent with, and explained by, the second-difference identity of Proposition 52.*

Proof. U inherits the travel poles (Proposition 1), and $\mathcal{N}_U(q_m) \ne 0$ once the gate (10) holds there. Corollary 61 closes the gate at *every* travel pole: on the uniform range $\tau \le 5 \cdot 10^{-3}$, $|P_{12}| \le 0.619\,\tau^{3/2}$ (Theorem 59) and $|S_e| \ge 0.63\sqrt{\tau}$ (Lemma 60) give $|s| = \frac{q}{1-q}\,|P_{12}|/|S_e| \le \frac{\tau}{e^\tau - 1} \cdot \frac{0.619}{0.63} \le 0.983 < 1$, and $b_0 > 0$ follows from the sign chain of the same lemmas; the six exceptional poles are checked directly. Hence U has an uncancelled pole at every q_m; these are infinitely many and accumulate at $q = 1$ (Lemma 8), so U has infinitely many poles in the unit disc and is transcendental over $\mathbb{Q}(x)$ (as in Theorem 14: an algebraic function over $\mathbb{Q}(x)$ has only finitely many poles in the unit disc). Transcendence does not consume the exceptional-pole enumeration of Theorem 59: the uniform range $\tau \le 5 \cdot 10^{-3}$ alone supplies infinitely many uncancelled poles (Lemma 8 produces them at $\tau \downarrow 0$); the enumeration only upgrades "at infinitely many travel poles" to "at every travel pole". For orientation: the computed values over $m = 5, \ldots, 71$ are $|P_{12}|/\tau^{3/2} \to 1/(4\sqrt{2}) = 0.1768$ and $s \to \frac{1}{4}$, well inside every bound above. □

Remark 44 (the assembly is machine-checked). The composition just performed (the gate arithmetic, the two halves of (10), the passage to the pole set, and the step from infinitely many poles to non-algebraicity) is verified in Lean 4 (`UAssembly.lean`, seven theorems, no `sorry`, axioms `propext`, `Classical.choice`, `Quot.sound`), each analytic input entering as an explicit hypothesis: Theorem 59, Lemma 60, Corollary 61, Lemma 8, the supplement's reduction, and the finiteness of an algebraic function's pole set in the disc. The file formalises *no* analysis: no q-series, no steepest descent, no numerics; its content is exactly that those hypotheses *compose*, so its hypothesis list is an audit of what Theorem 43 rests on. The quantitative step is isolated as `gate_ratio_lt_one`, whose hypothesis $K < c$ (numerator constant strictly below denominator constant) is what the gate needs and what a numerator bound alone cannot supply: with $K = 0.619$ from Theorem 59 and the constant 0.35 of Proposition 46 it is *false*, which is precisely why Lemma 60's 0.63 is required.

Remark 45 (what the turning-point atom does and does not give). Lemma 41 discharges the amplitude *through the turning point* $X = \frac{3}{2}$, which is the obstruction to any Liouville–Green treatment. It does *not* supply the amplitude at the *pole* $X = w$ to the $O(\tau^2)$-relative accuracy the numerator gate requires. There the two-Bessel form carries a relative defect whose $O(1)$ part is exactly $\frac{1}{36}$; i.e. the classical turning-point-to-pole connection factor $\frac{36}{35}$, which is itself derivable and, once applied, matches $Y_3(1/q)$ to $O(\tau)$-relative; and whose *residual* $O(\tau)$ part (the amplitude drift from the turning point to the pole) is not supplied by this route. Because $P_{12} = \frac{2q^3}{3(1-q^3)} Y_3(1/q) - \frac{2}{3} S_e$ is an $O(\sqrt{\tau})$ cancellation, that residual is amplified to the size of P_{12} itself. This residual is closed in Appendix B, not by resumming the amplitude at all, but by the second-difference identity of Proposition 52: P_{12} is the symmetric second difference of the q-cosine at its own zero, so the gate needs only $\max |\psi''| \le 3.5w$ on a window of width τ, which Theorem 59 proves; Lemma 60 and Corollary 61 then close the remaining halves of the gate ($|S_e| \ge 0.63\sqrt{\tau}$ and $b_0 > 0$). The two off-diagonal cocycle entries obey the Abel relation $x_N Y_N - X_N y_N = -1$ (the discrete q-Wronskian, conserved because $\det M_n \equiv 1$), whence

$$P_{12} = \frac{X_N\, S_e - 1}{x_N} \qquad \text{exactly,}$$

x_N, X_N being the leading components of the same three-term recursion. Hence P_{12} is fixed by the *same* family of Hahn–Exton blocks that controls S_e: one analytic class; that of Remark 5; not two. In this precise sense U sits on V's footing. What U additionally requires is the pole asymptotics of x_N, X_N; again Hahn–Exton, again governed by the exact q-Wronskian of Olde Daalhuis [7], $W_q(J_\nu, Y_\nu) = q^{\nu(\nu-1)}(1-q^2)/\Gamma_{q^2}(\frac{1}{2})^2$ for the numerically satisfactory pair, whose $q \to 1$ expansion is explicit via Moak's q-Stirling formula (e.g. $\Gamma_{q^2}(\frac{1}{2}) = \sqrt{\pi}(1 - \frac{3}{8}\tau + O(\tau^2))$). The block-asymptotic assembly is carried out *elementarily* in Remark 11: the numerator amplitude has a derived closed-form expansion with exact rational constants, and the phase-matching of the travel-pole equation against the amplitude's zero (both collapsing to the same sinh-product family) derives the gate value $1/(4\sqrt{2})$. U's analytic inputs thereby coincide with V's: the further pieces ($(\star)$ and the closure of the gate) are proved in Appendix B. (The orthogonal arithmetic route of §7 would settle U without any of this analysis, but is itself open.)

7 An orthogonal arithmetic route

For each prime p, $U \bmod p \in \mathbb{F}_p[[q]]$ is algebraic over $\mathbb{F}_p(q)$ iff its coefficient sequence is p-automatic (Christol [6]), iff its p-kernel is finite. We find the p-kernel to be *maximally* non-automatic: through the accessible depth its size tracks the full p-ary tree with *no* deficit. (An apparent deficit of 1 at level 3; $1, 4, 13, 39$ vs. the full $1, 4, 13, 40$ at $p = 3$; is a *comparison-length artifact*: a level-k decimation subsequence retains only $\sim N/p^k$ terms, so at a short prefix two distinct decimations can coincide; the deficit disappears once the comparison length reaches 9.) Two instances are machine-verified in Lean 4. For the true series itself, the 3-kernel of ($u_n \bmod 3$) through level 3 is the full ternary tree, deficit 0: all $1 + 3 + 9 + 27 = 40$ decimation words of length ≤ 3 are pairwise distinct: at comparison lengths 9 and 10 alike; on the 271-term prefix $n \le 270$ (`UKernel.lean`, from the validated `tools/u_modp_rust` output, whose u_n are self-checked against the 43 known terms and byte-identical to the earlier validated $n \le 180$ run on the overlap). For the sibling block Σ_1, already transcendental over $\mathbb{Q}(q)$ by Theorem 2, the same level-3 full tree is certified in `SigmaKernel.lean` (built from the A/C recursion over $\mathbb{F}_3$), with the short-prefix deficit-1 confirmed spurious. The Berlekamp–Massey linear complexity of ($u_n \bmod 3$) is $\approx N/2$. This is the strongest finite-data evidence for non-automaticity, and hence for transcendence over $\mathbb{Q}(x)$, independent of all of §§4–6. A proof for U would require a non-automaticity criterion valid for *dense*-support series (the coefficients carry no lacunary gaps), which the standard gap/Mahler methods do not provide.

8 Conclusion

The arithmetic of A396406 separates cleanly. The catalytic building blocks and the growth rate are unconditional (Pólya–Carlson; Corollary 15). **The relaxed series V is transcendental over** $\mathbb{Q}(x)$, and its non-cancellation input is Proposition 46, proved in Appendix A by an elementary chain with explicit numerical constants: an annulus representation, an Euler bound, a Poisson theta envelope, and a q-Wronskian invariant. No asymptotic theorem is cited on that path. Remark 5 gives the same conclusion by a different and shorter route (the saddles $z^*_\pm = \pm iW/2$ with all constants derived explicitly, the two saddles supplying the global cancellation that no partial-summation method can), but that route is a sketch only (Remark 7), and no theorem uses it. Both V and U take their pole set from Lemma 8, which consumes only the rectangle bound $|T_2| \to 0$ of Lemma 6; and V's non-cancellation input is the elementary Proposition 46, independent of either. The true series U stands on the same footing: its further inputs are proved in Appendix B: the gate block is the symmetric second difference of the q-cosine at its own zero (Proposition 52), the two-interval uniform certificate of Theorem 59 gives $|P_{12}| \le 0.619\,\tau^{3/2} < \tau^{3/2}/\sqrt{2}$ for $\tau \le 5 \cdot 10^{-3}$ with the six poles above that threshold checked directly, and the exact identity $S_e = \frac{qZ}{2}\mathfrak{s}(Z) - P_{12}$ turns the same certificate into the denominator bound and the sign input that close the gate ($|s| \le 0.983$, $b_0 > 0$; Corollary 61), consistent with the phase-matched leading values $|P_{12}|/\tau^{3/2} \to 1/(4\sqrt{2})$ and $s \to \frac{1}{4}$. U **is transcendental over** $\mathbb{Q}(x)$, unconditionally, with the same analytic standing as V. The turning-point amplitude atom on the U-side is the elementary rational bound of Lemma 41, formally verified in Lean 4. The independent mod-p route of §7 would settle U by a completely different mechanism, if a dense-support non-automaticity criterion can be found.

A An explicit lower bound for S_e at the travel poles

Throughout $z_0 = \sqrt{2(1-q)}$, $Z = z_0/\sqrt{q}$, $w = \sqrt{2/\tau}$, $f(z) = \cos(z;q^2) = C(z^2)$, $C(y) = \sum_{j\ge 0}(-1)^j q^{j^2+j} y^j/(q;q)_{2j}$, and $s(z) = \sin(z;q^2)$; by Proposition 12, $S_e = f(z_0)$ and the travel poles are the solutions of $f(Z) = 0$.

Proposition 46. *At every travel pole q_m with $\tau_m \le 1.29 \cdot 10^{-4}$ one has $|S_e(q_m)| \ge 0.35\sqrt{\tau_m}$. The finitely many (Lemma 53) poles with $\tau_m > 1.29 \cdot 10^{-4}$ (the dominant pole $q_1 = 0.449453\ldots$ ($|S_e|/\sqrt{\tau_1} = 0.581$; `star_gate_certificate.py`, Corollary 61) and the thirty-nine tabulated below it, $\tau_{41} = 1.2354 \cdot 10^{-4}$ being the first below the threshold) satisfy the same bound by direct high-precision computation.*

The proof occupies the rest of the appendix: an annulus representation turns all derivative bounds into cancellation-free integrals, and the pole condition supplies the value-level input.

Lemma 47. *For $0 < r < 1$, $C(y) = \frac{1}{2\pi}\int_0^{2\pi} \Theta(y/x^2)\,(x;q)_\infty^{-1}\,d\theta$ with $x = re^{i\theta}$ and $\Theta(t) = \sum_{j\in\mathbb{Z}}(-1)^j q^{j^2+j} t^j = (q^2;q^2)_\infty (q^2 t;q^2)_\infty (1/t;q^2)_\infty$.*

Proof. Euler's identity gives $1/(q;q)_{2j} = [x^{2j}]\,(x;q)_\infty^{-1}$; extract the coefficient by Cauchy's formula on $|x| = r$. For $j \le -1$ the integrand $x^{-2j}(x;q)_\infty^{-1}$ is analytic in $|x| < 1$ (the zeros of $(x;q)_\infty$ lie at $x = q^{-k}$, $|q^{-k}| \ge 1$), so the negative-index terms vanish and the sum completes to the bilateral one, which is the triple product. Equivalently this is the $p = q^2$ case of the q-Borel–q-Laplace inversion of [18] and [14, Lemma 1], the q^2-Borel transform of C being the even part of $(x;q)_\infty^{-1}$. □

Lemma 48. *For $0 < r \le \frac{1}{2}$: $(x;q)_\infty^{-1} \le \exp(w_r\cos\theta + K_r)$ in modulus, with $w_r = r/(1-q)$ and $K_r = r^2/(2(1-r)(1-q^2))$.*

Proof. $-\log|1-u| = \operatorname{Re} u + \operatorname{Re}\sum_{n\ge 2} u^n/n \le \operatorname{Re} u + \frac{1}{2}|u|^2/(1-|u|)$ for $|u| < 1$; summing over $u = xq^j$ gives $\sum_j \operatorname{Re}(xq^j) = w_r\cos\theta$ and $\sum_j r^2q^{2j}/(2(1-rq^j)) \le r^2/(2(1-r)(1-q^2)) = K_r$. $\square$

Lemma 49. *Let $p = q^2$, $t = \rho e^{i\varphi}$, $a = (\log\rho - \tau) + i\tilde\varphi$ with $\tilde\varphi \in (-\pi,\pi]$ representing $\varphi + \pi$. For $n \in \{0,1,2\}$,*

$$|\Theta^{(n)}(t)| \le \sqrt{\pi/\tau}\, e^{(\log\rho-\tau)^2/(4\tau)}\, \rho^{-n}\big(|a|/(2\tau) + \tau^{-1/2}\big)^n\, G(\tilde\varphi), \quad G(\tilde\varphi) = \sum_{k\in\mathbb{Z}} e^{-\pi^2(k-\tilde\varphi/2\pi)^2/\tau}.$$

Proof. $\Theta(t) = \theta_p(-pt)$ with $\theta_p(y) = \sum_n p^{n(n-1)/2}y^n$. Poisson summation gives the *exact* identity

$$S(a) := \sum_{j\in\mathbb{Z}} e^{-\tau j^2 + aj} = \sqrt{\pi/\tau}\sum_{k\in\mathbb{Z}} e^{a_k^2/(4\tau)}, \qquad a_k := a - 2\pi i k. \tag{13}$$

Since $\operatorname{Re}(a_k^2) = (\log\rho-\tau)^2 - (\tilde\varphi - 2\pi k)^2$, the imaginary part of a contributes *decay*, and $\sum_k e^{\operatorname{Re}(a_k^2)/(4\tau)} = e^{(\log\rho-\tau)^2/(4\tau)}G(\tilde\varphi)$; no cancellation is involved, the term $-(\tilde\varphi - 2\pi k)^2$ being precisely what constitutes G. Differentiating (13) in a, the j- and $j(j-1)$-weighted sums are S' and $S'' - S'$, and $\partial_a e^{a_k^2/(4\tau)} = (a_k/2\tau)e^{a_k^2/(4\tau)}$, so the k-th image carries the weight

$$W_1(a_k) = \frac{|a_k|}{2\tau}, \qquad W_2(a_k) = \frac{|a_k|^2}{4\tau^2} + \frac{1}{2\tau} + \frac{|a_k|}{2\tau}.$$

At the radius $\rho = e^\tau$ of Lemma 50 one has $a_k = i(\tilde\varphi - 2\pi k)$ exactly, so $|a_k| = |\tilde\varphi - 2\pi k|$. To pull the weight out of the k-sum, note that for $k \ne 0$ the excess $|a_k| - |a_0|$ is paid for by the Gaussian factor, which is smaller by $e^{-((\tilde\varphi-2\pi k)^2 - \tilde\varphi^2)/(4\tau)}$; the aggregate excess is maximised at the nearest image and is at most $\max_{\delta>0} 2\delta e^{-\pi\delta/\tau} = 2\tau/(\pi e) \le 0.234\,\tau$, which is below $\tau^{-1/2}$ for $\tau \le 0.0198$. Hence $\sum_k W_n(a_k)e^{\operatorname{Re}(a_k^2)/(4\tau)} \le \big(|a|/(2\tau) + \tau^{-1/2}\big)^n \sum_k e^{\operatorname{Re}(a_k^2)/(4\tau)}$, which is the stated bound. $\square$

Lemma 50. *For $\tau \le 0.0198$ and every $\xi \in [qZ, Z]$, with $r^2 = \xi^2 e^{-\tau}$: $M_2 := \max_{[qZ,Z]}|f''| \le 7w^4$.*

Proof. $f''(z) = 2C'(z^2) + 4z^2C''(z^2)$, and differentiating Lemma 47 under the integral, $|C^{(n)}(y)| \le (2\pi r^{2n})^{-1}\int_0^{2\pi}|\Theta^{(n)}(t)|\,(x;q)_\infty^{-1}d\theta$ at $\rho = y/r^2 = e^\tau$, which zeroes $(\log\rho - \tau)^2$. The key step is the exact unfolding: as θ runs over $[0,2\pi)$ the variable $u = \pi - 2\theta$ runs over two periods, consecutive branches carrying opposite signs of $\sin(u/2)$, so

$$\frac{1}{2\pi}\int_0^{2\pi} e^{w_r\cos\theta}G(\pi - 2\theta)\,d\theta = \frac{1}{2\pi}\int_{\mathbb{R}}\cosh\big(w_r\sin\tfrac{u}{2}\big)e^{-u^2/(4\tau)}\,du,$$

an identity; the derivative weights unfold with it, becoming polynomials in u, and $|\sin(u/2)| \le |u|/2$. The even weight gives an exact Gaussian moment, $\int v^2 e^{-(v-s)^2/(4\tau)}dv = \sqrt{4\pi\tau}(2\tau + s^2)$ with $s = \tau w_r \le \sqrt{2\tau}e^{\tau/2}$, and $\sqrt{\pi/\tau}\cdot\sqrt{4\pi\tau} = 2\pi$ cancels the prefactors; the odd weight does *not*, giving instead $\int|v|e^{-(v-s)^2/(4\tau)}dv = \sqrt{4\pi\tau}\,s\operatorname{erf}\big(s/(2\sqrt{\tau})\big) + 4\tau e^{-s^2/(4\tau)}$, which we bound by $\sqrt{4\pi\tau}\,s + 4\tau$. With the weights of Lemma 49 this assembles, for $\tau \le 0.0198$ (so $r \le \frac{1}{2}$, $K_r \le 0.631$, $e^{\tau w_r^2/4} \le e^{0.506}$), to

$$|C'(y)| \le e^{K_r - \tau + \tau w_r^2/4}\,(1.825\,\tau^{-1/2})r^{-2}, \qquad |C''(y)| \le e^{K_r - 2\tau + \tau w_r^2/4}\,(3.670/\tau)r^{-4}.$$

With $4\xi^2/r^4 \le (2/\tau)e^{3\tau}(1+2\tau)$ the total $M_2\tau^2$ is at most 25.0 at $\tau = 0.0198$, decreasing to 20.2 at $\tau = 1.29\cdot 10^{-4}$, in both cases below the target $7w^4\tau^2 = 28$. The bound is deliberately lossy: numerically $M_2 \le 0.07\,w^4$ over the same range. $\square$

Proof of Proposition 46. At a pole $f(Z) = 0$, so the Koornwinder–Swarttouw rules give the exact facts $f(qZ) = qZ\,s(Z)$ and $s(qZ) = s(Z)$; the mean value theorem on $[qZ, Z]$ then pins $f'(\zeta) = -q\,s(Z)/(1-q)$ at some $\zeta \in (qZ, Z)$, and a second mean value on $[z_0, Z] \subset [qZ, Z]$ gives

$$|S_e| = (Z - z_0)\,|f'(\tilde{\xi})| \;\geq\; \frac{q(1-\sqrt{q})}{1-q}\,Z\,|s(Z)| - (1-\sqrt{q})(1-q)Z^2 M_2.$$

The sine factor is bounded below by an invariant of Casoratian (q-Wronskian) type for the Hahn–Exton pair; the closed-form q-Wronskian is due to Swarttouw (PhD thesis, TU Delft, 1992, eq. (4.3.8)); see also [7, eq. (4.21)]. Concretely, with $c_n = \cos(Zq^n; q^2)$, $s_n = \sin(Zq^n; q^2)$ and $F_n = c_n^2 - Zq^{n+1}c_n s_n + q\,s_n^2$, the rules give the exact telescoping $F_n - F_{n+1} = -Zq^{n+1}(1-q)c_{n+1}s_{n+1}$ with $F_n \to 1$; the form has smallest eigenvalue $\lambda_n \geq q - Z/2$ (by $\sqrt{a^2+b^2} \leq a + b$), $\sum_{n\geq 0} Zq^{n+1}(1-q) = qZ$, and the discrete Gronwall gives $F_0 \geq e^{-0.61Z} \geq 1 - 0.61Z$ for $Z \leq \min(0.2, q/3)$; at the pole $F_0 = q\,s(Z)^2$, so $|s(Z)| \geq \sqrt{1-0.61Z}$. Inserting Lemma 50, the error fraction is $28\sqrt{2}\,(1.02)\sqrt{\tau} \leq 0.46$ for $\tau \leq 1.29 \cdot 10^{-4}$, a range containing every pole with $m \geq 41$, whence $|S_e(q_m)| \geq (1 - 0.02 - 0.46)\sqrt{\tau_m/2} \geq 0.35\sqrt{\tau_m}$ there. The tabulated poles are verified numerically at 60+ digits (`qtrig_verify.py`, mpmath, no interval arithmetic: $|S_e|/\sqrt{\tau} \in [0.698, 0.708]$ as asserted by the script; the tabulated set is poles $2, \ldots, 60$ in the indexing of §2 (the dominant pole is not among them, its value 0.581 coming from `star_gate_certificate.py`) and the table omits poles 53 and 60 (rows $52, 59$), whose $\tau < 7.7 \cdot 10^{-5}$ lies inside the proved range), and the two ranges overlap on every tabulated pole with $m \geq 41$; only poles $2, \ldots, 40$ (table rows $1, \ldots, 39$) are load-bearing. The polynomial identities underlying the telescoping and the pole reduction are machine-checked (`GramCasoratian.lean`: `F_shift`, `F_drift`, `pole_qsine`, `sigma1_term`; axioms `propext`, `Classical.choice`, `Quot.sound`, no `sorry`). □

On the intermediate window $1.29 \cdot 10^{-4} < \tau_m \leq 5 \cdot 10^{-3}$ the bound also follows, with margin, from Lemma 60 ($0.63 > 0.35$), independently of any enumeration of the poles there.

B Proof of the amplitude estimate: the gate holds at every travel pole

This appendix proves the estimate $(\star)$ of Theorem 43 (at every travel pole, $|P_{12}(q_m)|\,w_m^3 < 2$, $w_m = \sqrt{2/\tau_m}$; quantitatively $|P_{12}| \leq \frac{3.5\sqrt{2}}{8}\,\tau^{3/2} < 0.619\,\tau^{3/2}$ for $\tau \leq 5 \cdot 10^{-3}$, and $|P_{12}|w^3 \leq 0.75$ at the six poles above that threshold) and then closes the full gate (10): the same certificate yields the denominator bound $|S_e| \geq 0.63\sqrt{\tau}$ (Lemma 60) and the sign input $b_0 > 0$, hence $s < 1$ and $\mathcal{N}_U(q_m) \neq 0$ at every travel pole (Corollary 61). Throughout, $h = \tau/2$, $\sigma^2 = \tau/2$, $\psi(t) = \cos(Ze^t; q^2)$; by Proposition 52, $P_{12} = -\frac{1}{2}[\psi(h) + \psi(-h)]$ and $\psi(0) = 0$, so

$$|P_{12}| \;\leq\; \tfrac{h^2}{2}\max_{|t|\leq h}|\psi''(t)|. \tag{14}$$

The gate block is a second difference at a zero

The following exact identity is not needed for Theorem 14, but it is the sharpest available description of the object the U-gate must bound, and it makes the size $\tau^{3/2}$ structural rather than numerical.

Lemma 51 (the gate block in closed form). *Identically in $0 < q < 1$, with $c(u) = \cos(u; q^2)$ and $\mathfrak{s}(u) = q^{-1/2}\sin(q^{1/2}u; q^2)$:*

$$P_{12} = \tfrac{qZ}{2}\,\mathfrak{s}(Z) - c(z_0), \qquad S_o := \sum_{j\geq 0}\frac{(-1)^j[2(1-q)]^j\,q^{j^2+2j}\,(1-q)}{(q;q)_{2j+1}} = \tfrac{z_0}{2}\,\mathfrak{s}(z_0),$$

S_o being the odd bulk block of the supplement's closed form $b_0 = \frac{2q}{1-q} S_o/S_e$ ([20], §lifting, the telescoped first-order system).

Proof. From (12), $P_{12} = \frac{2q^3}{1-q^3}\sum_k d_k$ with $d_k = (-2)^k(1-q)^k q^{k^2+3k}/[(q^2;q^2)_k(q^5;q^2)_k]$; since $(q;q)_{2k+3} = (q;q^2)_{k+2}(q^2;q^2)_{k+1}$ and $(q;q^2)_{k+2} = (1-q)(1-q^3)(q^5;q^2)_k$, this is

$$P_{12} = \sum_{k\geq 1}(-1)^{k-1}\frac{[2(1-q)]^k\, q^{k^2+k+1}\,(1-q^{2k})}{(q;q)_{2k+1}}. \tag{15}$$

Since $q^{1/2}Z = z_0$ and $z_0^2 = 2(1-q)$, the j-th summand of $\frac{qZ}{2}\mathfrak{s}(Z) = \frac{z_0}{2}\sin(z_0;q^2)$ is $(-1)^j(1-q)[2(1-q)]^j q^{j^2+j}/(q;q)_{2j+1}$, and that of $c(z_0)$ is $(-1)^j[2(1-q)]^j q^{j^2+j}/(q;q)_{2j}$. Their difference carries the factor $(1-q)-(1-q^{2j+1}) = q^{2j+1}-q$: the $j=0$ terms cancel, and what remains is (15) term by term. The S_o identity is the same computation one half-step down: the j-th summand of $\frac{z_0}{2}\mathfrak{s}(z_0)$ is $(-1)^j(1-q)[2(1-q)]^j q^{j^2+2j}/(q;q)_{2j+1}$, which is S_o's summand verbatim. All three matchings are exponent bookkeeping of the kind machine-checked in `QTrigIdentities.lean`; the identities are verified off-pole (hence as identities) and at the six exceptional poles to 10^{-100} (`star_gate_certificate.py`); the collapse identities behind Proposition 52 are further verified at the 58 tabulated poles (`halfstep_verify.py`). ☐

Proposition 52. *At every travel pole, with $Z = z_0/\sqrt{q}$ the corresponding zero of $\cos(\,\cdot\,;q^2)$,*

$$\boxed{P_{12} \;=\; -\tfrac{1}{2}\left[\cos(z_0;q^2)+\cos(z_0/q;q^2)\right].}$$

Since $z_0 = Ze^{-\tau/2}$ and $z_0/q = Ze^{+\tau/2}$, and since $\cos(Z;q^2)=0$, the right-hand side is exactly the symmetric second difference of $\cos(\,\cdot\,;q^2)$ at its own zero, taken in the variable $\log u$ with step $\tau/2$.

Proof. Write $c(u) = \cos(u;q^2)$ and $\mathfrak{s}(u) = q^{-1/2}\sin(q^{1/2}u;q^2)$. The Hahn–Exton q-Wronskian (Swarttouw, thesis eq. (4.3.8); [7, eq. (4.21)]) reads $\mathfrak{s}(u)c(qu) - c(u)\mathfrak{s}(qu) = u$, and the shift rules give $c(qu) = c(u) + q^{3/2}u\,\mathfrak{s}(\sqrt{q}\,u)$ and $\mathfrak{s}(qu) = \mathfrak{s}(u) - u\,c(\sqrt{q}\,u)$. Taking $u = Z/\sqrt{q}$ in the first shift rule gives $c(z_0) - c(z_0/q) = qZ\,\mathfrak{s}(Z)$. Taking $u = Z$ in the q-Wronskian and in the first shift rule, and using $c(Z)=0$ twice, gives $\mathfrak{s}(Z)\,c(qZ) = Z$ and $c(qZ) = q^{3/2}Z\,\mathfrak{s}(z_0)$, whence the pole invariant $q^{3/2}\mathfrak{s}(z_0)\mathfrak{s}(Z) = 1$ and $c(qZ) = Z/\mathfrak{s}(Z)$. Substituting the closed form $P_{12} = \frac{qZ}{2}\mathfrak{s}(Z) - c(z_0)$ of Lemma 51 into $c(z_0)+c(z_0/q) = 2c(z_0) - qZ\,\mathfrak{s}(Z)$ yields the boxed identity. ☐

The two algebraic steps of the proof are machine-checked (`SelowerSkeleton.lean`): `shift_at_pole` is the instantiation of the half-step rule at $u = z_0/q$, written so that the index bookkeeping is explicit, and `P12_second_difference` is the elimination; both are stated in doubled form over an arbitrary commutative ring and depend only on `propext`. The identity is *verified* at all 58 tabulated poles to a worst relative deviation of $2\cdot 10^{-44}$ (`halfstep_verify.py`). The consequence, (14) above, is that the gate is a bound on a second derivative on a window of width τ, not an asymptotic evaluation.

Lemma 53 (The travel poles are locally finite)**.** *For every $\tau_\star > 0$ there are only finitely many travel poles with $\tau_m > \tau_\star$.*

Proof. By Proposition 12 the travel poles are the zeros in $q\in(0,1)$ of $g(q) := \cos(Z;q^2) = C(Z^2)$ with $Z^2 = 2(1-q)/q$, i.e. of

$$g(q) = \sum_{j\geq 0}(-1)^j\,\frac{2^j\,q^{j^2}(1-q)^j}{(q;q)_{2j}}.$$

First, g has no zero near $q = 0$. For $0 < q \le \frac{1}{10}$ one has $(q;q)_{2j} \ge \prod_{i\ge1}(1-q^i) \ge 1 - \frac{q}{1-q} \ge \frac{8}{9}$ and $(1-q)^j \le 1$, so $|g(q)-1| \le \frac{9}{8}\sum_{j\ge1} 2^j q^{j^2} \le \frac{9}{8}(2q + 4q^4/(1-q)) \le 0.24 < 1$, whence $g(q) > 0$ there.

Second, g is real-analytic on $(0,1)$: the series converges locally uniformly, since on $q \le 1-\delta$ the j-th term is bounded by $2^j q^{j^2}/(q;q)_\infty$ and q^{j^2} dominates 2^j. Hence the zeros of g are isolated in $(0,1)$, and any zero set with an accumulation point in the *open* interval would force $g \equiv 0$, contradicting $g(\frac{1}{10}) > 0$.

Now fix $\tau_\star > 0$. The poles with $\tau_m > \tau_\star$ lie in $q \in [\frac{1}{10}, e^{-\tau_\star}]$, a compact subinterval of $(0,1)$. An analytic function on a domain whose zeros are isolated has only finitely many zeros on any compact subset; hence there are finitely many such poles. □

Remark 54. Only local finiteness is proved here; that the poles *do* accumulate at $q = 1$ (equivalently $\tau_m \downarrow 0$) is Lemma 8. The present lemma says merely that they cannot accumulate anywhere inside $(0,1)$, so that any threshold $\tau_\star$ leaves a finite set to be checked directly. It is used four times: to make the splice in Proposition 46 well posed, to license the phrase "for all large m" in Theorem 14, to make the exceptional set of Theorem 59 finite, and in Lemma 8's accumulation clause (its proof also shows $g > 0$ on $(0, \frac{1}{10}]$, so every travel pole has $q_m > \frac{1}{10}$).

Lemma 55. *For $0 < \rho < 1$, with $\theta \sim N(\pi/2, \tau/2)$: $\mathbb{E}[(\rho e^{i\theta};q)_\infty^{-1}] = \sum_{M\ge0} q^{M^2/4}(i\rho)^M/(q;q)_M =: \Phi(\rho)$, and $\operatorname{Re}\Phi(\rho) = \cos(\rho/\sqrt{q};q^2)$. The sum is the q-Airy function of Kajiwara–Masuda–Noumi–Ohta–Yamada at base $\sqrt{q}$, and the representation is Zhang's q-Laplace transform of the second kind at level $\frac{1}{2}$ [18]; only the probabilistic packaging is ours.*

Proof. Euler's identity $(x;q)_\infty^{-1} = \sum_M x^M/(q;q)_M$ integrated termwise (absolute uniform convergence on $|x| = \rho < 1$), with $\mathbb{E}[e^{iM\theta}] = i^M q^{M^2/4}$; splitting M by parity and matching exponents ($(2j+1)^2 = 4(j^2+j)+1$; Lean `parity_exponent`) identifies $\operatorname{Re}\Phi$ with the cosine series of Proposition 12. □

Lemma 56. *Write $A(\xi) = (\rho e^{i(\pi/2+\xi)};q)_\infty^{-1}$. Then $\log A(\xi) = \sum_{k\ge1}(i\rho)^k e^{ik\xi}/(k(1-q^k))$; the cumulants $\Lambda_n = i^n\sum_k k^{n-1}(i\rho)^k/(1-q^k)$ satisfy $|\Lambda_n| \le (n-1)!\,\rho/((1-q)(1-\rho)^n)$; Gaussian integration by parts gives, at radius $\rho(t) = z_0e^t$,*

$$\psi'(t) = \tfrac{2}{\tau}\ \operatorname{Im}\ \mathbb{E}[\xi A], \qquad \psi''(t) = \tfrac{4}{\tau^2}\ \operatorname{Re}\ \mathbb{E}[(\tfrac{\tau}{2} - \xi^2)A];$$

and for complex a, b with $\operatorname{Re}(1-b\sigma^2) > 0$, $\mathbb{E}[\xi^m e^{a\xi+b\xi^2/2}] = \beta^{-1/2}e^{a^2\sigma^2/(2\beta)}M_m$, $\beta = 1-b\sigma^2$, $M_0 = 1$, $M_1 = \mu = a\sigma^2/\beta$, $M_m = \mu M_{m-1} + (m-1)(\sigma^2/\beta)M_{m-2}$.

Proof. $-\log(1-u) = \sum u^m/m$ summed over $u = \rho e^{i\theta}q^j$; $1-q^k \ge 1-q$ and a binomial estimate; $\rho\partial_\rho A = -i\partial_\theta A$ with $\mathbb{E}[f'] = \mathbb{E}[f\xi/\sigma^2]$ once and twice; completion of the square and analytic continuation in (a,b). The exponent bookkeeping is machine-checked (`QTrigIdentities.lean`). □

Lemma 57. *At every travel pole with $\tau \le 5\cdot10^{-3}$ and every $t \in [-\tau, h]$: $|\psi'(t)| \le 1.10\,w$.*

Lemma 58. *At every travel pole with $\tau \le 5\cdot10^{-3}$, writing $\mathfrak{s}(u) = q^{-1/2}\sin(q^{1/2}u;q^2)$: $\cos(qZ;q^2) = Z/\mathfrak{s}(Z)$, and $|\mathfrak{s}(Z)| \ge Z/(\tau\max_{-\tau\le t\le0}|\psi'|) \ge 0.90$. Moreover, for $t \in [-\tau, 0]$, $|\Phi(\rho(t))| \ge q^{3/4}(|\mathfrak{s}(Z)| - 0.49\,\tau)$.*

Proof of Lemmas 57 and 58. The identity is the pole reduction underlying Proposition 52 (the Hahn–Exton q-Wronskian at the zero, cited there). The bound in Lemma 58 is the mean value theorem on $[-\tau, 0]$, where $qZ = Ze^{-\tau}$ exactly and all radii satisfy $\rho(t) \le z_0$. For the last bound: $\operatorname{Im}\Phi(\rho) = q^{3/4}\,\mathfrak{s}(\rho/\sqrt{q})$ (the odd half of the parity split in Lemma 55), and the drift of $\mathfrak{s}(\rho(t)/\sqrt{q})$

over $t \in [-\tau, 0]$ is the even/odd G_4 term of the certificate, $\leq 0.49\,\tau$ (`budget.py`, `G4_coef`). For Lemma 57: the principal term is the $m = 1$ tilted moment of Lemma 56 at $a = \Lambda_1$, $b = \Lambda_2$, at most $e^{c_1\sqrt{\tau}}|\beta|^{-1/2}L_1(\operatorname{Re}\beta)^{-1}\cdot w \leq 1.088\,w$, where L_1 is the interval constant bounding $|\Lambda_1|/w$ (Lemma 56, $n = 1$) and $c_1 \leq 0.177$ bounds the exponent residual per $\sqrt{\tau}$ (the two leading halves of the exponent cancel exactly; every residual carries a geometric tail); the tail (linear orders 3..10 with ξ-weighted moments, the product convolution, and the degree-≥ 11 remainder) is at most 0.131 *absolute*, i.e. $\leq 0.0066\,w$ at the top endpoint $w = 20$. All constants are produced by the committed certificate `code/zeta_probe/budget.py` (function `uniform_stack`), which works in 30-digit floating point with the interval constants realised as endpoint evaluations. That evaluation is *re-certified in directed-rounding interval arithmetic* by `budget_iv.py`: the same stack is evaluated with τ entering as an *interval*, over 160 cells covering $[1.2\cdot10^{-4}, 5\cdot10^{-3}]$ and 160 covering $[10^{-8}, 1.2\cdot10^{-4}]$ (contiguous, with both outer endpoints clamped exactly), giving enclosures valid for *every* τ in each cell; so on those ranges neither the floating-point arithmetic nor the monotonicity of any atom is assumed. The certified maxima are 3.349 and 2.650, against the budget 3.5. On the remaining range $(0, 10^{-8}]$, which no finite set of cells can cover, the power-counting argument below (every term a positive power of τ) is what proves the bound. The companion `star_gate_certificate.py` works in the floating-point model at 45–260 digits with explicit truncation control. □

Theorem 59 (the amplitude estimate $(\star)$)**.** *At every travel pole, $|P_{12}(q_m)|\,w_m^3 < 2$. Quantitatively: for $\tau \leq 5\cdot10^{-3}$, $\max_{|t|\leq h}|\psi''(t)| \leq 3.5\,w$, so* (14) *gives $|P_{12}| \leq \frac{3.5\sqrt{2}}{8}\tau^{3/2} < 0.619\,\tau^{3/2} < \tau^{3/2}/\sqrt{2}$; and the six poles with $\tau > 5\cdot10^{-3}$,*

$$\tau_1 = 0.79972,\quad \tau_2 = 9.049\cdot10^{-2},\quad \tau_3 = 3.248\cdot10^{-2},\quad \tau_4 = 1.656\cdot10^{-2},\quad \tau_5 = 1.001\cdot10^{-2},\quad \tau_6 = 6.701\cdot10^{-3},$$

satisfy $|P_{12}|w^3 = 0.745,\ 0.530,\ 0.510,\ 0.505,\ 0.503,\ 0.502$ by direct evaluation at 120-digit precision with explicit truncation control (`star_gate_certificate.py`; the underlying identities are independently verified by `halfstep_verify.py` at the 58 tabulated poles, 47–54 digits on the five it shares with this list: the dominant pole is covered by `star_gate_certificate.py` alone). The first is the dominant pole $q_1 = 1/\beta_2^2$ of Corollary 15. Enumeration of the exceptional set: every travel pole has $q_m > \frac{1}{10}$ and the set is finite (Lemma 53); the count of six is established by a derivative-majorant subdivision for $\tau \geq 0.03$ (three poles; in the arithmetic model disclosed above) and needs no enumeration for $\tau \leq 5\cdot10^{-3}$, where the uniform certificate covers every pole; on the window $5\cdot10^{-3} < \tau < 0.03$ the three listed poles are located by a $\Delta w \approx 5\cdot10^{-4}$ scan (they sit on the $(k+\frac{1}{2})\pi$ ladder to three decimals, and $|\cos(Z; q^2)|$ swings to 1.00 between them with minimum 0.019 outside the root cells). The scan is the weakest step of the enumeration and is stated as such; transcendence in Theorem 43 does not consume it.

Proof. By Lemma 56, $\psi'' = \frac{4}{\tau^2}\operatorname{Re}\big[-e^{\Lambda_0}G_Q(\Lambda_2\sigma^4/\beta + \mu^2) + e^{\Lambda_0}\mathbb{E}[(\frac{\tau}{2} - \xi^2)e^Q(T-1)]\big]$, three terms, with $Q = \Lambda_1\xi + \Lambda_2\xi^2/2$, $G_Q = \mathbb{E}[e^Q] = \beta^{-1/2}e^{\Lambda_1^2\sigma^2/(2\beta)}$, $\beta = 1 - \Lambda_2\sigma^2$, $\mu = \Lambda_1\sigma^2/\beta$ (Lemma 56 at $m = 0, 1$), and $T = e^{\Omega}$, $\Omega = \sum_{n\geq3}\Lambda_n\xi^n/n!$, the exponential of the higher cumulants. The μ^2 term's phase is pinned by the pole: $|\cos\arg\Phi(\rho(t))| = |\psi(t)|/|\Phi(\rho(t))| \leq \max|\psi'|\,|t|/|\Phi|$ with $\psi(0) = 0$ (Lemma 57) and $|\Phi| \geq q^{3/4}(|\mathfrak{s}(Z)| - 0.49\tau)$ (Lemma 58); its argument's sine obeys $|\sin 2\arg\mu| \leq 13.3\,\tau^{3/2}$, the two $O(1/w)$ tilts of Λ_1 and β cancelling to $O(\tau)$ because they are the same Lambert data shifted by one index. Every remaining quantity is bounded by an interval constant in scaled variables, each resting on a one-line monotonicity ($(1-e^{-\tau})/\tau$ decreasing; $e^{c\tau}$, $\rho^2 = 2(1-q)e^{\tau}$ increasing; w, q decreasing), with tails via $1 - q^k \geq (1-q)kq^{k-1}$ and the signed cancellations via the alternating-series bound $\tau^2/6 - \tau^3/12 \leq 1 + q - 2(1-q)/\tau \leq \tau^2/6$. The degree-$\geq 11$ remainder is handled on two intervals: on $[1.2\cdot10^{-4}, 5\cdot10^{-3}]$ by an exponential majorant whose profile $e^{0.097/\sqrt{\tau}}\tau^{4.5}$ is increasing exactly there (threshold $1.16\cdot10^{-4}$), total $\leq 3.03\,w$; on

$(0, 1.2\cdot 10^{-4}]$ with no exponential device (linear remainder geometric with ratio $\frac{1}{2}$ on the cutoff region; all products by $|e^{\Omega} - 1 - \Omega| \le \frac{1}{2}|\Omega|^2 e^{|\Omega|}$ with $|\Omega| \le 0.45$ on $|\xi| \le t_0\kappa^{-1/3}$; the intermediate zone suppressed by the Gaussian; here $t_0 = 1-\rho$ and $\kappa = \rho/(1-q)$), total $\le 2.10\,w$. Both are endpoint evaluations of quantities proved monotone; on $[10^{-8}, 5\cdot 10^{-3}]$ the same conclusion is obtained without any monotonicity assumption by the interval-arithmetic certification `budget_iv.py` described above, and the monotonicity argument is what carries the remaining range $(0, 10^{-8}]$. The assembly is the committed script `budget.py`. □

Lemma 60 (the denominator at the pole)**.** *At every travel pole, exactly: $S_e = \frac{qZ}{2}\,\mathfrak{s}(Z) - P_{12}$. Consequently, for $\tau \le 5\cdot 10^{-3}$,*

$$|S_e| \;\ge\; 0.90\cdot \tfrac{qZ}{2} - |P_{12}| \;\ge\; \Big[\tfrac{0.90}{\sqrt{2}}\sqrt{q\,v} - 0.619\,\tau\Big]\sqrt{\tau} \;\ge\; 0.63\,\sqrt{\tau}, \qquad v = \tfrac{1-q}{\tau},$$

and S_e has the sign of $\mathfrak{s}(Z)$.

Proof. The identity is Lemma 51 rearranged, since $S_e = \cos(z_0; q^2)$ (Proposition 12). For the bound: $|\mathfrak{s}(Z)| \ge 0.90$ (Lemma 58), $\frac{qZ}{2} = \sqrt{q(1-q)/2} = \sqrt{\tau/2}\,\sqrt{qv}$, $|P_{12}| \le 0.619\,\tau^{3/2}$ (Theorem 59), and $\sqrt{qv} \ge \sqrt{(1-\tau)(1-\tau/2)} \ge 1 - \frac{4}{5}\tau$ for $\tau \le \frac{1}{2}$ (since $(1-\frac{4}{5}\tau)^2 \le (1-\tau)(1-\frac{\tau}{2})$ there); at $\tau = 5\cdot 10^{-3}$ the bracket is ≥ 0.630 (exact endpoint value 0.6309) and it increases as τ decreases. The sign statement holds because $0.90\cdot\frac{qZ}{2} > |P_{12}|$ with the same margin. □

Corollary 61 (the gate closes)**.** *At every travel pole the gate* (10) *holds, so $\mathcal{N}_U(q_m) \ne 0$ (the enumeration standing of the six-pole list is that of Theorem 59). Quantitatively, for $\tau \le 5\cdot 10^{-3}$:*

$$|s| \;=\; \frac{q}{1-q}\,\frac{|P_{12}|}{|S_e|} \;\le\; \frac{\tau}{e^{\tau}-1}\cdot\frac{0.619}{0.63} \;\le\; 0.983 \;<\; 1,$$

and $b_0 > 0$: by the supplement's closed form $b_0 = \frac{2q}{1-q}\,S_o/S_e$ [20] with $S_o = \frac{z_0}{2}\,\mathfrak{s}(z_0)$ (Lemma 51), the pole invariant $q^{3/2}\mathfrak{s}(z_0)\mathfrak{s}(Z) = 1$ (Proposition 52's proof) gives $\operatorname{sign} S_o = \operatorname{sign}\mathfrak{s}(Z) = \operatorname{sign} S_e$ *(Lemma 60), so $S_o/S_e > 0$. At the six poles with $\tau > 5\cdot 10^{-3}$ the gate quantities are evaluated directly (`star_gate_certificate.py`): $|s| = 0.296, 0.256, 0.252, 0.251, 0.251, 0.250$ and $b_0\tau =$ 2.174, 2.001, 2.000, 2.000, 2.000, 2.000; moreover $|S_e|/\sqrt{\tau} \ge 0.58$ at all six, so the bound $0.35\sqrt{\tau}$ of Proposition 46 holds at the dominant pole, as its statement records.*

Machine verification of the q-trigonometric layer

Every series identity of the q-trigonometric sections and of this appendix is proved by matching the j-th summand; the matching is exponent bookkeeping, which is where such arguments fail in practice. That bookkeeping, and the Casoratian constancy behind the half-step invariant, are machine-checked in Lean 4 (`QTrigIdentities.lean`; the last four rows of the table below live in `SelowerSkeleton.lean`); convergence and the interchange of summation are analytic and are not formalised.

Paper step	Lean declaration	axioms
half-step rule for c, exponent match	`cshift_exponent`	`propext`
half-step rule for s, exponent match	`sshift_exponent`	`propext`
parity split of Φ	`parity_exponent`	`propext`
functional equation of F	`qairy_exponent`	`propext`
P_{12} k-sum split, low branch	`P12_split_lo`	`propext`
P_{12} k-sum split, high branch	`P12_split_hi`	`propext`
Casoratian constancy	`casoratian_re_const`	standard
constancy $\Rightarrow$ every rung	`casoratian_re_eq_zero`	standard
shift rule at the pole	`shift_at_pole`	`propext`
P_{12} as a second difference	`P12_second_difference`	`propext`
scaling $\tau^2 w = \sqrt{2}\,\tau^{3/2}$	`tau_sq_mul_w_sq`	standard
gate threshold $2/w^3 = \tau^{3/2}/\sqrt{2}$	`gate_threshold`	standard

The algebraic skeleton of Proposition 46 is likewise machine-checked (`SelowerSkeleton.lean`): `pole_facts` derives the two exact facts $s(qZ) = s(Z)$ and $c(qZ) = qZ\,s(Z)$ from the two shift rules together with $c(Z) = 0$; `F_at_pole` collapses the invariant to $q\,s(Z)^2$ at a pole; `form_lower_bound` is the quadratic-form estimate behind $\lambda_n \geq q - Z/2$; and `prod_one_sub_ge` is the Weierstrass bound $\prod(1 - \delta_i) \geq 1 - \sum \delta_i$ used in the discrete Gronwall step. The analytic inputs (the annulus representation, the Poisson envelope and the mean value theorem) are hypotheses there, not formalised.

"standard" means `propext`, `Classical.choice`, `Quot.sound`; no file in the accompanying development contains `sorry`, and each ends with a `#print axioms` audit. The realness of the ladder coefficient is what makes the added term in `casoratian_re_const` purely imaginary, and is carried as an explicit hypothesis.

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